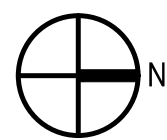
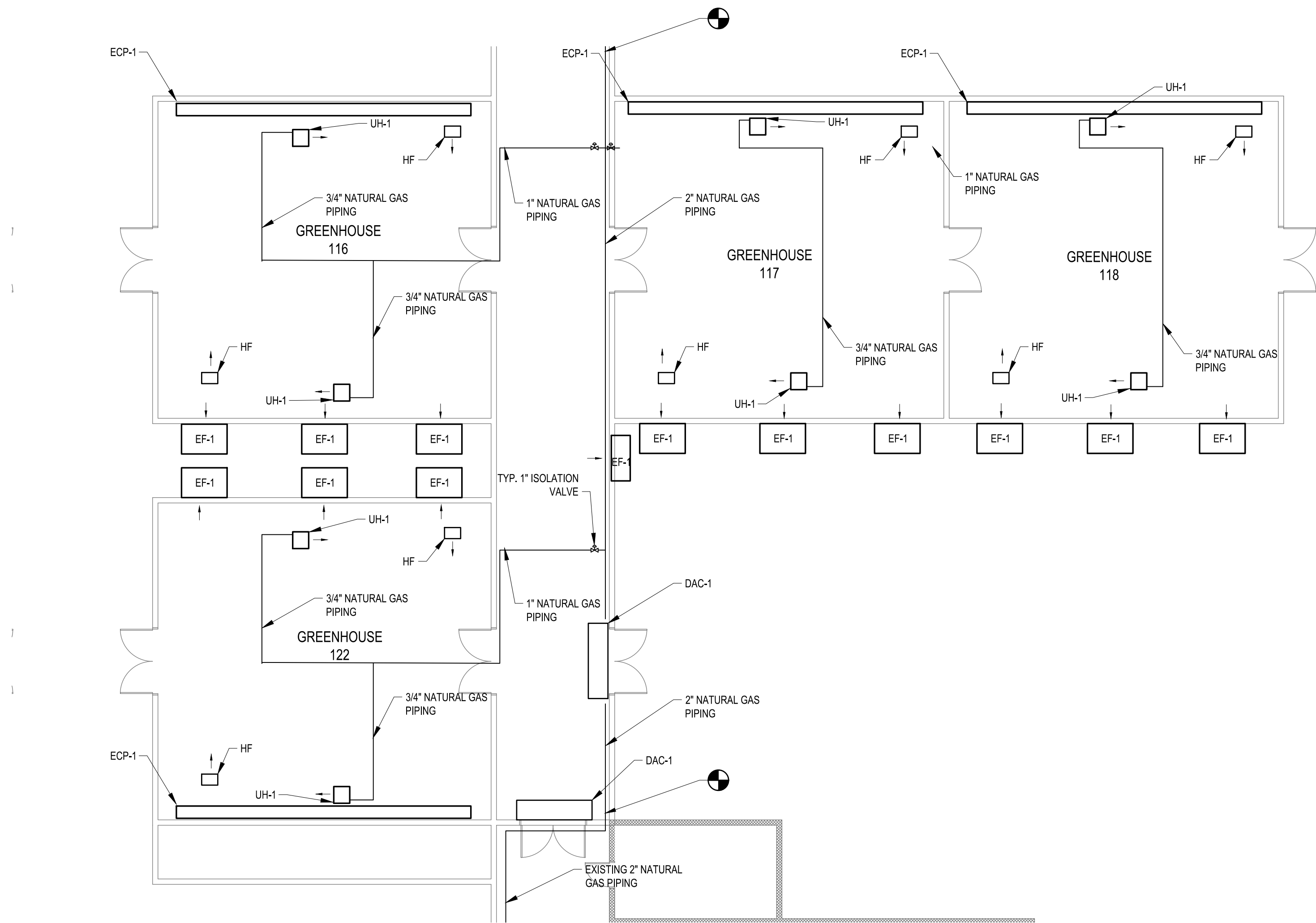


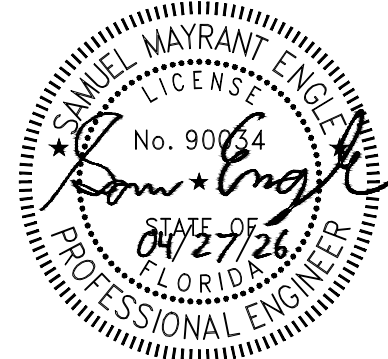
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M-103

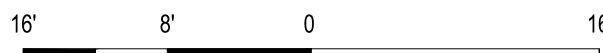
MECHANICAL PLAN: RENOVATION - BID OPTION #2

SCALE: 3/32"=1'-0"



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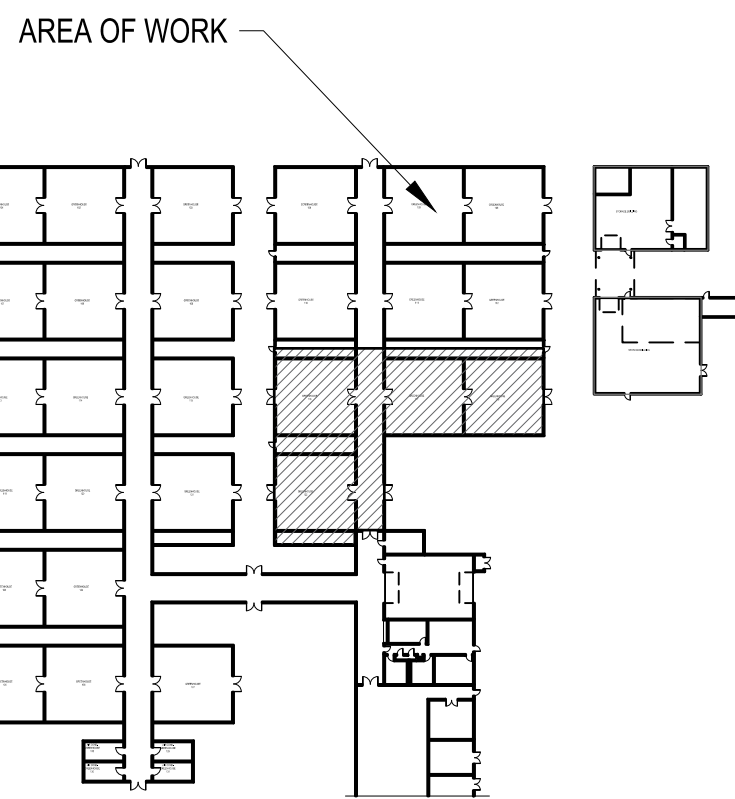
POINT OF CONNECTION



SCALE: 3/32" = 1'-0"

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10. DEMOLITION AND NEW CONSTRUCTION SHALL BE PHASED IN ACCORDANCE WITH THE PHASING PLAN.



KEYPLAN

NTS

THE JOHNSON-McADAMS FIRM, INC.
P.O. BOX 915, GREENWOOD, MISSISSIPPI 39202-4585

A-E FIRM

USDA

PROJECT MANAGER

EPM

BP

XXX

DESIGNER

PPM

JH

XXX

CHECKED BY

SAFETY & HEALTH

JGM

XXX

JOB #

REAL PROPERTY

4825

XXX

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USDA Agricultural Research Service

USDA AGRICULTURAL RESEARCH SERVICE
US HORTICULTURAL RESEARCH LABORATORY
REPAIR TORNADO DAMAGE
FT. PIERCE, FL

MECHANICAL PLAN
RENOVATION - BID OPTION #2

PROJECT NO.

XXX

DATE

04/27/26

AGRICULTURAL RESEARCH SERVICE

SOUTHEAST AREA

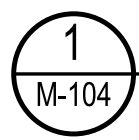
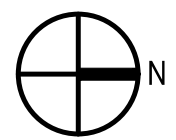
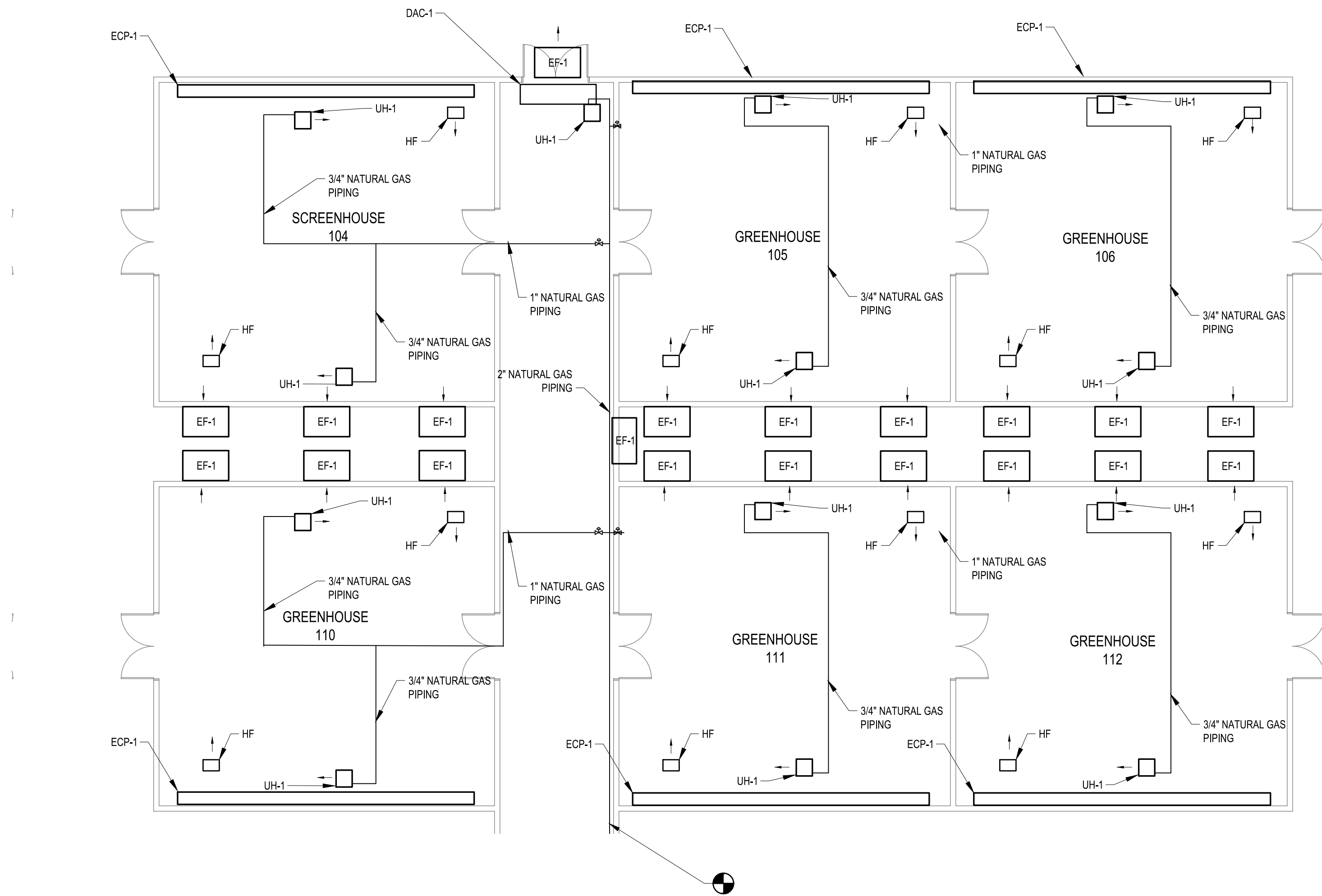
STONEVILLE, MS

SHEET

M-103

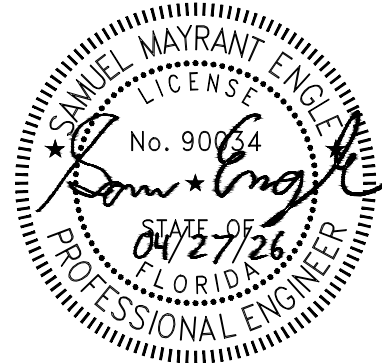
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Apr 27, 2025 - 10:02am

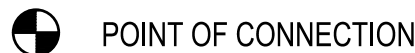


MECHANICAL PLAN: RENOVATION - BID OPTION #3

SCALE: 3/32\"/>



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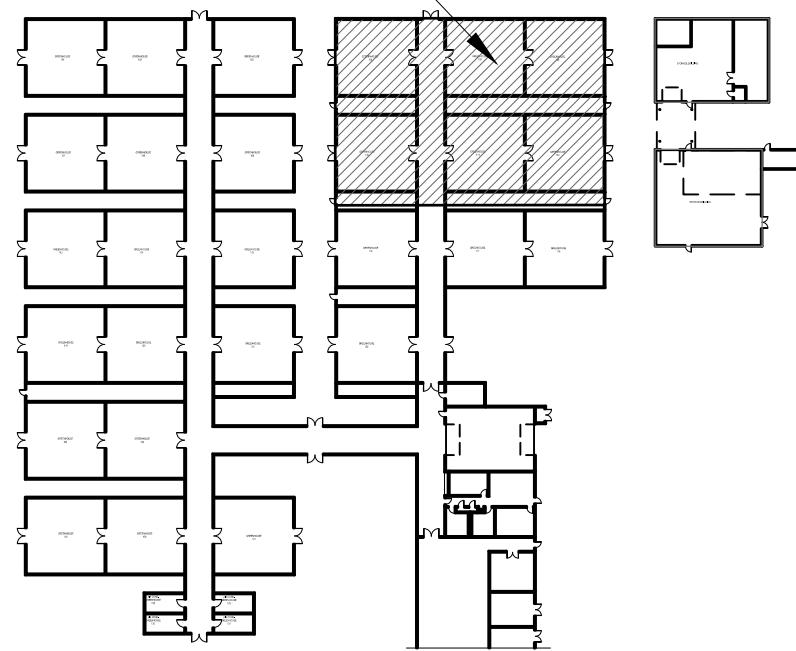


SCALE: 3/32\"/>

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10. DEMOLITION AND NEW CONSTRUCTION SHALL BE PHASED IN ACCORDANCE WITH THE PHASING PLAN.
11. PROVIDE TEMPORARY GAS AND DOMESTIC WATER. UTILITY PATH IS SHOWN ON P-100.

AREA OF WORK



KEYPLAN

NTS



A-E FIRM

USDA

PROJECT MANAGER

EPM

DESIGNER

PPM

CHECKED BY

SAFETY & HEALTH

JGM

XXX

JOB #

REAL PROPERTY

4825

XXX

USDA



USDA AGRICULTURAL RESEARCH SERVICE
US HORTICULTURAL RESEARCH LABORATORY
REPAIR TORNADO DAMAGE
FT. PIERCE, FL

MECHANICAL PLAN
RENOVATION - BID OPTION #3

PROJECT NO.

XXX

DATE

04/27/26

AGRICULTURAL RESEARCH SERVICE

SOUTHEAST AREA

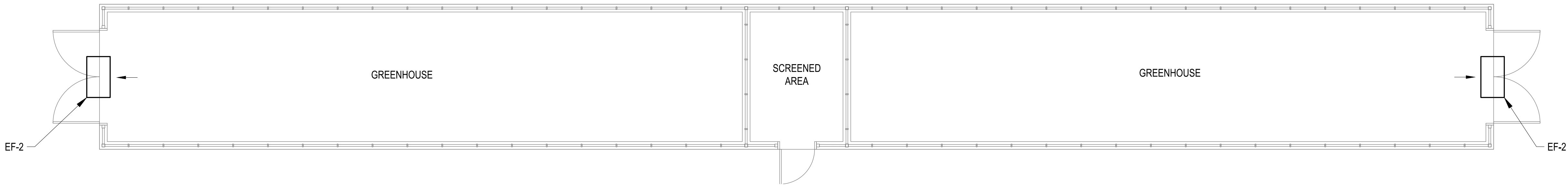
STONEVILLE, MS

SHEET

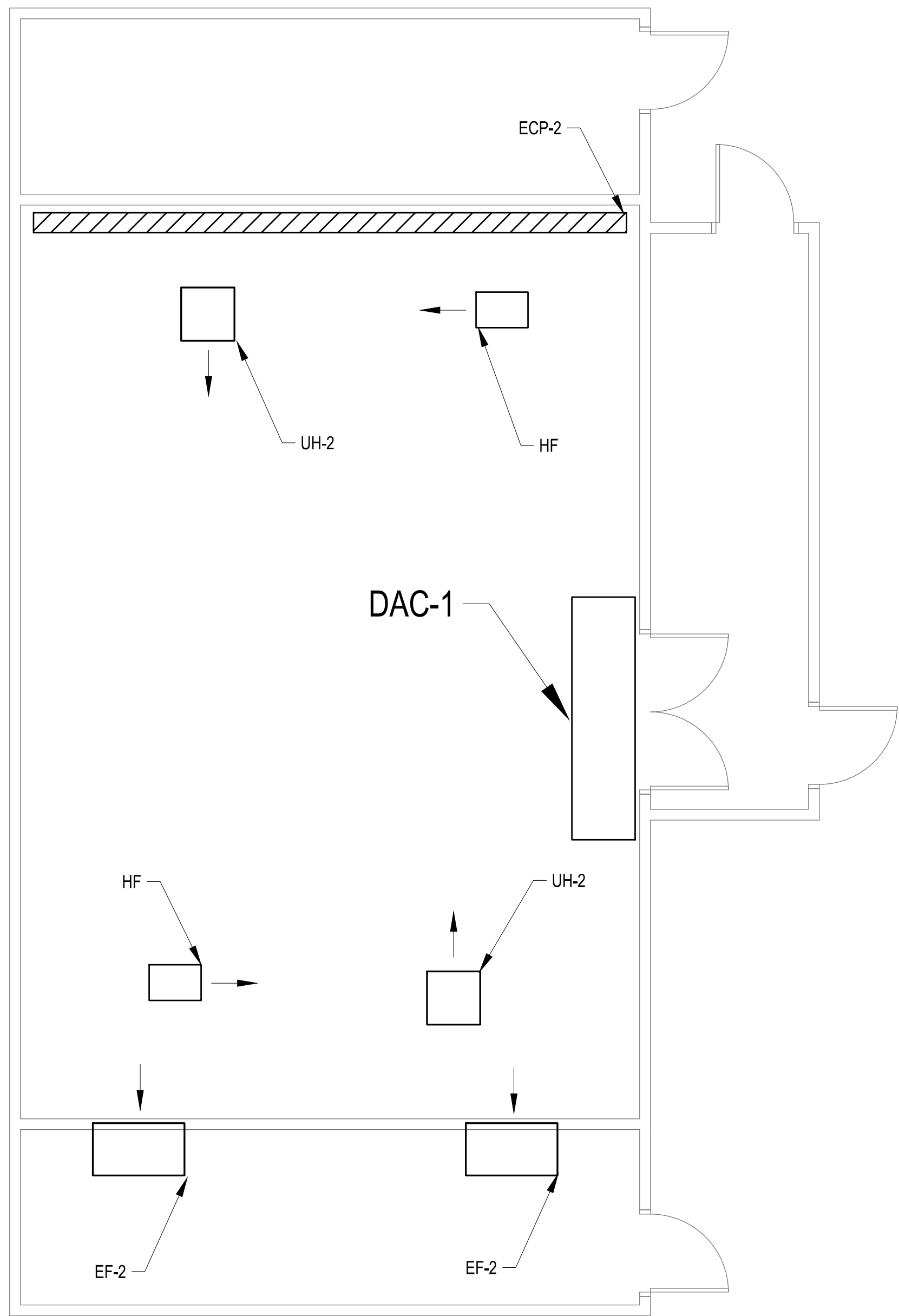
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81 OF 108

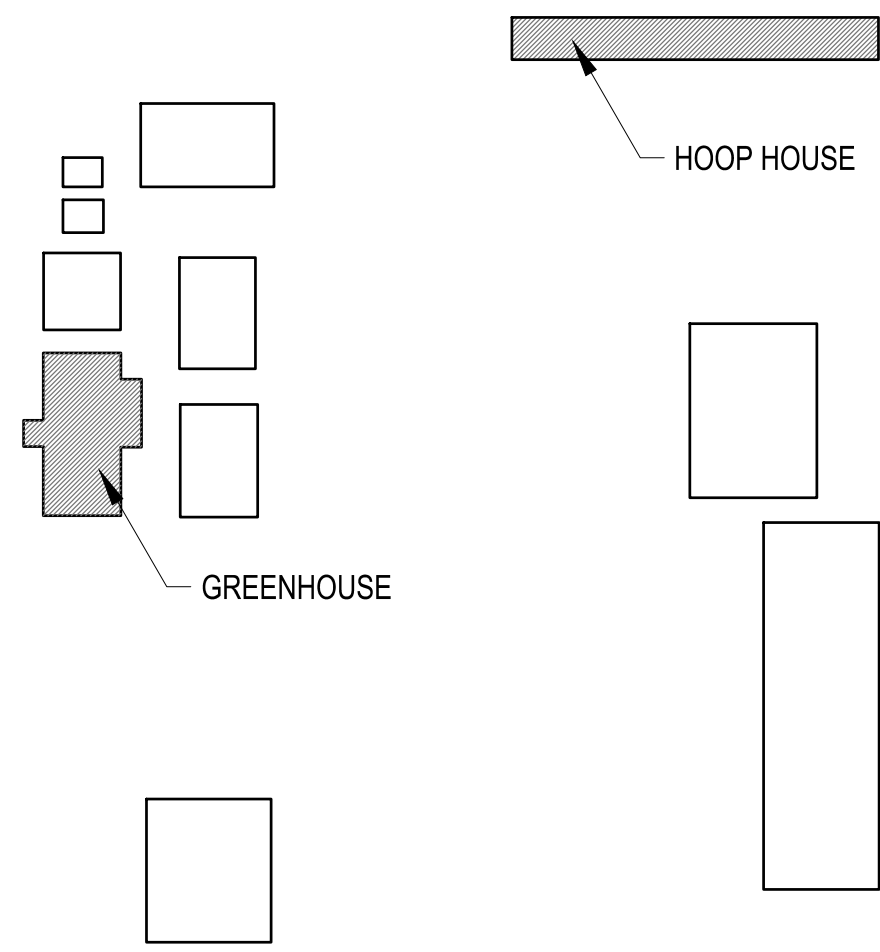
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Apr 27, 2025 - 10:02am



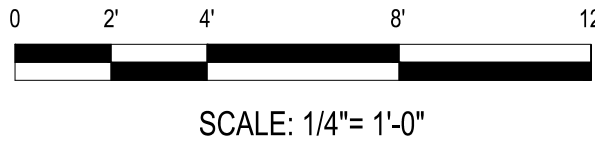
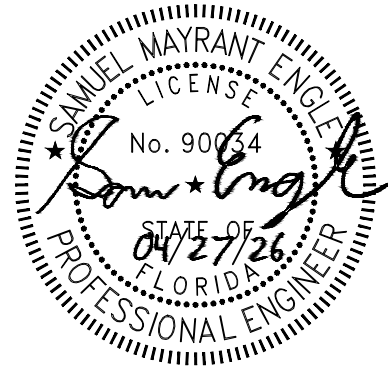
2 BID OPTION #4 MECHANICAL FLOOR PLAN HOOP HOUSE
M-105 SCALE: 1/4"=1'-0"



1 GREENHOUSE MECHANICAL PLAN
BID OPTION #4
M-105 SCALE: 1/4"=1'-0"

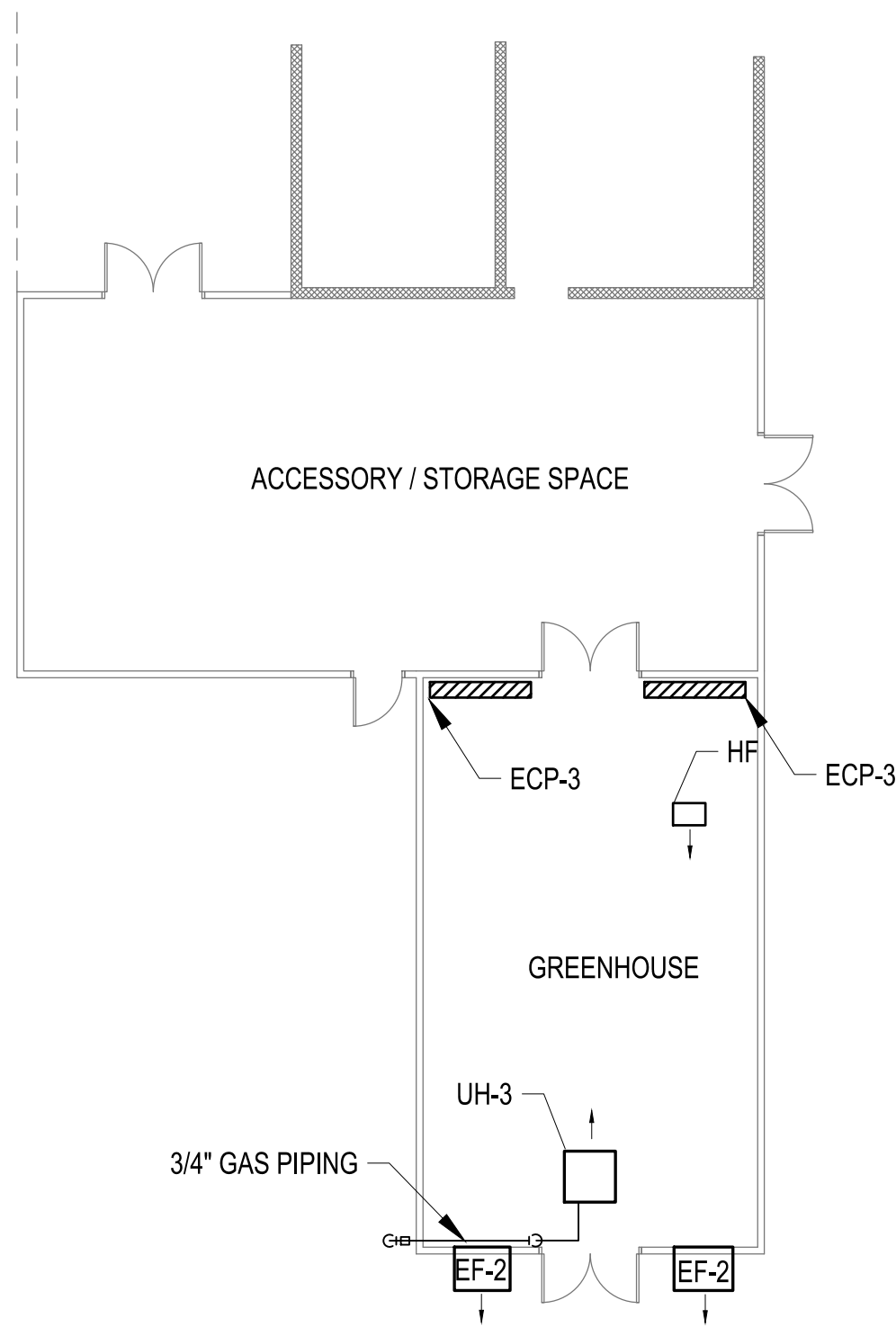


KEYPLAN - PICOS FARM SITE
NTS

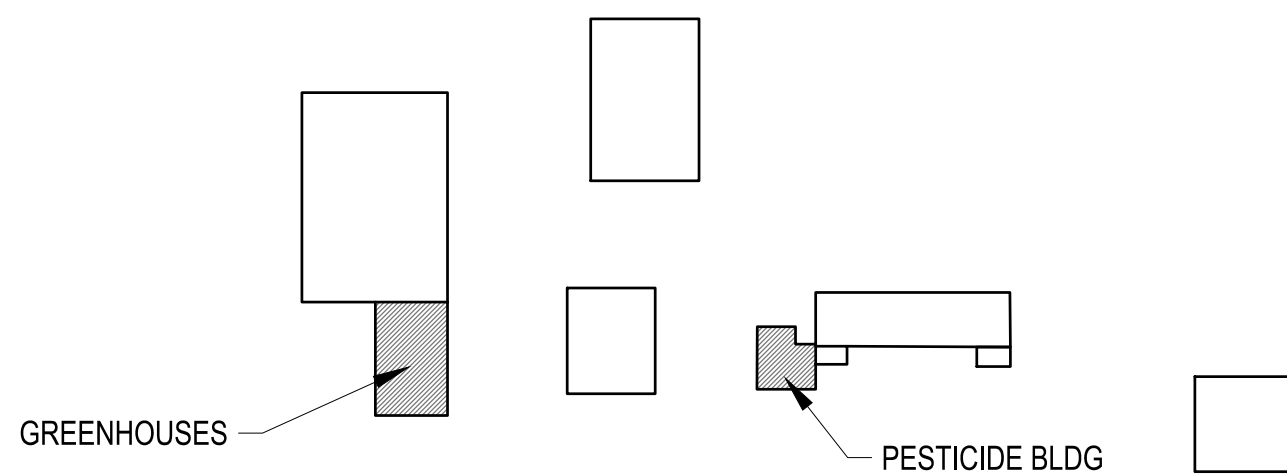
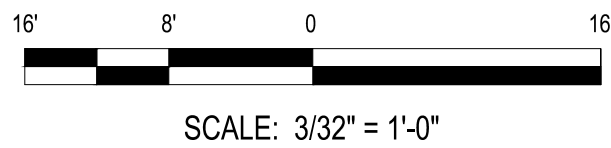
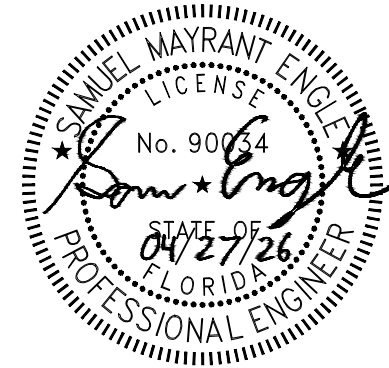


 THE JOHNSON-McADAMS GROUP P.O. BOX 515, GREENWOOD, MISSISSIPPI 39202-0515											
		NO.		DATE		BY		DESCRIPTION			
		REVISIONS									
A-E FIRM		USDA				Agricultural Research Service					
PROJECT MANAGER		EPM		USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL							
BP		XXX									
DESIGNER		PPM		MECHANICAL PLAN							
JH		XXX		RENOVATION - BID OPTION #4							
CHECKED BY		SAFETY & HEALTH		PROJECT NO.		AGRICULTURAL RESEARCH SERVICE				SHEET	
JGM		XXX		XXX		SOUTHEAST AREA				M-105	
JOB #		REAL PROPERTY		DATE		STONEVILLE, MS				82 OF 108	
4825		XXX		04/27/26							

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Apr 27, 2025 - 10:02am



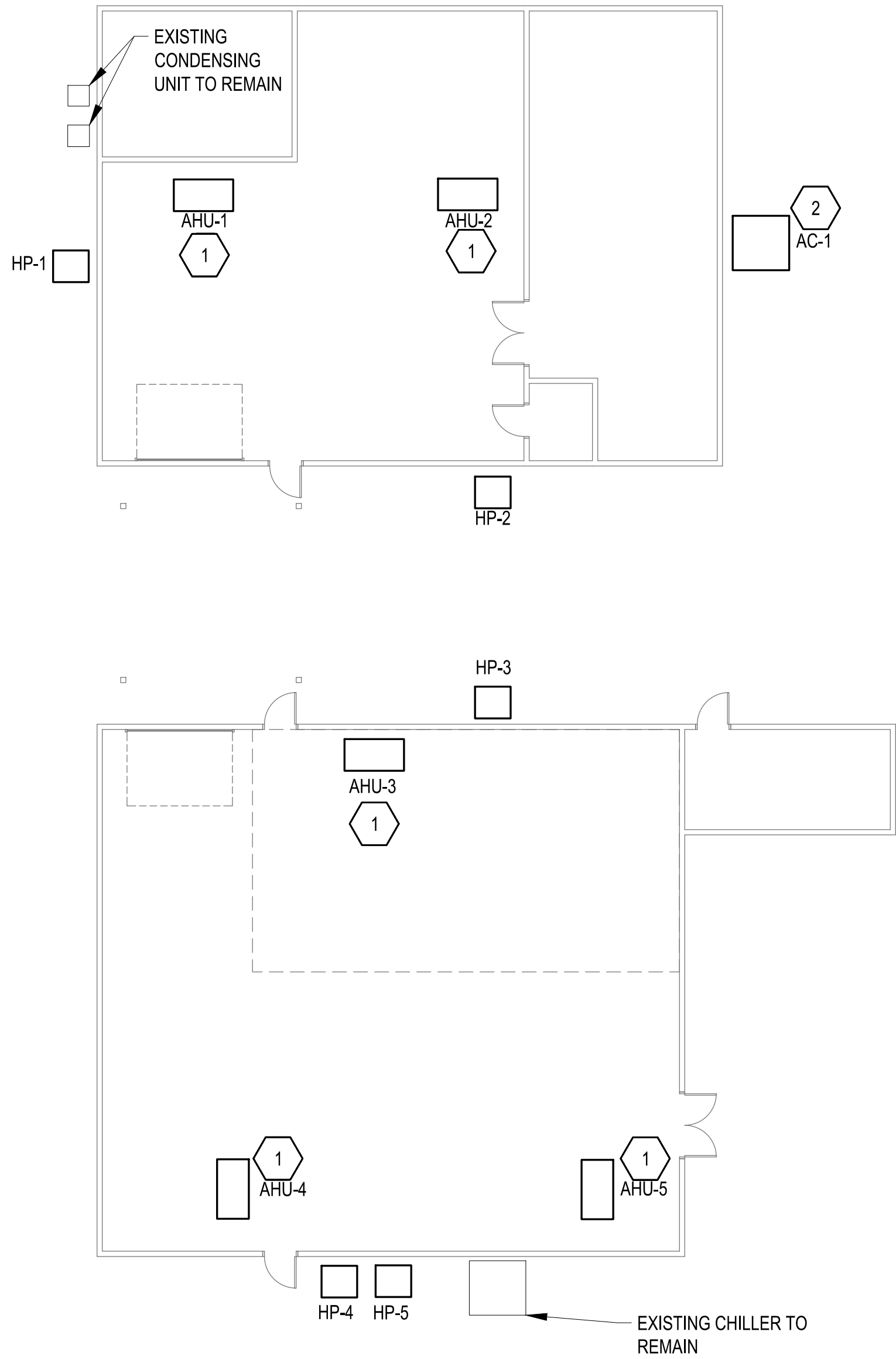
**GREENHOUSE MECHANICAL PLAN
- BID OPTION #5**
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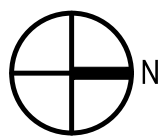


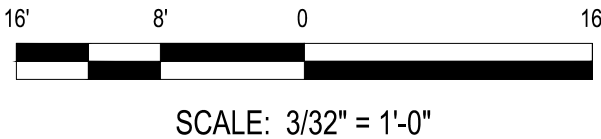
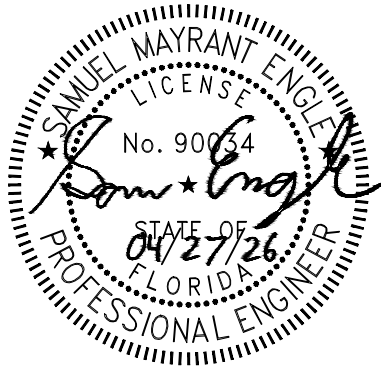
KEYPLAN
NTS

		NO.	DATE	BY	DESCRIPTION
		REVISIONS			
USDA		USDA Agricultural Research Service			
		USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL			
		MECHANICAL PLAN RENOVATION - BID OPTION #5			
		PROJECT NO.	AGRICULTURAL RESEARCH SERVICE		SHEET
A-E FIRM		USDA		SOUTHEAST AREA	
PROJECT MANAGER	BP	XXX	DATE		M-106
DESIGNER	JH	XXX	04/27/26		83 OF 108
CHECKED BY	JGM	SAFETY & HEALTH XXX	DATE		
JOB #	4825	REAL PROPERTY XXX			

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Apr 27, 2025 - 10:02am



  **ENLARGED MECHANICAL PLAN - BID OPTION #1**
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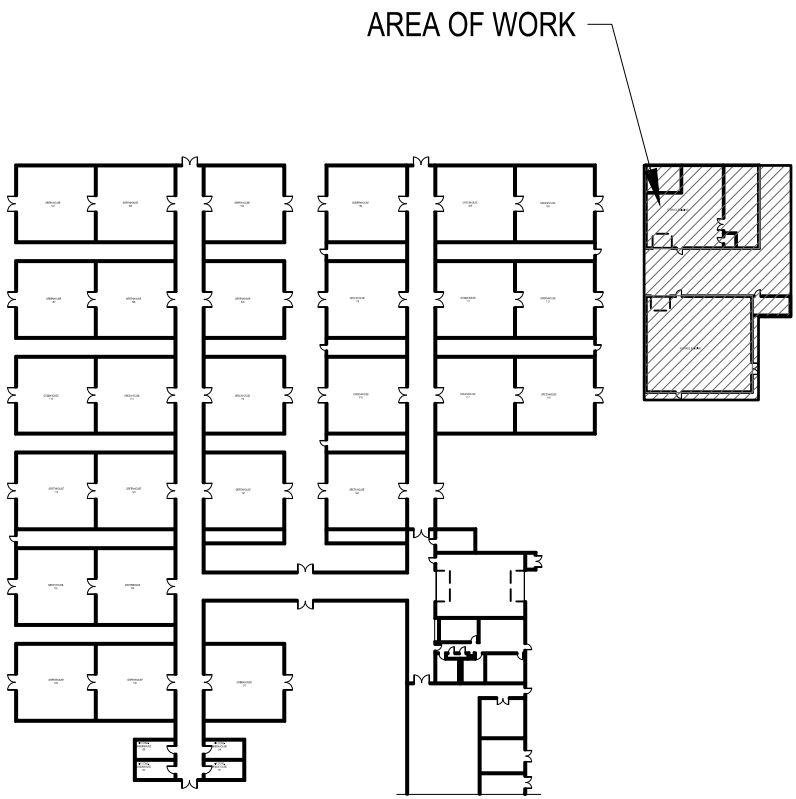
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10. DEMOLITION AND NEW CONSTRUCTION SHALL BE PHASED IN ACCORDANCE WITH THE PHASING PLAN.



KEYED NOTES

1. PROVIDE AIR HANDLING UNIT MOUNTED FROM ABOVE. MOUNT UNIT AT THE SAME HEIGHT OF EXISTING EQUIPMENT. RECONNECT TO EXISTING DUCTWORK AND CONDENSATE PIPING.
2. PROVIDE GROUND MOUNTED PACKAGED HEAT PUMP. RECONNECT TO EXISTING DUCT WORK.



 **The Johnson-McAdams Group, Inc.**
P.O. BOX 515, GREENWOOD, MISSISSIPPI 39202-0515

PROJECT MANAGER	EPM
DESIGNER	PPM
CHECKED BY	SAFETY & HEALTH
JOB #	REAL PROPERTY
4825	XXX

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USDA Agricultural
Research
Service

USDA AGRICULTURAL RESEARCH SERVICE
US HORTICULTURAL RESEARCH LABORATORY
REPAIR TORNADO DAMAGE
FT. PIERCE, FL

**ENLARGED MECHANICAL
PLAN- BID OPTION #1**

PROJECT NO.	XXX	AGRICULTURAL RESEARCH SERVICE		SHEET	M-401
JOB #	4825	DATE	04/27/26	STONEVILLE, MS	84 OF 108

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Apr 27, 2025 - 10:22am

UNIT HEATER SCHEDULE (EACH)									
MARK	TYPE	INPUT (BTUH)	OUTPUT (BTUH)	THERMAL EFFICIENCY	FAN (CFM)	FAN (HP)	VOLTAGE	AMPS	REMARKS
UH-1	NATURAL GAS	75,000	62,250	83%	961	2/3	120/1	3.7	SEE NOTES
UH-2	ELECTRIC	10,200	10,200	N/A	380	1/40	208/3	8.3	SEE NOTES
UH-3	PROPANE	75,000	62,250	83%	961	2/3	120/1	3.7	SEE NOTES
NOTES: 1. INTERGRATE WITH GREENHOUSE CONTROL SYSTEM. 2. PROVIDE WITH MOUNTING KIT. 3. EQUIPMENT SHALL COMPLY WITH THE "BUY AMERICAN ACT".									

GREENHOUSE FAN SCHEDULE											
MARK	MATERIAL	DIAMETER	TYPE	AIRFLOW (CFM)	RPM	E.S.P. (IN W.G.)	DRIVE	MOTOR			REMARKS
								HP	VOLTS	PHASE	
EF-1	ALUMINUM	36"	SIDEWALL PROP	10500	495	0.1"	BELT	1/2	208	3	SEE NOTE 1,2,3
EF-2	ALUMINUM	36"	SIDEWALL PROP	10500	495	0.1"	BELT	1/2	208	1	SEE NOTE 1,2,3
EF-3	ALUMINUM	30"	SIDEWALL PROP	5770	450	.1"	BELT	1/4	208	3	SEE NOTE 1,2,3
HF	ALUMINUM	18"	CEILING MOUNTED HORIZONTAL	2000	1100	NA	DIRECT	1/15	115	1	SEE NOTE 1
NOTES:											
1. PROVIDE SINGLE-SPEED MOTOR.											
2. PROVIDE WITH SLANT WALL HOUSING											
3. PROVIDE WITH ALUMINUM BACKDRAFT SHUTTER.											

EVAPORATIVE COOLING PAD												
MARK	CONSTRUCTION	TROUGH	LENGTH	HEIGHT	GPM	EAT		LAT		MEDIA THICKNESS	MAX. FACE VELOCITY	REMARKS
						DB(°F)	WB(°F)	DB(°F)	WB(°F)			
ECP-1	STAINLESS STEEL AND PVC	ONE PIECE PVC	36'	4'	68.3	95	78	83	83	6"	125	SEE NOTES
ECP-2	STAINLESS STEEL AND PVC	ONE PIECE PVC	22'	4'	68.3	95	78	83	83	6"	125	SEE NOTES
ECP-3	STAINLESS STEEL AND PVC	ONE PIECE PVC	6'	4'	68.3	95	78	83	83	6"	125	SEE NOTES
NOTES: 1. INTERLOCK ECP CONTROL WITH GREENHOUSE MASTER CONTROLLER. 2. INSTALL SYSTEM IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS. 3. PROVIDE WITH SUMP PUMP 110 VAC. 4. CONNECT TO ADJACENT DOMESTIC WATER LINE FOR MAKE UP WATER SUPPLY												

GREENHOUSE AIR CURTAIN SCHEDULE									
MARK	NOZZLE WIDTH	NOZZLE VELOCITY	AIR VOLUME	POWER	WEIGHT	MOTOR			REMARKS
						HP	VOLTS	PHASE	
DAC-1	111"	3,300 FPM	3500 CFM	1.5 KW	120 LBS	3 @ 1/2	115	1	ALL
NOTES: 1. PROVIDE SINGLE-SPEED MOTOR. 2. PROVIDE WITH DOOR ACTIVATION.									

HEAT PUMP SCHEDULE										
MARK	SYSTEM TYPE	COOLING CAPACITY (BTU/HR)	SENSIBLE CAPACITY (BTU/HR)	SEER	OUTDOOR TEMPERATURE (F)	VOLTS/ PHASE	MCA	MOCP	WEIGHT (LBS)	REMARKS
HP-1	AIR-COOLED	46180	34476	15.2	95	230/3	18.1	30	251	ALL
HP-2	AIR-COOLED	56927	42229	14.30	95	230/3	28.2	35	251	ALL
HP-3	AIR-COOLED	23030	16647	16.00	95	230/1	13	20	174	ALL
HP-4	AIR-COOLED	23030	16647	16.00	95	230/1	13	20	174	ALL
HP-5	AIR-COOLED	46180	34476	15.2	95	230/3	18.1	30	251	ALL
REMARKS 1. PROVIDE AND INSTALL REFRIGERANT SUPPLY AND RETURN LINES PER MANUFACTURER'S RECOMMENDATIONS.										

PACKAGED UNIT SCHEDULE																					
MARK	TYPE	SUPPLY AIRFLOW DATA				DX COOLING				ELECTRIC HEATING			FILTER	EER	SEER2	VOLTS/ PHASE	OPER. WEIGHT (LBS)	MOP	MCA	REMARKS	
		SA FLOW (CFM)	ESP (IN H2O)	TSP (IN H2O)	FAN SPEED (RPM)	CAPACITY		AIR SIDE		CAPACITY		AIR SIDE									
						TOTAL MBH	SENS MBH	EAT °F (DB/WB)	LAT °F (DB/WB)	KW	EAT °F	LAT °F									
AC-1	PACKAGED	1200	2.0	2.5	1241	36.8	25.5	80/67	59.0/57.0	11.27	60	90	MERV 8	11.0	13.8	208/1	400	70	68	SEE NOTES	
NOTES: 1. PROVIDE DOUBLE WALL CONSTRUCTION 2. PROVIDE WITH HORIZONTAL DISCHARGE. 3. PROVIDE WITH UNIT MICROPROCESSOR CONTROLLER CAPABLE OF STANDALONE OPERATION WITH INTEGRAL SCHEDULE AND OPERATIONS SETPOINTS (ADJ.). LOCATE MICROPROCESSOR CONTROL ADJACENT TO THERMOSTAT. 4. PROVIDE BAROMETRIC RELIEF.																					

AIR HANDLING UNIT SCHEDULE																
MARK	UNIT TYPE	AIRFLOW (CFM)	COOLING						HEATING			UNIT WT (LBS)	ELECTRICAL DATA			REMARKS
			CAPACITY		AIRSIDE				CAPACITY		AIRSIDE		VOLTS/ PHASE	MCA	MOCP	
			TOTAL (BTU/HR)	SENSIBLE (BTU/HR)	EAT (db °F)	EAT (wb °F)	LAT (db °F)	LAT (wb °F)	TOTAL (BTU/HR)	EAT (db °F)	LAT (db °F)					
AHU-1	HORIZONTAL DUCTED	1600	46184	34476	80	67	59.70	57.6	46170	70	86	170	230/1	9.0	15	ALL
AHU-2	HORIZONTAL DUCTED	1900	56927	42229	80	67	59.1	57.2	52500	70	86	170	230/1	9.0	15	ALL
AHU-3	HORIZONTAL DUCTED	700	23030	16647	80	67	57.6	56.1	22200	70	88	140	230/1	5.0	15	ALL
AHU-4	HORIZONTAL DUCTED	700	23030	16647	80	67	57.6	56.1	22200	70	88	140	230/1	5.0	15	ALL
AHU-5	HORIZONTAL DUCTED	1600	46184	34476	80	67	59.70	57.6	46170	70	86	170	230/1	9.0	15	ALL
REMARKS 1. AHU SHALL HAVE MERV 8 RATED FILTERS. 2. PROVIDE WITH THERMOSTAT BY UNIT MANUFACTURER.																

PACKAGED WALL MOUNT UNIT SCHEDULE										
MARK	UNIT TYPE	AIRFLOW (CFM)	COMPRESSOR TYPE	COOLING CAPACITY (BTU/HR)	EER	UNIT WT (LBS)	ELECTRICAL DATA			REMARKS
							VOLTS/ PHASE	MCA	MOCP	
WMU-1	WALL MOUNTED	1700	VARIABLE CAPACITY ECU	67000	10	600	230/3	39	60	ALL
REMARKS 1. AHU SHALL HAVE MERV 8 RATED FILTERS. 2. PROVIDE WITH THERMOSTAT BY UNIT MANUFACTURER.										

GREENHOUSE BAY CONTROLLER INPUT/ OUTPUT SUMMARY

	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT
TEMPERATURE SENSOR	X			
HUMIDITY SENSOR	X			
FAN #1 START/STOP				X
FAN #2 START/STOP (LOW SPEED)				X
FAN #2 START/STOP (HIGH SPEED)				X
VENT WALL MOTOR (OPEN)				X
VENT WALL MOTOR (CLOSE)				X
VENT ROOF MOTOR (OPEN)				X
VENT ROOF MOTOR (CLOSE)				X
EVAPORATING COOLING WALL START/STOP				X
UNIT HEATER		X		
WALL VENT WALL OPEN LIMIT			X	
WALL VENT CLOSE WALL LIMIT			X	
ROOF VENT WALL OPEN LIMIT			X	
ROOF VENT CLOSE WALL LIMIT			X	
ADD ALT #1 CIRC, FANS START/STOP				X

NOTE: PROVIDE CONTROLLERS AS REQUIRED FOR THE INPUT/OUTPUT/OUTPUT SUMMARY ABOVE. PROVIDE CENTRAL MONITORING STATION IN THE GREENHOUSE MANAGERS OFFICE.

WEATHER STATION CONTROLLER INPUT/ OUTPUT SUMMARY

	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT
RAIN			X	
WIND DIRECTION	X			
WIND SPEED	X			
SOLAR K-LUX	X			
OUTDOOR TEMPERATURE	X			

NOTE: PROVIDE CONTROLLERS AS REQUIRED FOR THE INPUT/OUTPUT/OUTPUT SUMMARY ABOVE. WEATHER STATION CONTROLLER TO BE BY SAME MANUFACTURER TO GREENHOUSE BAY CONTROLLER.

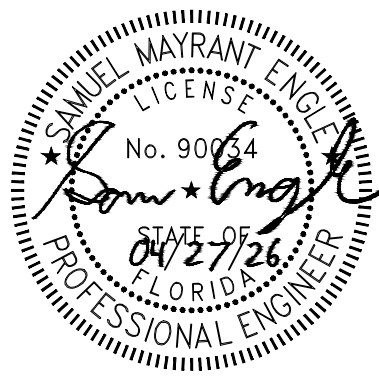
SEQUENCE OF OPERATION

OCCUPIED COOLING: STEPS TO MAINTAIN ADJUSTABLE SETPOINT AS

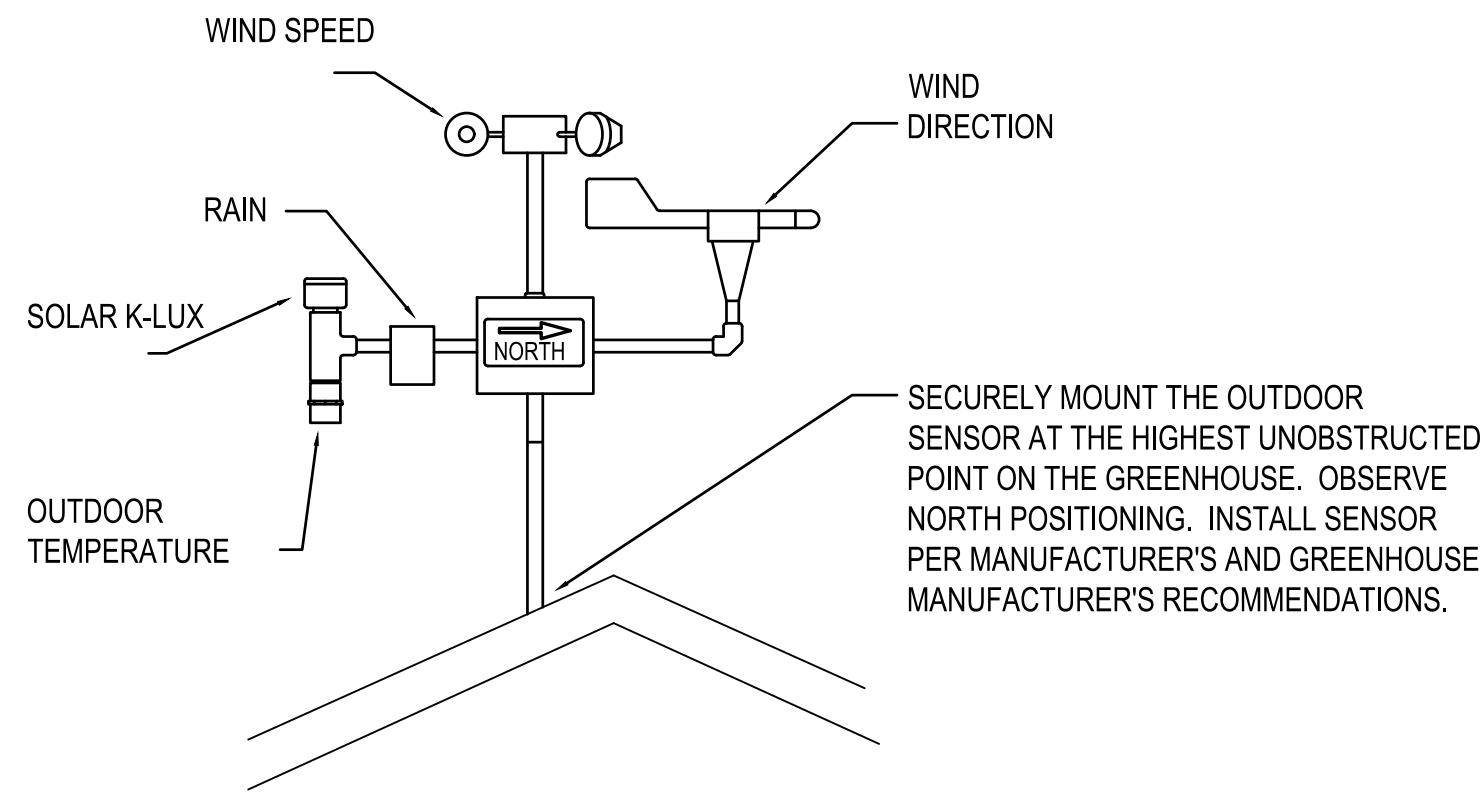
MEASURED BY TEMPERATURE SENSOR
STEP 1: OPEN SIDE VENTS. STEP 2: COOLING 1 - ACTIVATE LOW SPEED OF TWO SPEED FANS.
STEP 3: COOLING 2 - ACTIVATE HIGH SPEED OF TWO SPEED FANS.
STEP 4: COOLING 3 - ACTIVATE EVAPORATIVE COOLING WALL SYSTEM.

HEATING: STEPS TO MAINTAIN ADJUSTABLE SETPOINT
STEP 1: CLOSE ROOF AND SIDE VENTS.
STEP 2: HEATING 1 - MODULATE THE NEW ELECTRONIC VALVE ACTUATOR TO MAINTAIN TEMPERATURE SETPOINT.

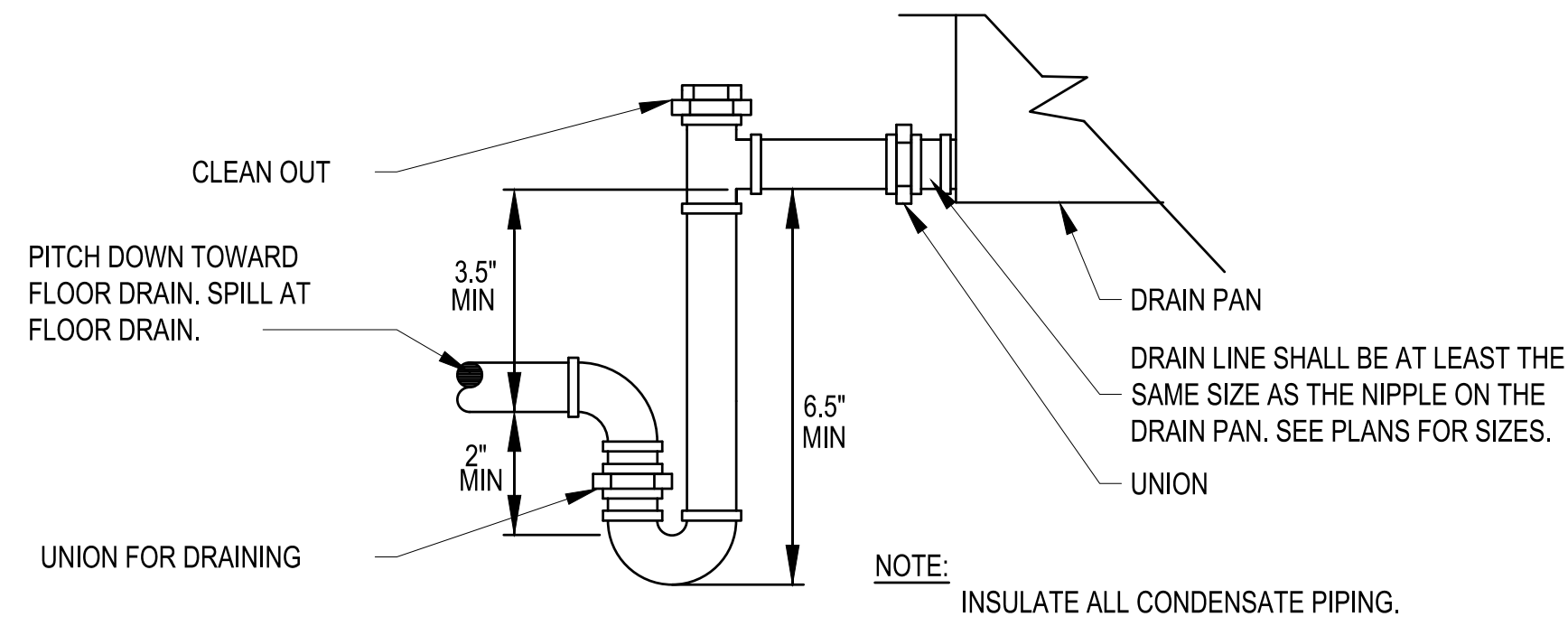
CONTROL SYSTEM SHALL MONITOR INDOOR AIR TEMPERATURE AND HUMIDITY, OUTDOOR AIR TEMPERATURE, RAIN CONDITIONS, WIND DIRECTION AND SPEED, AND SOLAR KLUX. CONTROL SYSTEM SHALL PROTECT GREENHOUSE VENTILATION SYSTEMS FROM HIGH WINDS AND RAIN DAMAGE. PROVIDE MANUAL OVERRIDE FOR ROOF VENTS. PROVIDE ALL WIRING AND CONTROLS FOR A COMPLETE OPERATIONAL SYSTEM.



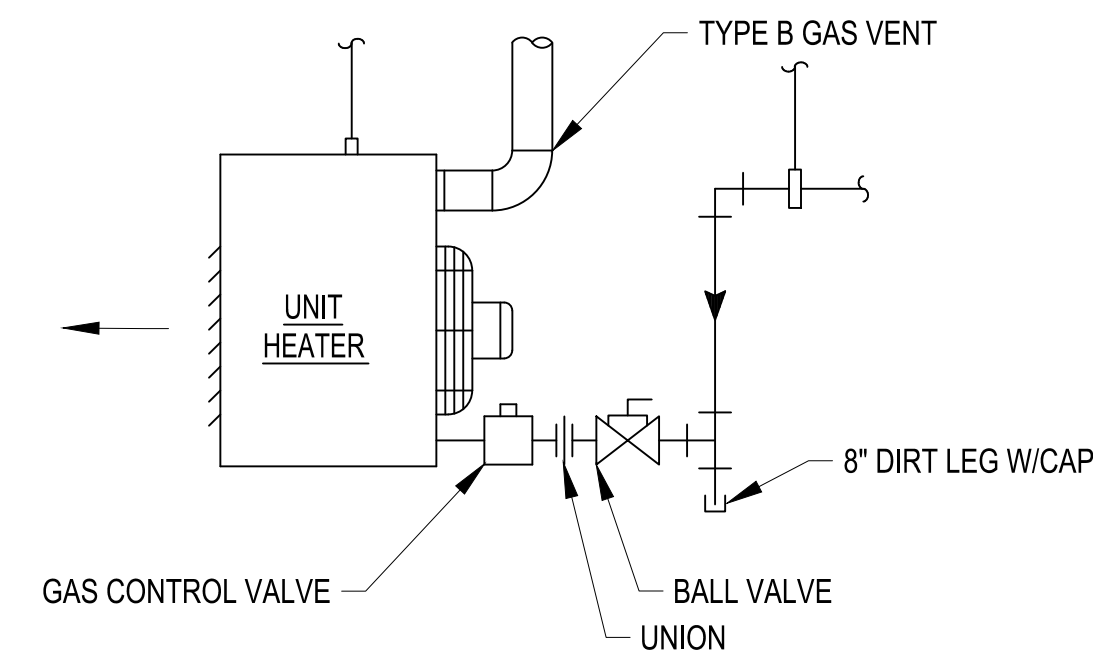
 The Johnson-MAdams Group P.O. BOX 515, GREENWOOD, MISSISSIPPI 39202-0515					
		NO.	DATE	BY	DESCRIPTION
REVISIONS					
A-E FIRM	USDA				
	PROJECT MANAGER	EPM	XXX		
	DESIGNER	JH	XXX		
	CHECKED BY	JGM	SAFETY & HEALTH XXX		
JOB #	REAL PROPERTY XXX				
	4825				
PROJECT NO.		AGRICULTURAL RESEARCH SERVICE		SHEET	M-601
DATE		04/27/26		SOUTHEAST AREA STONEVILLE, MS	



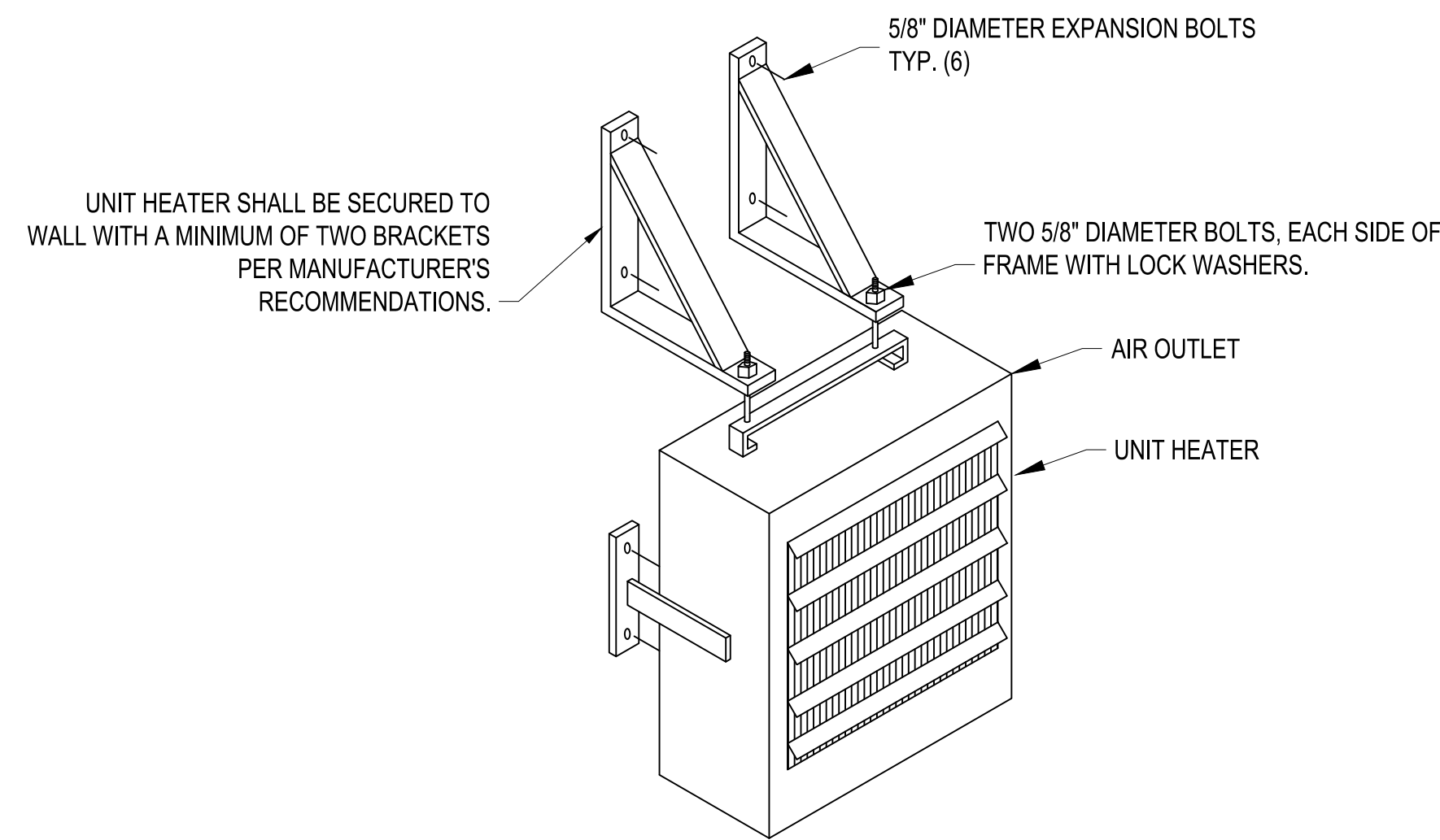
1 WEATHER STATION DETAIL
M602 SCALE: NOT TO SCALE



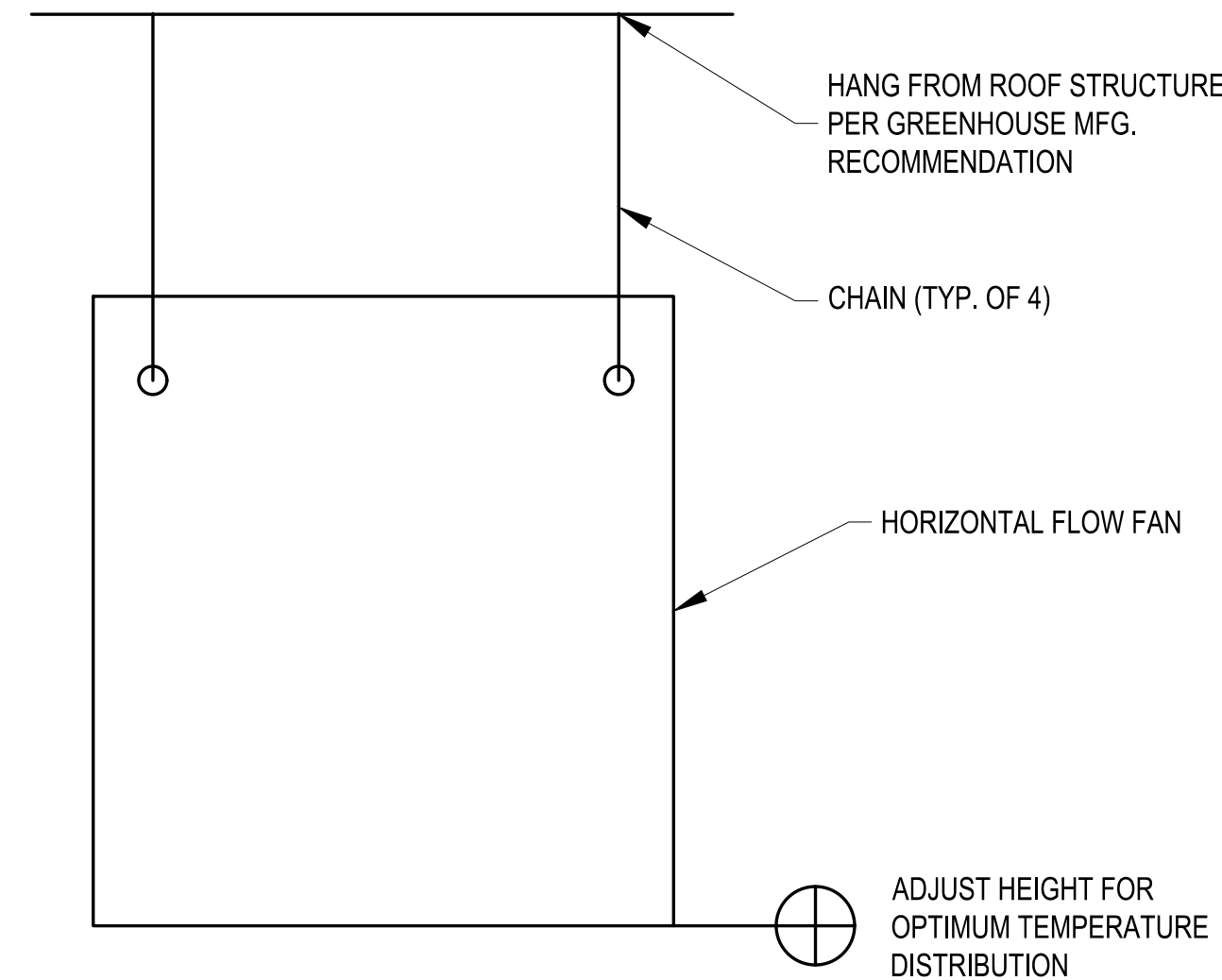
2 CONDENSATE DRAIN P-TRAP DETAIL
M602 SCALE: NOT TO SCALE



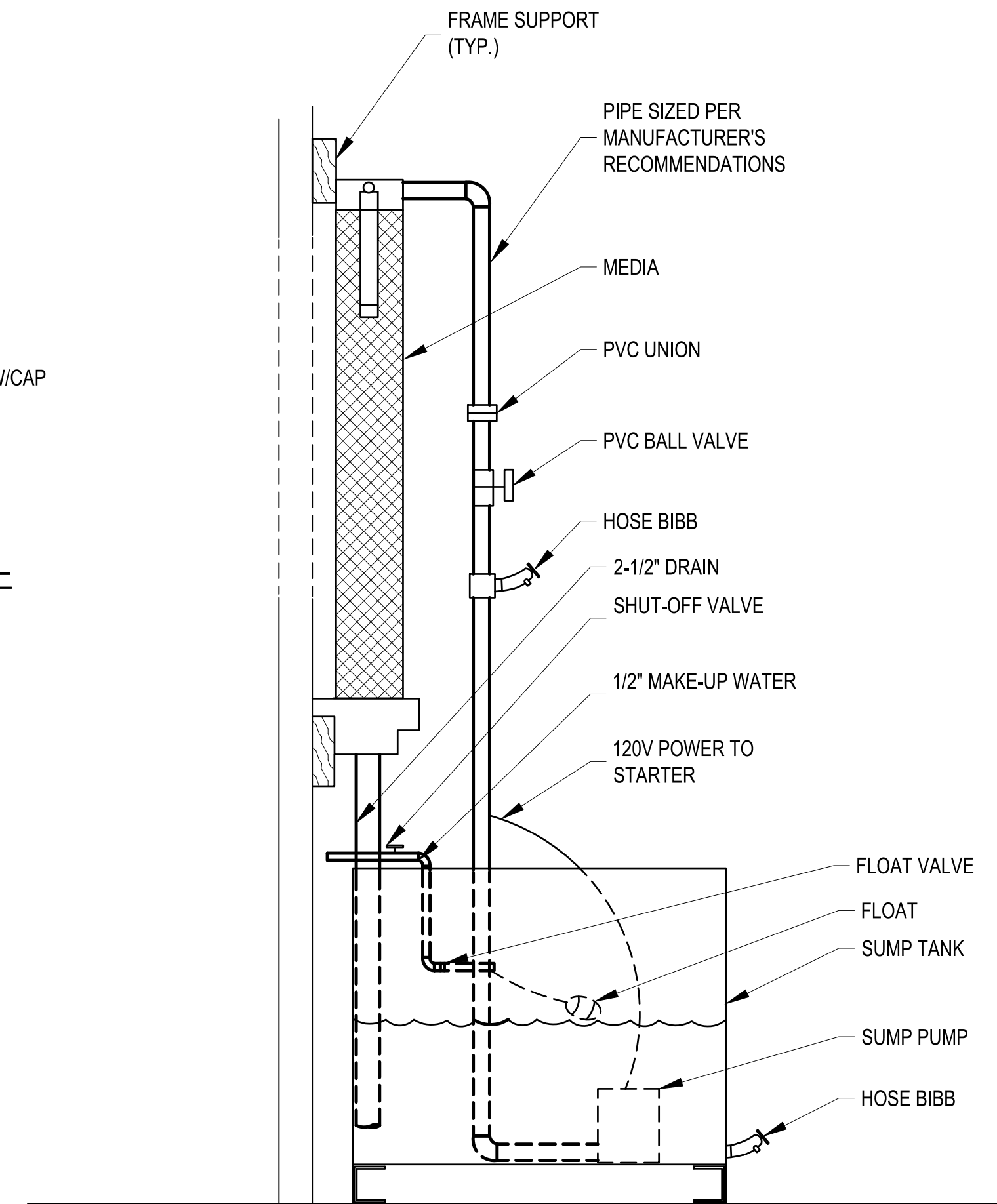
3 GAS FIRED UNIT HEATER DETAIL
M602 SCALE: NOT TO SCALE



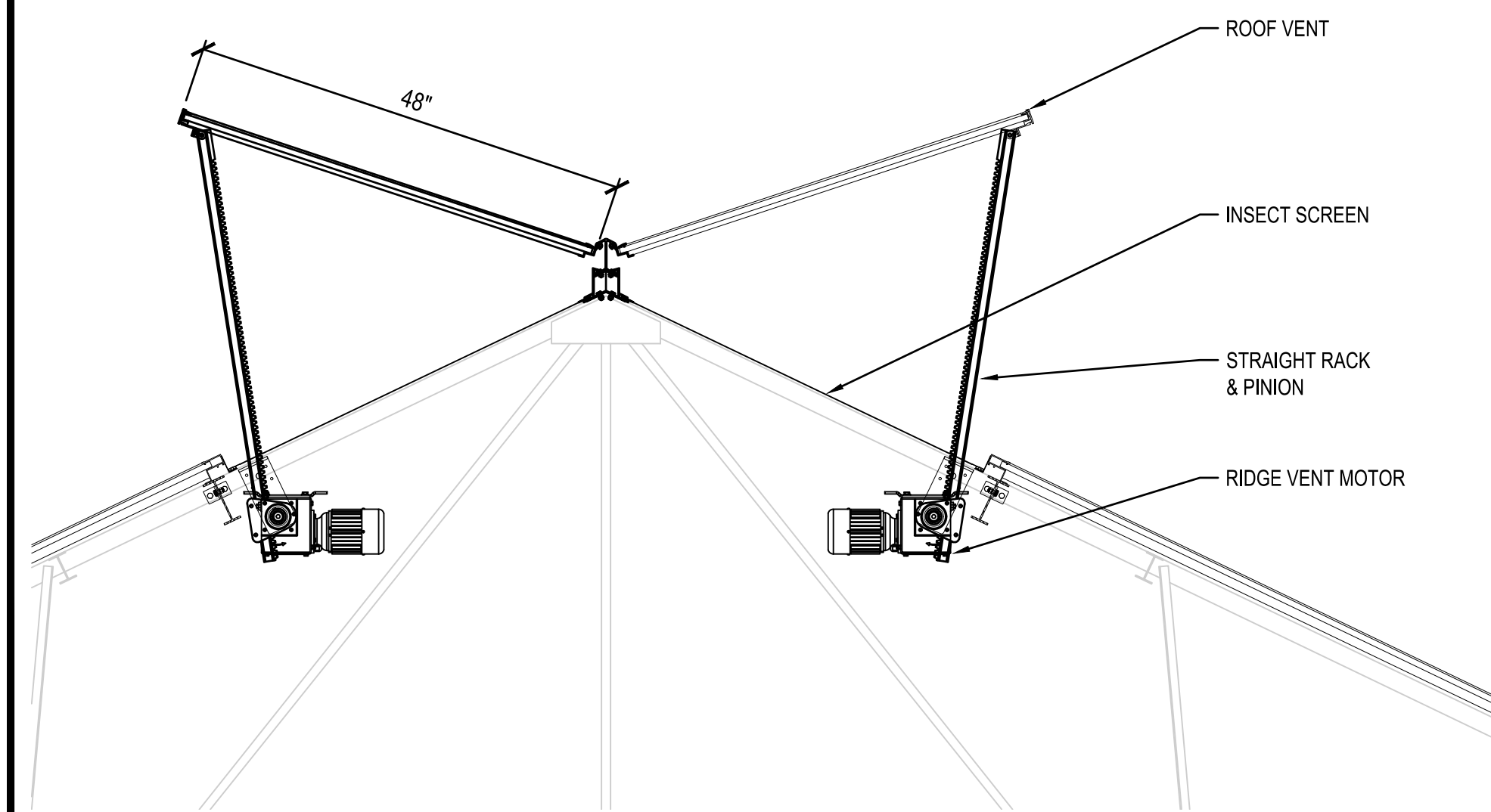
4 ELECTRIC UNIT HEATER DETAIL
M602 SCALE: NOT TO SCALE



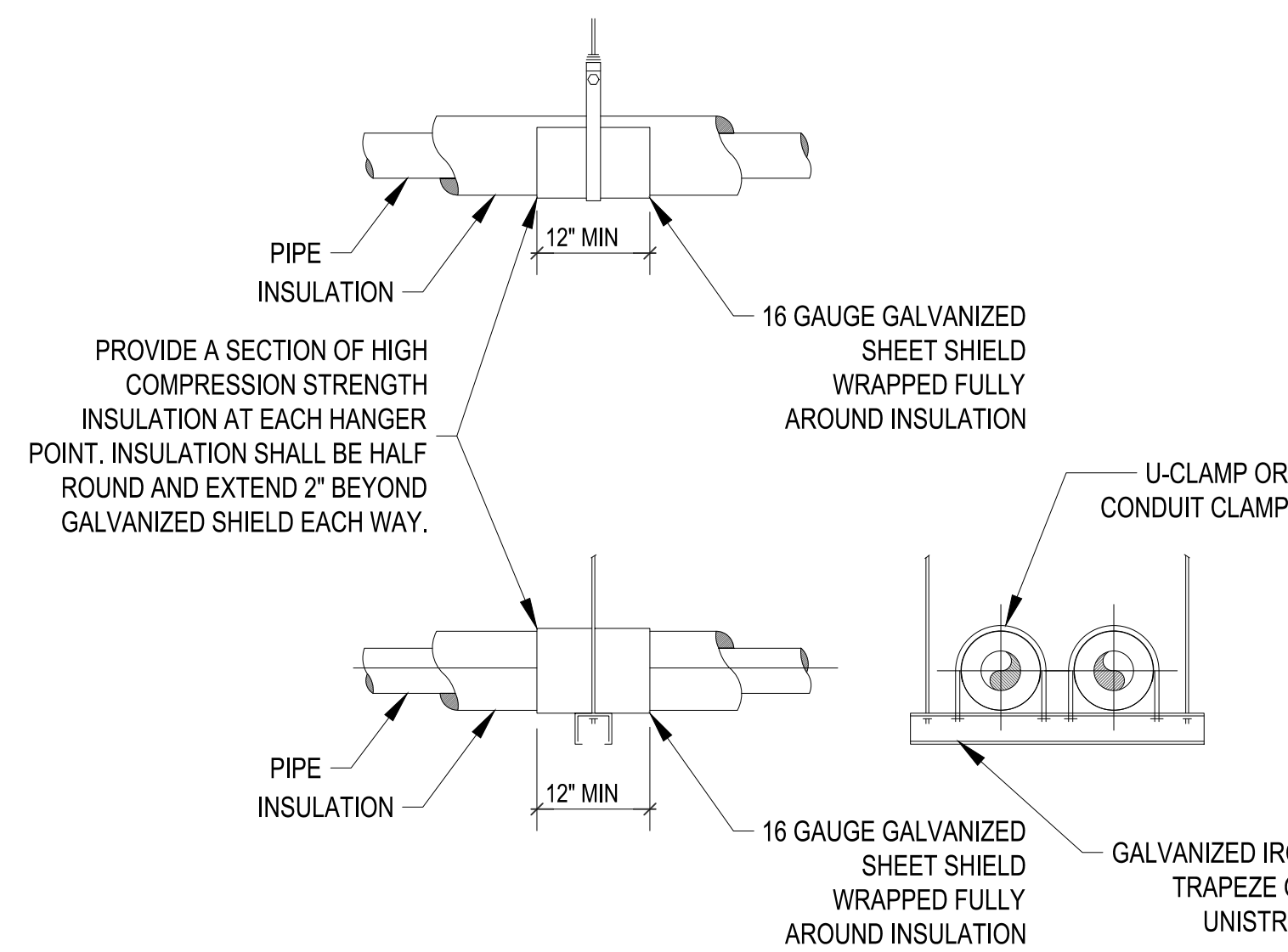
5 HAF DETAIL
M602 SCALE: NOT TO SCALE



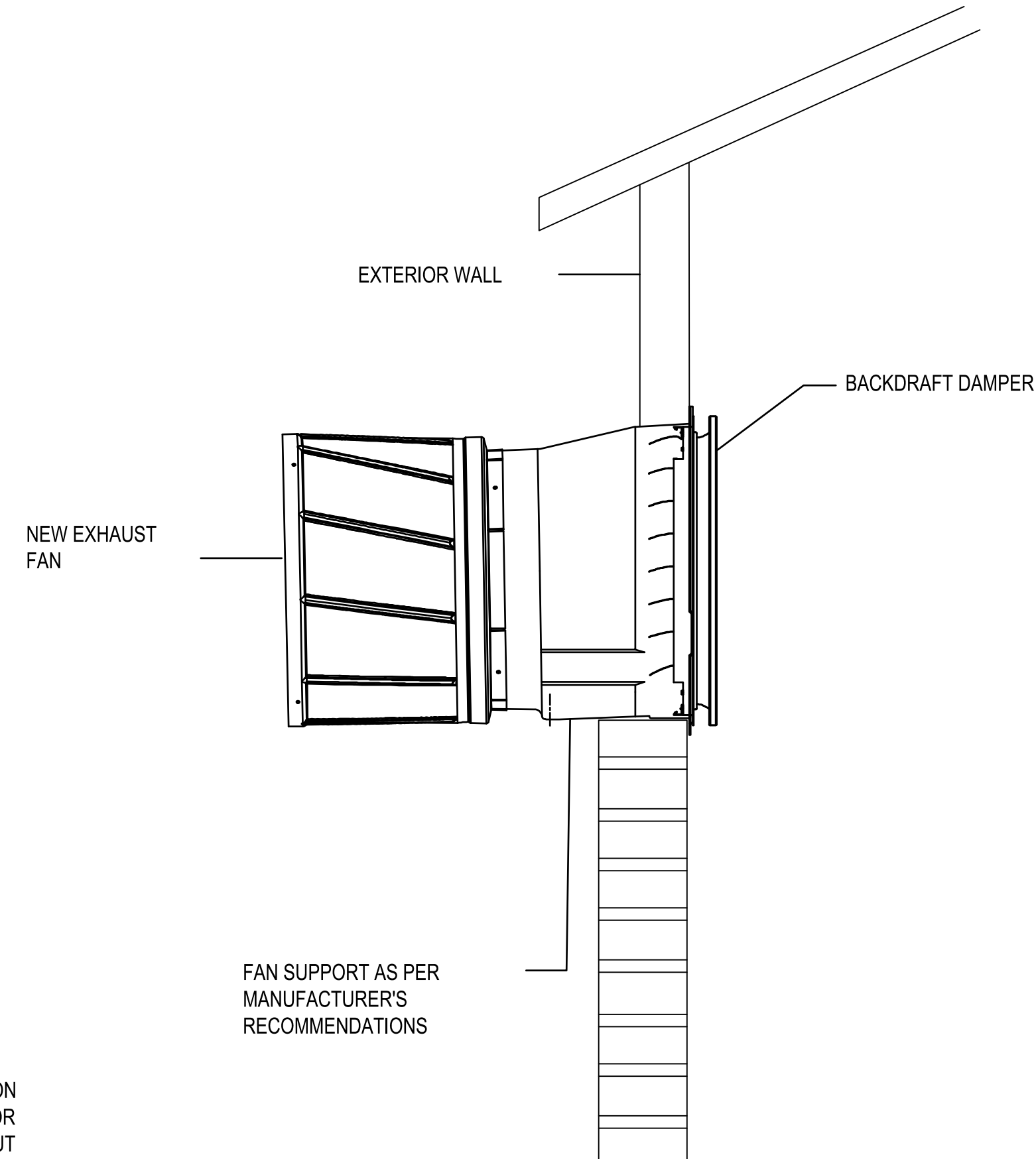
6 EVAPORITIVE COOLING PAD & PUMP DETAIL
M602 SCALE: NOT TO SCALE



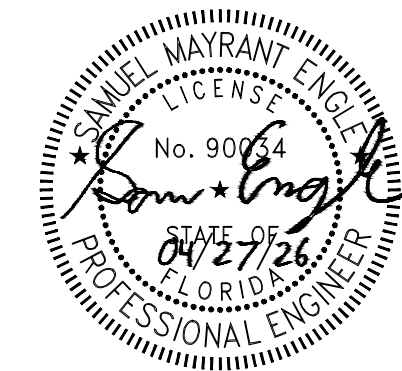
7 ROOF VENT DETAIL
M602 SCALE: NOT TO SCALE



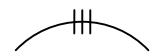
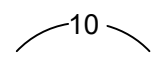
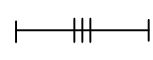
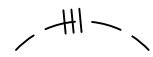
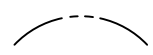
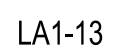
8 PIPE HANGER DETAILS
M-602 SCALE: N.T.S



9 WALL FAN MOUNTING DETAIL
M-602 SCALE: NOT TO SCALE



 The Johnson-McAdams FIRM PROJECT MANAGER - MARIANNE JOHNSON-McADAMSDISTRICT P.O. BOX 915, GREENWOOD, MISSISSIPPI - (662) 461-0463							
		NO.		DATE	BY	DESCRIPTION	
		REVISIONS					
A-E FIRM USDA		U S D A	 Agricultural Research Service				
			USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL				
			MECHANICAL DETAILS				
PROJECT MANAGER BP	EPM XXX	PROJECT NO. XXX	AGRICULTURAL RESEARCH SERVICE SOUTHEAST AREA STONEVILLE, MS				SHEET M-602 86 OF 108
DESIGNER JH	PPM XXX						
CHECKED BY JGM	SAFETY & HEALTH XXX	DATE 4/27/26					
JOB # 4825	REAL PROPERTY XXX						

GENERAL SYMBOLS LEGEND	
(NOT ALL SYMBOLS MAY APPLY, TYP.)	
SYMBOL / KEY	DESCRIPTION
	KEY NOTE DESIGNATION.
	REVISION TAG DESIGNATION.
	TOP DESIGNATION IS PLAN / ENLARGED PLAN / DETAIL NUMBER.
	BOTTOM DESIGNATION IS SHEET LOCATION (STANDARD SYMBOL SCALED DOWN FOR LEGEND).
	CONDUCTORS IN CONDUIT CONCEALED WITHIN WALL OR CEILING. TIC MARKS INDICATE NUMBER OF CONDUCTORS. THE EQUIPMENT GROUNDING CONDUCTOR IS NOT SHOWN, BUT SHALL BE PROVIDED. SIZE THE EQUIPMENT GROUNDING CONDUCTOR AND THE CONDUIT PER THE NEC. THE ABSENCE OF TIC MARKS SIGNIFIES THAT TWO CONDUCTORS PLUS AN EQUIPMENT GROUNDING CONDUCTOR SHOULD BE PROVIDED. FOR EXAMPLE, THE MARKINGS TO THE LEFT SIGNIFY THAT THREE CONDUCTORS PLUS AN EQUIPMENT GROUNDING CONDUCTOR SHOULD BE PROVIDED.
	THE TEXT INSIDE THE ARC INDICATES THE AWG SIZE OF THE CONDUCTORS THAT SHALL BE RUN IN THE CONDUIT. THE ABSENCE OF TEXT SIGNIFIES THAT THE CONDUCTORS SHOULD BE #12 AWG.
	CIRCUITRY RUN IN STRAIGHT LINE SEGMENTS SIGNIFIES EXPOSED SURFACE-MOUNTED RACEWAY (SEE SPECIFICATIONS).
	CONDUCTORS IN CONDUIT CONCEALED BELOW GRADE OR FLOOR. TIC MARKS INDICATE NUMBER OF CONDUCTORS. THE EQUIPMENT GROUNDING CONDUCTOR IS NOT SHOWN, BUT SHALL BE PROVIDED. SIZE THE EQUIPMENT GROUNDING CONDUCTOR AND THE CONDUIT PER THE NEC. THE ABSENCE OF TIC MARKS SIGNIFIES THAT TWO CONDUCTORS PLUS AN EQUIPMENT GROUNDING CONDUCTOR SHOULD BE PROVIDED. THE MARKINGS TO THE LEFT SIGNIFY THAT THREE CONDUCTORS PLUS AN EQUIPMENT GROUNDING CONDUCTOR SHOULD BE PROVIDED.
	HOMERUN TO PANELBOARD. ARC DENOTES CONCEALED CIRCUITRY. TEXT DENOTES PANELBOARD NAME WITH CIRCUIT NUMBER. DEVICES HAVING CIRCUIT NUMBERS LOCATED BESIDE THEM MAY NOT SHOW THE CIRCUIT NUMBERS AT THE HOMERUN ARROWS.
	PARTIAL HOMERUN TO PANELBOARD. COMBINE ALL PARTIAL HOMERUNS THAT ARE ON THE SAME CIRCUIT IN A JUNCTION BOX PRIOR TO ENTERING THE PANELBOARD.
	LOW VOLTAGE CONDUCTORS USED FOR MOTION DETECTOR CIRCUITRY. SEE MANUFACTURER'S RECOMMENDATIONS FOR CONDUCTOR REQUIREMENTS.
	PANELBOARD AND CIRCUIT DESIGNATION

POWER OUTLET AND COMMUNICATIONS LEGEND

SYMBOL / KEY	DESCRIPTION
 ?	SINGLE RECEPTACLE, NEMA 5-30R, MOUNTED 18" A.F.F. TO CENTERLINE OF BOX UNLESS NOTED OTHERWISE.
 ?	DUPLEX RECEPTACLE, NEMA 5-20R, MOUNTED 18" A.F.F. TO CENTERLINE OF BOX UNLESS NOTED OTHERWISE.
 (42)	ELECTRICAL OR COMMUNICATION DEVICE, MH = AT ELEVATION AFF INDICATED (42" AFF SHOWN)
 ?	DUPLEX RECEPTACLE, NEMA 5-20R, SUSPENDED FROM CEILING. SEE DETAIL 3/E-002.
 ?	DUPLEX RECEPTACLE, NEMA 5-20R, MOUNTED WITH BOTTOM OF BOX 2" ABOVE COUNTER BACKSPLASH. WHERE THERE IS NO BACKSPLASH MOUNT 6" ABOVE COUNTER. WHERE RECEPTACLE IS SHOWN IN AN AREA WITH NO COUNTER, MOUNT 45" A.F.F. TO CENTERLINE OF BOX.
 T	TELEPHONE MOUNTED IN BOX WITH HINGED DOOR
	JUNCTION BOX FOR FUTURE TELEPHONE
	CELLULAR AMPLIFIER ANTENNA
	TELEPHONE TERMINAL CABINET OR BACKBOARD

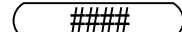
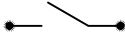
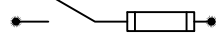
LIGHTING AND SWITCHING LEGEND

SYMBOL / KEY	DESCRIPTION	SYMBOL / KEY	DESCRIPTION
	SURFACE MOUNTED OR SUSPENDED FIXTURE.	\$	SINGLE -POLE, SINGLE-THROW SWITCH.
	SURFACE MOUNTED OR SUSPENDED EMERGENCY FIXTURE.	\$ ^T	HORSEPOWER RATED SWITCH WITH THERMAL OVERLOADS (MANUAL MOTOR STARTER).
	PASSIVE INFRARED AND ULTRASONIC DUAL TECHNOLOGY OCCUPANCY SENSOR WITH A 26° RADIAL COVERAGE. SURFACE MOUNTED TO CEILING.	\$ ²	DOUBLE POLE.
		\$ ³	THREE WAY.
	POWER PACK SURFACE MOUNTED TO CEILING.	\$ ^{3P}	THREE POLE.

UPPERCASE INDICATES TYPE OF FIXTURE → **A1** ← LOWERCASE INDICATES FIXTURES TO BE CONTROLLED TOGETHER. **c** ← LOWERCASE INDICATES GROUP OF FIXTURES TO BE CONTROLLED BY SWITCH. **c**

FIRE ALARM AND ACCESS LEGEND			
SYMBOL / KEY		DESCRIPTION	
	MANUAL PULL STATION. MOUNT 48" A.F.F. TO CENTERLINE OF BOX.		MOTION DETECTOR TO RELEASE MAGNETIC LOCK.
	STROBE. MOUNT 80" A.F.F. TO BOTTOM OF BOX.		CARD READER.
	COMBINATION HORN & STROBE. MOUNT 80" A.F.F. TO BOTTOM OF BOX.		PUSH TO EXIT BUTTON.
	ACCESS CONTROL PANEL.		PUSH BUTTON.
			MAGNETIC LOCK.
			POWER SUPPLY.

ONE-LINE DIAGRAM, SCHEDULE, AND DISTRIBUTION LEGEND

SYMBOL / KEY		DESCRIPTION	
	FEEDER KEY		CEILING MOUNTED JUNCTION BOX.
	DISCONNECT SWITCH DIST BOARD MTD.		WALL MOUNTED JUNCTION BOX.
	CKT BREAK DIST BOARD MTD.		FLEX. CONNECTION TO EQUIPMENT.
	FUSE DIST BOARD MTD.		PANELBOARD.
	FUSED DISC SWITCH 600V & UNDER.		MAGNETIC MOTOR STARTER.
	GROUND		NON-FUSED DISCONNECT SWITCH. TEXT INDICATES AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE.
	XFMR		FUSED DISCONNECT SWITCH. TEXT INDICATES AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE; F-(RATING OF FUSES).
	EQUIPMENT		CONTINUATION, UPSTREAM LABEL
	GENERATOR		LOCAL CONTACTOR
	METER		PANELBOARD (ONE-LINE)
	SURGE PROTECTION DEVICE.		TRANSFER SWITCH (ATS : AUTOMATIC) (MTS: MANUAL)
	GROUND FAULT INTERRUPTER.		MOTOR
	TIMECLOCK		AIR TERMINAL
	PHOTOELECTRIC SWITCH		
	JUNCTION BOX		
	PULL BOX		
	TRANSFORMER (PLAN VIEW)		
	3/4" X 10' COPPER CLAD STEEL GROUND ROD DRIVEN W/ TOP ATLEAST 18" BELOW GRADE.		

VOLTAGE DROP CHART FOR 20A, 1Ø CIRCUITS

VOLTAGE	CIRCUIT LENGTH	CONDUCTOR SIZE (AWG)
120	<50'	#12
120	>50'	#10
120	>90'	#8
120	>140'	#6
277	<130'	#12
277	>130'	#10
277	>200'	#8
277	>330'	#6

VOLTAGE DROP CHART NOTES

- 1) CIRCUIT SIZES INDICATED ON THE DRAWINGS ARE MINIMUM REQUIREMENTS. REFER TO THIS CHART FOR UPSIZING CONDUCTORS AS NEEDED.
- 2) DO NOT CONNECT CONDUCTORS LARGER THAN #10 DIRECTLY TO A RECEPTACLE OR A SWITCH. PROVIDE A JUNCTION BOX TO DOWNSIZE THE CONDUCTOR TO #12 AT THE DEVICE.
- 3) FOR CIRCUITS LONGER THAN THOSE LISTED ABOVE, CONSULT WITH THE ENGINEER FOR CONDUCTOR SIZES.

MASTER NOTES - ELECTRICAL:

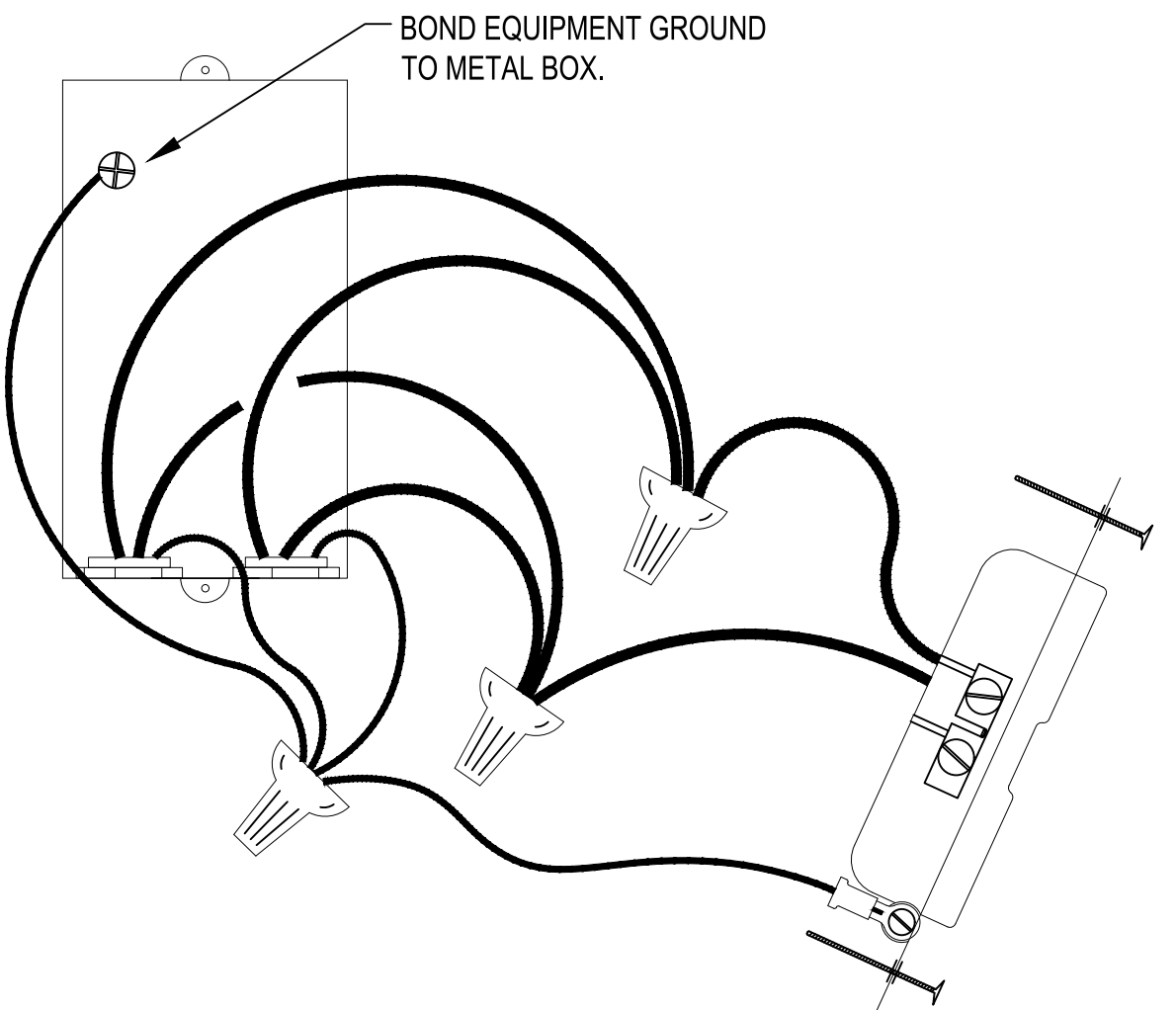
1. THE CONTRACTOR MUST ABIDE BY ALL FEDERAL, STATE, AND/OR LOCAL CODES. IF A DISCREPANCY BETWEEN CODES OCCURS, THE MOST STRINGENT SHALL PREVAIL.
 2. THE CONTRACTOR MUST FIELD VERIFY EXISTING CONDITIONS PRIOR TO THE COMMENCEMENT OF ANY WORK. SHOULD DISCREPANCIES BE DISCOVERED, THE CONTRACTOR MUST VERIFY INTENT WITH THE ENGINEER/CONTRACTING OFFICER BEFORE PROCEEDING.
 3. COORDINATE LOCATIONS OF ALL CEILING MOUNTED DEVICES WITH OTHER TRADES PRIOR TO INSTALLATION.
 4. COORDINATE ALL ROUGH-IN REQUIREMENTS FOR GOVERNMENT FURNISHED EQUIPMENT WITH THE OWNER PRIOR TO BEGINNING WORK. THESE DRAWINGS ARE BASED ON EXISTING BUILDING CONSTRUCTION DRAWINGS, SITE SURVEYS, AND OWNER FURNISHED EQUIPMENT SPECIFICATIONS.
 5. RECEPTACLES MUST NOT BE CONNECTED IN A FEED-THRU MANNER. WIRE CONNECTIONS IN RECEPTACLE BOXES MUST BE MADE IN A PIGTAIL MANNER AS SHOWN IN DETAIL 1/E-001.
 6. REFER TO ARCHITECTURAL DETAIL 1/G-103 FOR INFORMATION REGARDING AREAS OF WORK FOR EACH BID OPTION AND PHASING.
1. THE CONTRACTOR SHALL PROVIDE SEPARATE PRICING, INCLUDING ALL DEMOLITION AND NEW WORK, BASED ON THE AREAS SHOWN ON THE BID OPTION KEY PLAN. NO WORK SHALL BE EXCLUDED BASED ON AREA BOUNDARIES; THE CONTRACTOR IS RESPONSIBLE FOR COORDINATING ALL SCOPE NECESSARY TO DELIVER A COMPLETE AND FUNCTIONAL RESULT FOR EACH BID OPTION.

BID OPTION NOTES:

1. THE CONTRACTOR SHALL PROVIDE SEPARATE PRICING, INCLUDING ALL DEMOLITION AND NEW WORK, BASED ON THE AREAS SHOWN ON THE BID OPTION KEY PLAN. NO WORK SHALL BE EXCLUDED BASED ON AREA BOUNDARIES; THE CONTRACTOR IS RESPONSIBLE FOR COORDINATING ALL SCOPE NECESSARY TO DELIVER A COMPLETE AND FUNCTIONAL RESULT FOR EACH BID OPTION.

GENERAL NOTES:

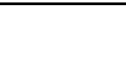
1. DEVICES NOTED AS 'GF' SHALL BE GROUND FAULT CIRCUIT INTERRUPTING DEVICES.
2. DEVICES NOTED AS 'WP' SHALL BE WEATHERPROOF WHILE-IN-USE.
3. PROVIDE UNSWITCHED POWER TO THE EMERGENCY BATTERY PACKS.
4. 'W/E' DENOTES PROVIDED WITH EQUIPMENT.
5. MINIMUM RACEWAY SIZE SHALL BE 3/4". INCREASE RACEWAY SIZE AS REQUIRED TO LIMIT RACEWAY FILL RATIO TO LESS THAN 40% FULL.
6. THE ELECTRICAL CONTRACTOR SHALL CAREFULLY COORDINATE WORK WITH OTHER TRADES THROUGH THE GENERAL CONTRACTOR AND SHALL BE RESPONSIBLE FOR SECURING SPACE REQUIREMENTS FOR ELECTRICAL EQUIPMENT AND CORRECT ROUGH-IN LOCATIONS OF ELECTRICAL CONNECTIONS.
7. VERIFY DOOR SWINGS WITH ARCHITECTURAL DRAWINGS BEFORE ROUGHING IN WALL SWITCHES. SWITCHES IN THE SAME LOCATION SHALL BE GANGED TOGETHER IN ONE COMMON BACKBOX AND SHALL HAVE ONE COMMON FACE PLATE.
8. ALL FEEDERS AND BRANCH CIRCUITS SHALL INCLUDE A GREEN INSULATED GROUND CONDUCTOR, SIZED PER THE NATIONAL ELECTRICAL CODE, OR AS SHOWN, CONNECTED TO EACH DEVICE AND OUTLET BOX ON THE CIRCUIT AND TO THE PANELBOARD GROUND BUS. PROVIDE A DEDICATED NEUTRAL CONDUCTOR FOR EACH BRANCH CIRCUIT. MULTIPLE BRANCH CIRCUITS IN ONE RACEWAY REQUIRE ONLY ONE GROUND CONDUCTOR.
9. VERIFY LUMINAIRE, CEILING MOUNTED OCCUPANCY SENSOR LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATIONS AND DIMENSIONS PRIOR TO INSTALLATION. VERIFY EXACT LOCATIONS OF MOTORS AND EQUIPMENT BEFORE ROUGHING-IN.
10. "PROVIDE" IS AN INCLUSIVE TERM USED TO DESCRIBE ASPECTS OF THE WORK TO BE ACCOMPLISHED, AND IS HEREBY DEFINED TO REQUIRE TO STORE, FURNISH, INSTALL, MOUNT, CONNECT, CONTROL AND POWER EQUIPMENT INDICATED, AS WELL AS ALL APPURTENANCES REQUIRED TO MAKE ELECTRICAL SYSTEMS OPERATE AS INDICATED WITHIN THESE DRAWINGS AND SPECIFICATIONS AND TO FULFILL THE SCOPE OF WORK.
11. ALL CONDUCTORS SHALL BE COPPER
12. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS, DIMENSIONS AND ELEVATIONS BEFORE PROCUREMENT OF ANY MATERIALS AND DEVELOPMENT OF ANY SHOP DRAWINGS OR SUBMITTALS.
13. PROVIDE LABELS ON ALL RECEPTACLES, WALL MOUNTED LIGHT SWITCHES/OCCUPANCY SENSORS AND JUNCTION BOXES INDICATING THE SOURCE PANEL & CIRCUIT(S). TEXT SHALL BE BLACK, 1/4". LABELS ABOVE FINISHED CEILING CAN BE HANDWRITTEN WITH AN INDELIBLE MARKER.
14. CIRCUIT SIZES INDICATED ON THE DRAWINGS ARE MINIMUM REQUIREMENTS.
15. DO NOT CONNECT CONDUCTORS LARGER THAN #10 DIRECTLY TO A RECEPTACLE OR A SWITCH. PROVIDE A JUNCTION BOX TO DOWNSIZE THE CONDUCTOR TO #12 AT THE DEVICE.
16. FOR INSTALLATION OF LIGHTING SWITCHES, MOUNT CENTERLINE OF BOX AT 46" A.F.F., UNLESS OTHERWISE INDICATED.



RECEPTACLE DETAIL NOTES:

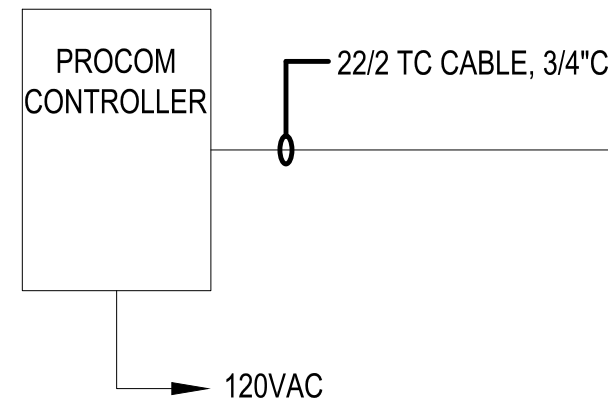
- A. THESE DRAWINGS ARE BASED ON THE BEST INFORMATION AVAILABLE AT THE TIME OF DESIGN. COORDINATE WITH THE MILLWORK CONTRACTOR TO DETERMINE THE EXACT LOCATION OF OUTLETS BEING PLACED IN AND AROUND MILLWORK.
- B. RECEPTACLES SHOWN AS GFI MAY BE NON GFI TYPE RECEPTACLES IF FED FROM A 20/1 GFI BREAKER OR THE LOAD SIDE OF A GFI RECEPTACLE IN THE SAME ROOM, ON THE SAME CIRCUIT AND RATED 20 AMP FEED-THRU CAPACITY. COVER PLATES SHALL BE CLEARLY MARKED GFI.
- C. NON GFI RECEPTACLES SHALL NOT BE CONNECTED IN A FEED-THRU MANNER. WIRE CONNECTIONS IN RECEPTACLE BOXES SHALL BE MADE IN A PIGTAIL MANNER AS SHOWN IN RECEPTACLE DETAIL.



 The Johnson M Adams Group 10000 Greenwood, Suite 100, Greenwood, Mississippi 39206 P.O. BOX 110, GREENWOOD, MISSISSIPPI 39202-0110						
		NO.	DATE	BY	DESCRIPTION	
		REVISIONS				
A-E FIRM		USDA				
			USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL			
PROJECT MANAGER	EHM	U S D A				
BP	XXX					
DESIGNER	PPM					
EH	XXX					
CHECKED BY	SAFETY & HEALTH	ELECTRICAL LEGEND				
BTD	XXX	PROJECT NO.	AGRICULTURAL RESEARCH SERVICE		SHEET	
		XXX			E-001	
JOB #	REAL PROPERTY	DATE	FT. PIERCE, FL		87 OF 108	
4825	XXX	04/27/2026				

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Apr 27, 2026 9:55am

NOTE: CONTROL CIRCUITS MAY BE COMBINED INTO ONE CONDUIT. CONTROL CIRCUITRY CONDUIT MUST HAVE LESS THAN 40% FILL RATIO PER THE NATIONAL ELECTRICAL CODE.



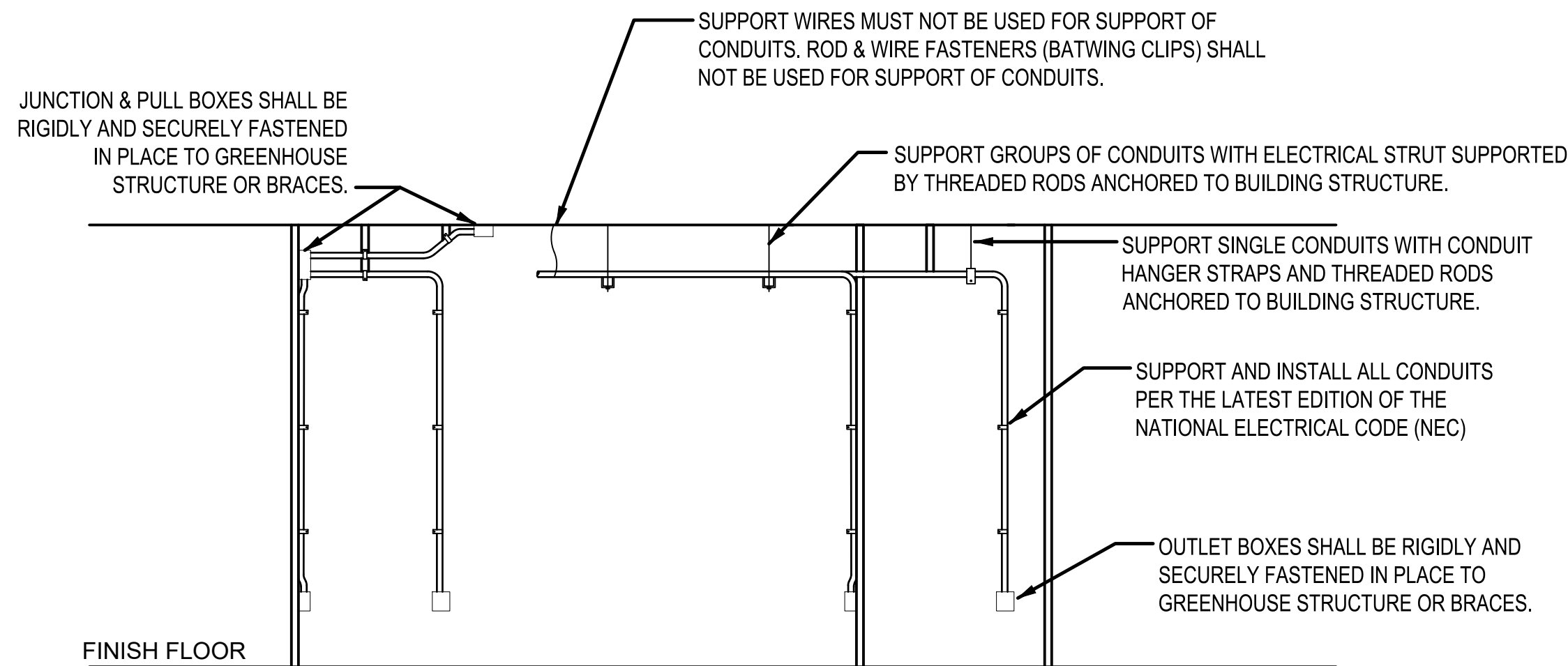
LOCATED IN OFFICE. SEE EP101 FOR EXACT LOCATION

LOCATED IN OFFICE. SEE EP101 FOR EXACT LOCATION

NOTE: THE GREENHOUSE ENVIRONMENTAL CONTROL SYSTEM (MICROGROW OR APPROVED EQUAL) FURNISHED BY THE MECHANICAL CONTRACTOR AND INSTALLED BY THE ELECTRICAL CONTRACTOR SHALL BE PROGRAMMED AND COMMISSIONED AS A COMPLETE AND FULLY OPERATIONAL SYSTEM. THE CONTRACTOR SHALL COORDINATE DIRECTLY WITH THE GREENHOUSE CONTROL SYSTEM MANUFACTURER AND/OR AUTHORIZED REPRESENTATIVE TO VERIFY ALL EQUIPMENT REQUIREMENTS, WIRING METHODS, NETWORK ARCHITECTURE, ANNUNCIATION, AND REMOTE MONITORING FEATURES.

1 GREENHOUSE CONTROLLER, ANNUNCIATOR, AND REMOTE VIEWING DETAIL

E-002 SCALE: NONE



NOTE: JUNCTION, PULL AND OUTLET BOXES SHALL BE 4" x 4" x 2-1/8" OR LARGER.

NOTE: ALL CONDUIT SHALL BE UV-RESISTANT SCH. 80 PVC. ALL JUNCTION BOXES SHALL BE WATERTIGHT POLYCARBONATE BOXES WITH WEATHERPROOF GASKET AND FITTINGS.

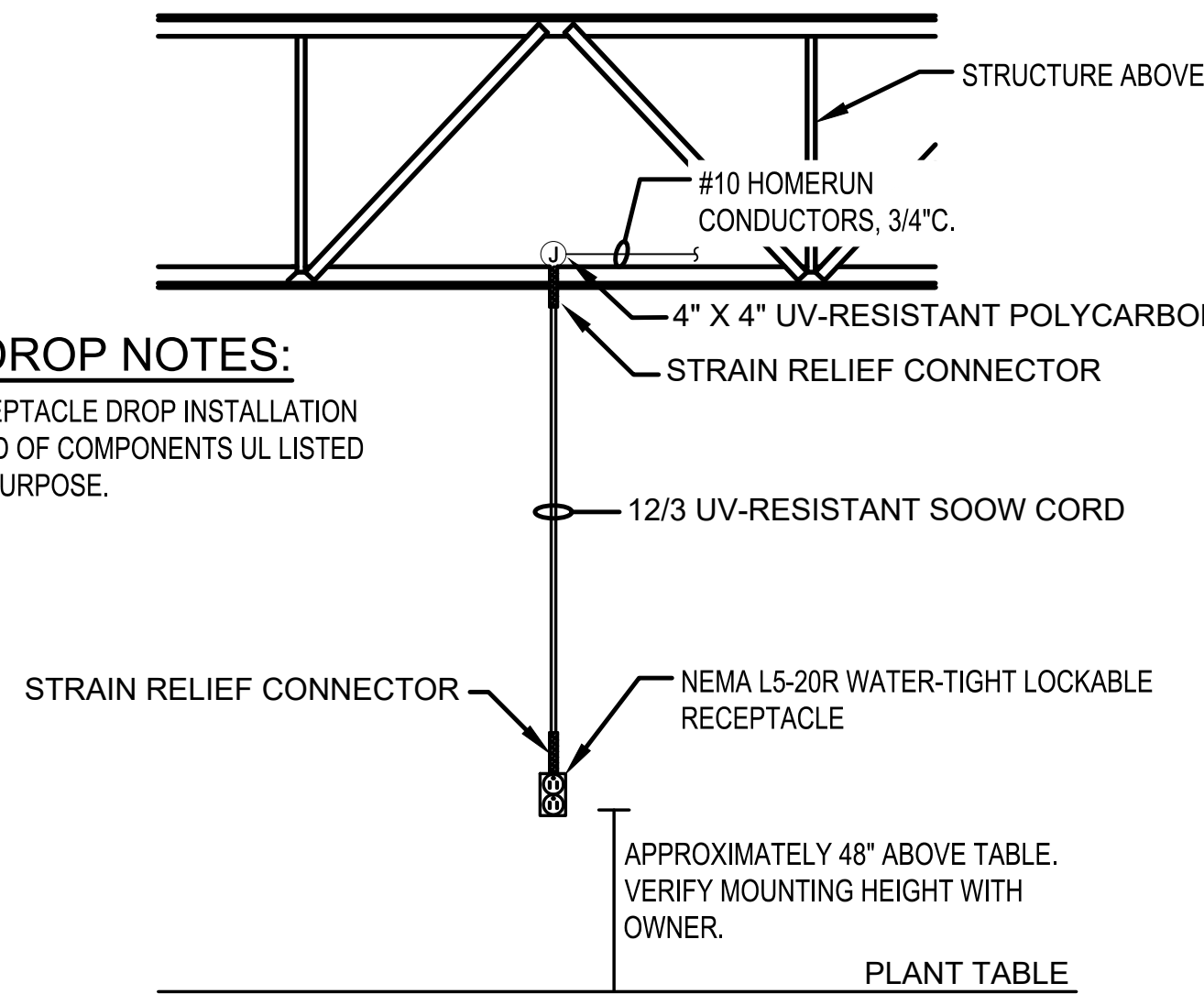
CONDUIT INSTALLATION SHALL COMPLY WITH THE REQUIREMENTS LISTED IN ARTICLE 352 OF THE NATIONAL ELECTRICAL CODE.

2 JUNCTION AND OUTLET BOX DETAIL

E-002 SCALE: NONE

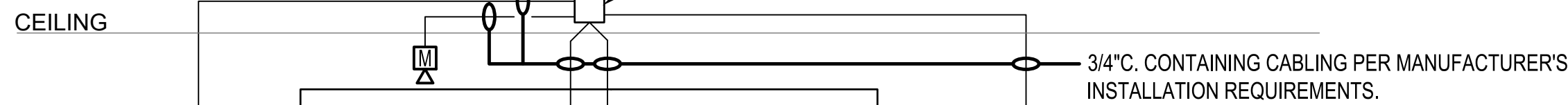
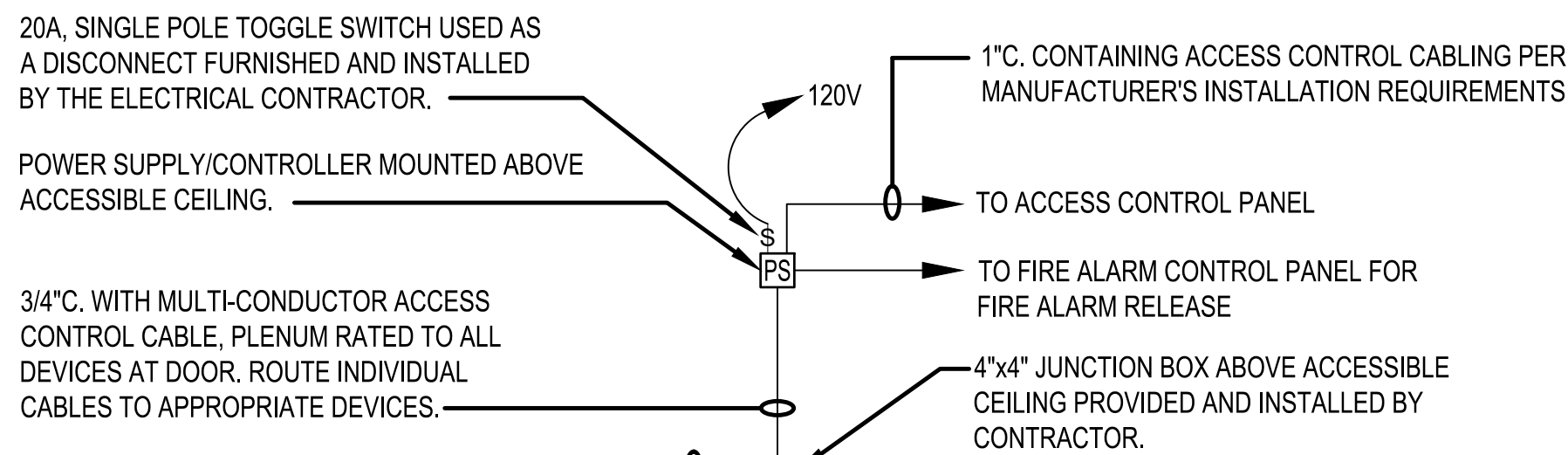
RECEPTACLE DROP NOTES:

A. THE COMPLETE RECEPTACLE DROP INSTALLATION SHALL BE COMPRISED OF COMPONENTS UL LISTED FOR THE INTENDED PURPOSE.



3 CEILING MOUNTED SUSPENDED RECEPTACLE DETAIL

E-002 SCALE: NONE



NOTES:
COORDINATE ACCESS CONTROL REQUIREMENTS WITH USDA MAINTENANCE PERSONNEL PRIOR TO INSTALLATION. UTILIZE 20/1 CIRCUIT BREAKER LOCATED IN PANEL '1GE'.

THE CONTRACTOR SHALL REMOVE THE EXISTING ACCESS CONTROL SYSTEM AT GREENHOUSE 101 AND 102 IN ITS ENTIRETY AND PROVIDE A COMPLETE NEW ACCESS CONTROL SYSTEM FOR GREENHOUSE 101 AND 102. THE SYSTEM SHALL INCLUDE, BUT IS NOT LIMITED TO CARD READERS, ELECTROMAGNETIC LOCKS, DOOR POSITION SWITCHES, REQUEST-TO-EXIT DEVICES, POWER SUPPLIES, CONTROL MODULES, ASSOCIATED WIRING, AND ALL REQUIRED ACCESSORIES.

PUSH-TO-EXIT PUSHBUTTON AS INDICATED ON FLOOR PLAN, MOUNTED ON EXIT SIDE OF DOOR AT 48" A.F.F.

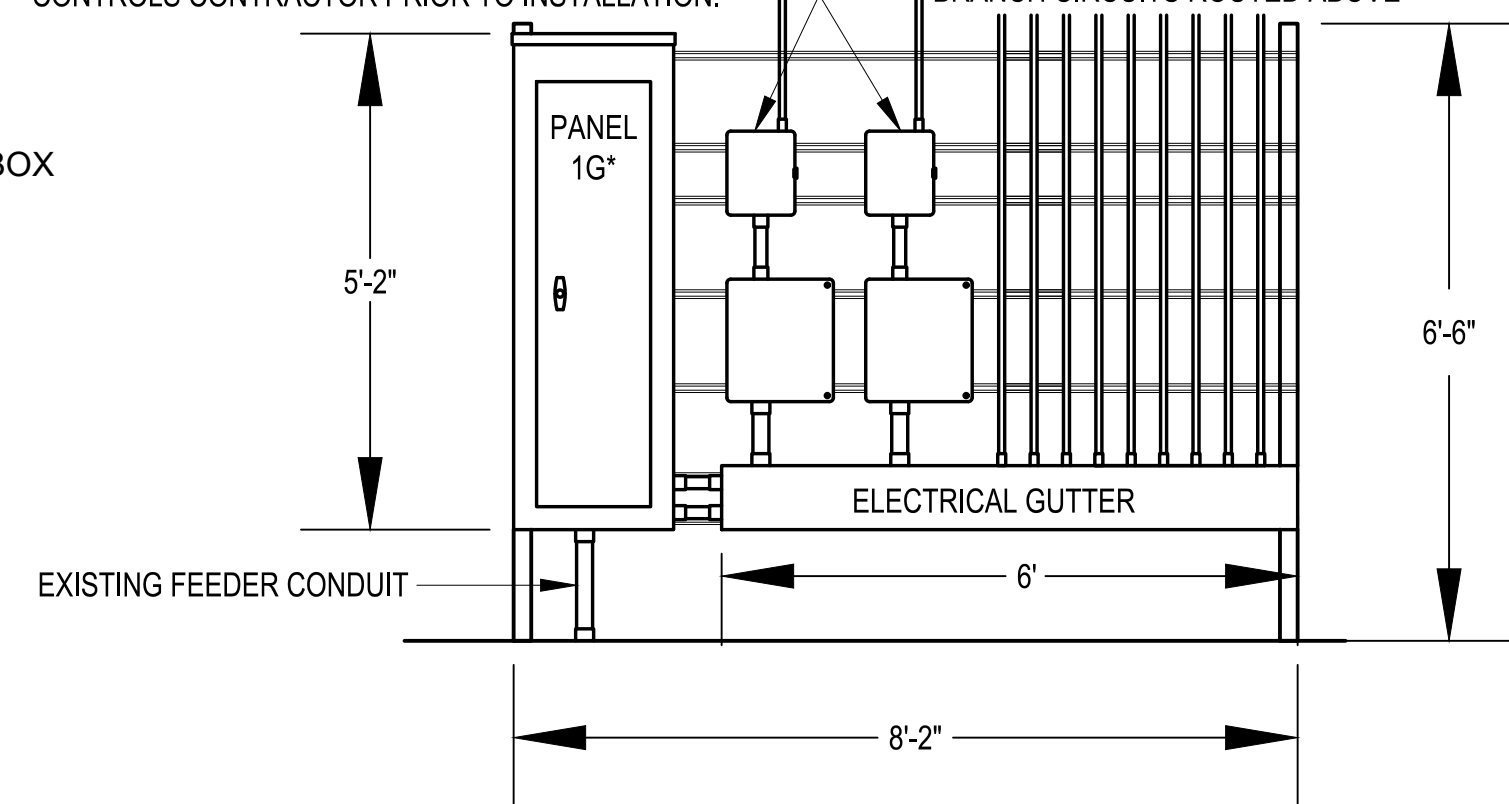
CARD READER AS INDICATED ON FLOOR PLAN, MOUNTED ON ENTRANCE SIDE OF DOOR AT 48" A.F.F.

6 MAG-LOCK ACCESS CONTROL DETAIL

E-002 SCALE: NONE

MICROGROW CONTROLLER (ONE FOR EACH GREENHOUSE), MECHANICAL EQUIPMENT CONTACTORS/MOTOR STARTERS MOUNTED IN 12"x12"x8"D GASKETED POLYCARBONATE BOX WITH HINGED COVER BELOW. COORDINATE CONNECTION REQUIREMENTS WITH THE MECHANICAL CONTRACTOR AND GREENHOUSE CONTROLS CONTRACTOR PRIOR TO INSTALLATION.

TO GREENHOUSE ANNUNCIATOR PANEL. COORDINATE EXACT REQUIREMENTS WITH GREENHOUSE CONTROLS CONTRACTOR PRIOR TO INSTALLATION.



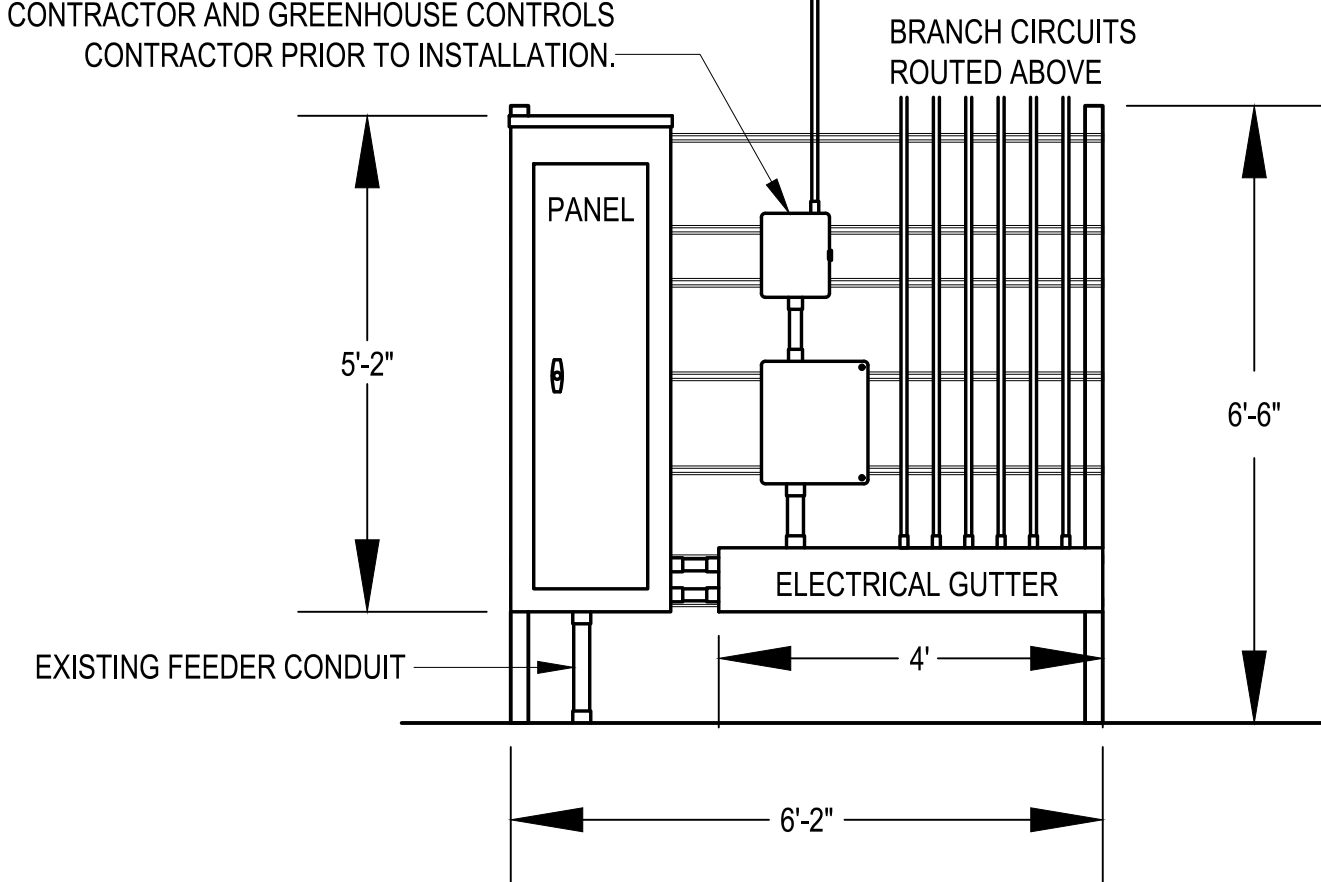
NOTE: THE CONTRACTOR SHALL UTILIZE INSULATED POLARIS LUGS TO EXTEND EXISTING FEEDER CIRCUITRY AS REQUIRED.

4 DOUBLE GREENHOUSE PANEL DETAIL

E-002 SCALE: NONE

TO GREENHOUSE ANNUNCIATOR PANEL. COORDINATE EXACT REQUIREMENTS WITH GREENHOUSE CONTROLS CONTRACTOR PRIOR TO INSTALLATION.

MICROGROW CONTROLLER, MECHANICAL EQUIPMENT CONTACTORS/MOTOR STARTERS MOUNTED IN 12"x12"x8"D GASKETED POLYCARBONATE BOX WITH HINGED COVER BELOW. COORDINATE CONNECTION REQUIREMENTS WITH THE MECHANICAL CONTRACTOR AND GREENHOUSE CONTROLS CONTRACTOR PRIOR TO INSTALLATION.

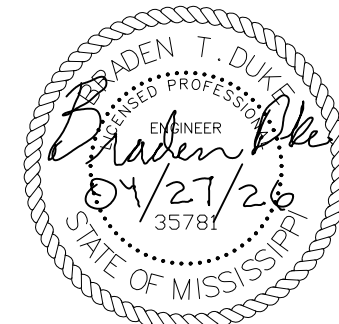


NOTE: THE CONTRACTOR SHALL UTILIZE INSULATED POLARIS LUGS TO EXTEND EXISTING FEEDER CIRCUITRY AS REQUIRED.

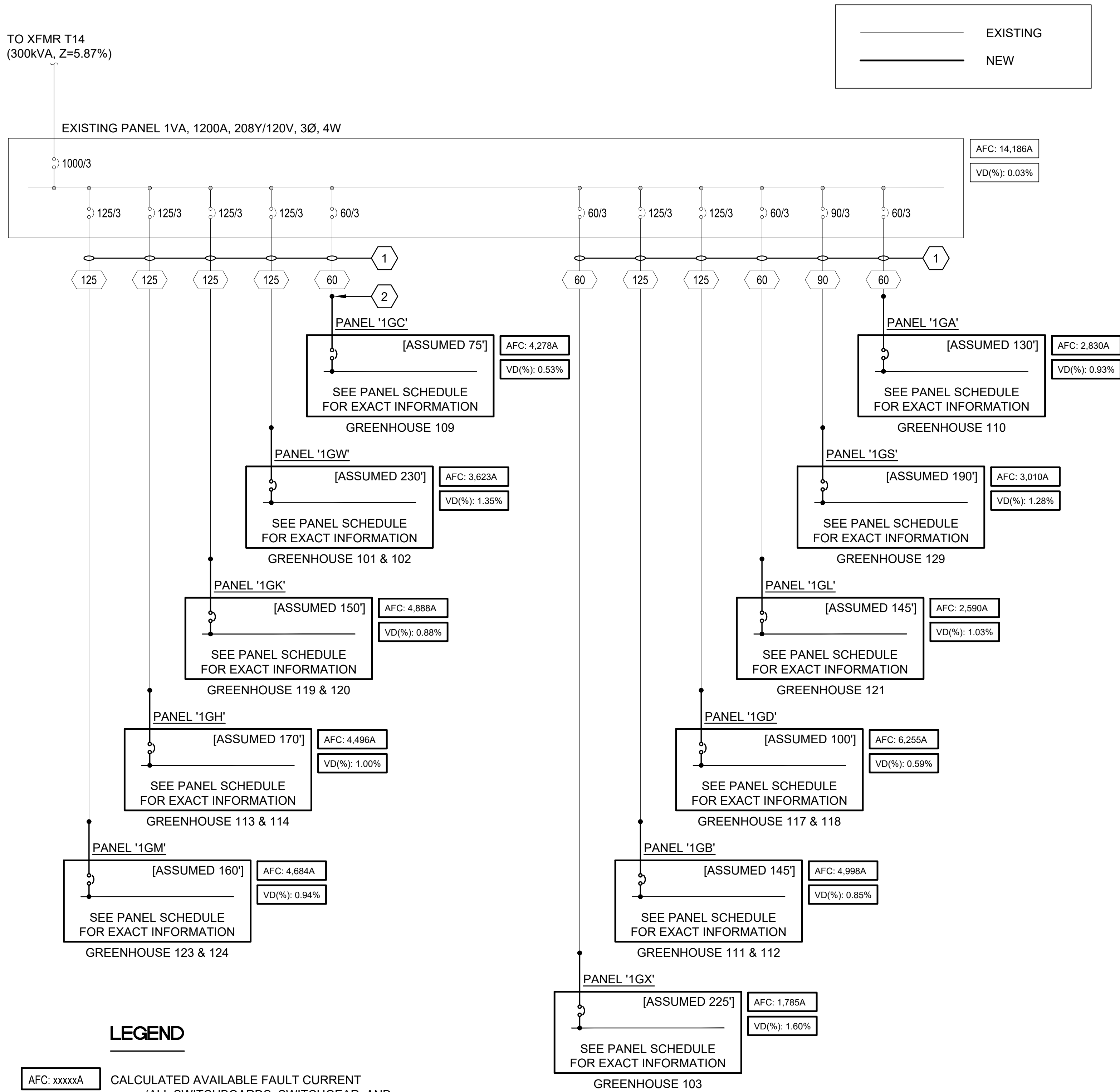
5 SINGLE GREENHOUSE PANEL DETAIL

E-002 SCALE: NONE

		NO.	DATE	BY	DESCRIPTION
		REVISIONS			
A-E FIRM		USDA			
PROJECT MANAGER	BP	EPM	XXX	 USDA Agricultural Research Service	
DESIGNER	EH	PPM	XXX		
CHECKED BY	BTD	SAFETY & HEALTH	XXX	USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL	
JOB #	4825	REAL PROPERTY	XXX		
PROJECT NO.		AGRICULTURAL RESEARCH SERVICE		SHEET	
DATE		04/27/2026		E-002	
JOB #		4825		88 OF 108	



\\jagnework\2025_jba\4825_USDA_Tornado_Damage_Repair_FL_Pierce_FL\07_Wiring_Folder\Elec\CAD\E-601.dwg
Apr 27, 2026 9:55am



LEGEND

AFC: xxxxA

CALCULATED AVAILABLE FAULT CURRENT
(ALL SWITCHBOARDS, SWITCHGEAR, AND PANELBOARDS SHALL HAVE THE AVAILABLE FAULT CURRENT AND THE DATE THE CALCULATION WAS PERFORMED FIELD MARKED ON THE ENCLOSURE PER NEC ARTICLES 408.6 AND 110.21(B)(3).) THE CALCULATIONS SHALL BE UPDATED BY THE ELECTRICAL ENGINEER ONCE ALL CIRCUITRY IS INSTALLED AND CIRCUITRY LENGTHS DOCUMENTED ON THE "AS-BUILT" DRAWINGS. CALCULATIONS SHOWN ARE BASED ON THE ASSUMPTIONS LISTED.

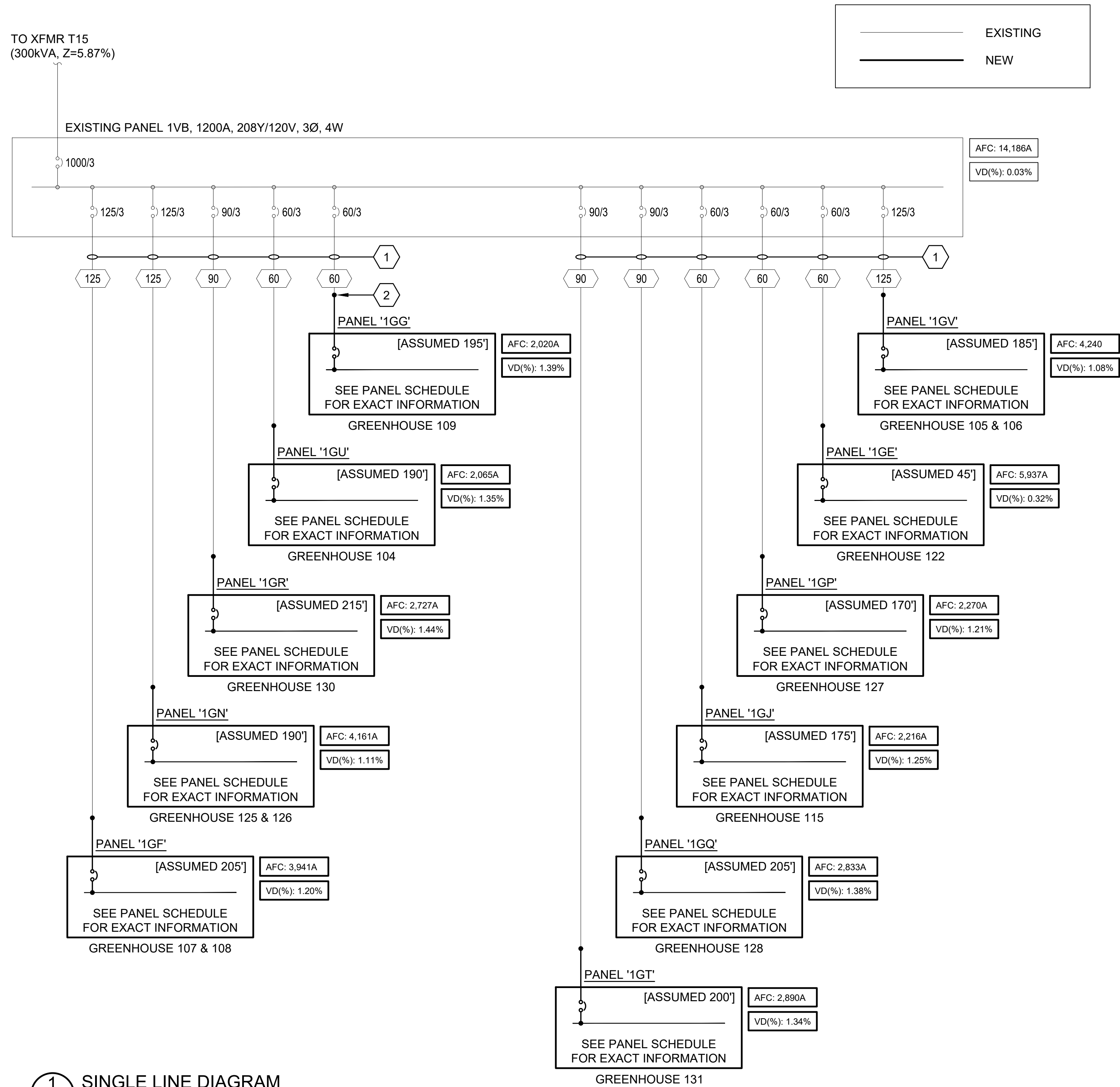
VD: X.XX%

VOLTAGE DROP
THE VOLTAGE DROP IS CALCULATED BASED ON THE ASSUMPTIONS LISTED.

KEYED NOTES

- EXISTING CONDUIT AND CIRCUITRY TO REMAIN AND BE UTILIZED TO FEED NEW INSTALLATION.
- EXTEND CIRCUITRY TO NEW PANEL AS REQUIRED. UTILIZE INSULATED POLARIS LUGS OR EQUAL (TYPICAL OF ALL PANELS). SPLICE MUST OCCUR IN ACCESSIBLE LOCATION. PROVIDE NEMA 4X BOX BELOW PANEL WHERE SPLICE IS REQUIRED. SEE FEEDER SCHEDULE FOR WIRE SIZE

FEEDER SCHEDULE		
ID	FEEDER AMPS	FEEDER
60	60	4#4, #8(G)
90	90	4#2, #8(G)
125	125	4#1/0, #6(G)



1 SINGLE LINE DIAGRAM
E601 SCALE: NO SCALE



 P.O. BOX 1515, GREENWOOD, MISSISSIPPI 39202-4515					
		NO.	DATE	BY	DESCRIPTION
A-E FIRM		USDA		REVISIONS	
PROJECT MANAGER	BP	EPM	XXX		
DESIGNER	EH	PPM	XXX		
CHECKED BY	BTD	SAFETY & HEALTH	XXX		
JOB #	4825	REAL PROPERTY	XXX		
		PROJECT NO.	XXX		
		DATE	04/27/2026		
		AGRICULTURAL RESEARCH SERVICE		SHEET	
		FT. PIERCE, FL		E-601	
				89 OF 108	

**USDA** Agricultural Research Service
USDA AGRICULTURAL RESEARCH SERVICE
US HORTICULTURAL RESEARCH LABORATORY
REPAIR TORNADO DAMAGE
FT. PIERCE, FL
SINGLE LINE DIAGRAM

C:\Users\JAWHANE\OneDrive\Local\Temp\JAWHANE_1746E-701.dwg
Apr 27, 2026 9:53am

PANEL 1GA			LOCATION: GREENHOUSE CORRIDOR VOLT: 208Y/120V, 3Ø, 4W BUS: 100A		LUG LOCATION: BOTTOM FEED MAIN BUS: 60A MAIN BREAKER MOUNTING: SURFACE			NEMA 3R ENCLOSURE PANELBOARD SCCR RATING (A):			10,000
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS	POLES	CIRCUIT NO.	
				A	B	C					
1	20	1	LIGHTING	0.3	0.3		GREENHOUSE 110 EF-1	15	3	2	
3	20	1	GREENHOUSE 110 CONTROL PANEL		0.5	0.3	-	-	-	4	
5	20	1	SPARE			0.5	0.3	-	-	6	
7	20	1	GREENHOUSE 110 RECS.	0.5	0.3		GREENHOUSE 110 EF-1	15	3	8	
9	20	1	GREENHOUSE 110 RECS.		0.5	0.3	-	-	-	10	
11	20	1	GREENHOUSE 110 RECS.			0.7	0.3	-	-	12	
13	20	1	GREENHOUSE 110 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 110 EF-1	15	3	14	
15	20	1	GREENHOUSE 110 OVERHEAD RECS.		0.5	0.3	-	-	-	16	
17	20	1	GREENHOUSE 110 OVERHEAD RECS.			0.5	0.3	-	-	18	
19	20	1	GREENHOUSE 110 OVERHEAD RECS.	0.5	0.2		GREENHOUSE 110 UH-1	20	1	20	
21	20	1	GREENHOUSE 110 OVERHEAD RECS.		0.4	0.2	GREENHOUSE 110 UH-1	20	1	22	
23	20	1	GREENHOUSE 110 OVERHEAD RECS.			0.4	0.5	GREENHOUSE 110 HF	20	1	24
25	20	1	GREENHOUSE 110 ECP SUMP PUMP	0.5	0.5		GREENHOUSE 110 HF	20	1	26	
27	20	1	GREENHOUSE 110 ROOF VENT		0.2	0.0	SPARE	30	1	28	
29	20	1	GREENHOUSE 110 SIDE VENT			0.2	0.0	SPARE	20	1	30
31	20	1	SPARE	0.0	0.0		SPARE	20	1	32	
33	20	1	SPARE		0.0	0.0	SPARE	20	1	34	
35	20	1	SPARE			0.0	0.0	SPARE	20	1	36
37	20	1	SPARE	0.0	0.0		SPARE	20	1	38	
39	20	1	SPARE		0.0	0.0	SPARE	20	1	40	
41	20	1	SPARE			0.0	0.0	SPARE	20	1	42
TOTAL				4.0	3.2	3.7	* GFCI BREAKER				

PANEL 1GC			LOCATION: GREENHOUSE CORRIDOR VOLT: 208Y/120V, 3Ø, 4W BUS: 100A		LUG LOCATION: BOTTOM FEED MAIN BUS: 60A MAIN BREAKER MOUNTING: SURFACE			NEMA 3R ENCLOSURE PANELBOARD SCCR RATING (A):			10,000
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS	POLES	CIRCUIT NO.	
				A	B	C					
1	20	1	LIGHTING	0.3	0.3		GREENHOUSE 116 EF-1	15	3	2	
3	20	1	GREENHOUSE 116 CONTROL PANEL		0.5	0.3	-	-	-	4	
5	20	1	SPARE			0.5	0.3	-	-	6	
7	20	1	GREENHOUSE 116 RECS.	0.5	0.3		GREENHOUSE 116 EF-1	15	3	8	
9	20	1	GREENHOUSE 116 RECS.		0.5	0.3	-	-	-	10	
11	20	1	GREENHOUSE 116 RECS.			0.7	0.3	-	-	12	
13	20	1	GREENHOUSE 116 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 116 EF-1	15	3	14	
15	20	1	GREENHOUSE 116 OVERHEAD RECS.		0.5	0.3	-	-	-	16	
17	20	1	GREENHOUSE 116 OVERHEAD RECS.			0.5	0.3	-	-	18	
19	20	1	GREENHOUSE 116 OVERHEAD RECS.	0.5	0.2		GREENHOUSE 116 UH-1	20	1	20	
21	20	1	GREENHOUSE 116 OVERHEAD RECS.		0.4	0.2	GREENHOUSE 116 UH-1	20	1	22	
23	20	1	GREENHOUSE 116 OVERHEAD RECS.			0.4	0.5	GREENHOUSE 116 HF	20	1	24
25	20	1	GREENHOUSE 116 ECP SUMP PUMP	0.5	0.5		GREENHOUSE 116 HF	20	1	26	
27	20	1	GREENHOUSE 116 ROOF VENT		0.2	0.0	SPARE	30	1	28	
29	20	1	GREENHOUSE 116 SIDE VENT			0.2	0.0	SPARE	20	1	30
31	20	1	SPARE	0.0	0.0		SPARE	20	1	32	
33	20	1	SPARE		0.0	0.0	SPARE	20	1	34	
35	20	1	SPARE			0.0	0.0	SPARE	20	1	36
37	20	1	SPARE	0.0	0.0		SPARE	20	1	38	
39	20	1	SPARE		0.0	0.0	SPARE	20	1	40	
41	20	1	SPARE			0.0	0.0	SPARE	20	1	42
TOTAL				4.0	3.2	3.7	* GFCI BREAKER				

PANEL 1GE			LOCATION: GREENHOUSE CORRIDOR VOLT: 208Y/120V, 3Ø, 4W BUS: 100A		LUG LOCATION: BOTTOM FEED MAIN BUS: 60A MAIN BREAKER MOUNTING: SURFACE			NEMA 3R ENCLOSURE PANELBOARD SCCR RATING (A):			10,000
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS	POLES	CIRCUIT NO.	
				A	B	C					
1	20	1	LIGHTING	0.3	0.3		GREENHOUSE 122 EF-1	15	3	2	
3	20	1	GREENHOUSE 122 CONTROL PANEL		0.5	0.3	-	-	-	4	
5	20	1	SPARE			0.5	-	-	-	6	
7	20	1	GREENHOUSE 122 RECS.	0.5	0.3		GREENHOUSE 122 EF-1	15	3	8	
9	20	1	GREENHOUSE 122 RECS.		0.5	0.3	-	-	-	10	
11	20	1	GREENHOUSE 122 RECS.			0.7	-	-	-	12	
13	20	1	GREENHOUSE 122 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 122 EF-1	15	3	14	
15	20	1	GREENHOUSE 122 OVERHEAD RECS.		0.5	0.3	-	-	-	16	
17	20	1	GREENHOUSE 122 OVERHEAD RECS.			0.5	-	-	-	18	
19	20	1	GREENHOUSE 122 OVERHEAD RECS.	0.5	0.2		GREENHOUSE 122 UH-1	20	1	20	
21	20	1	GREENHOUSE 122 OVERHEAD RECS.		0.4	0.2	GREENHOUSE 122 UH-1	20	1	22	
23	20	1	GREENHOUSE 122 OVERHEAD RECS.			0.4	GREENHOUSE 122 HF	20	1	24	
25	20	1	GREENHOUSE 122 ECP SUMP PUMP	0.5	0.5		GREENHOUSE 122 HF	20	1	26	
27	20	1	GREENHOUSE 122 ROOF VENT		0.2	1.5	DAC-1	30	1	28	
29	20	1	GREENHOUSE 122 SIDE VENT			0.2	FOYER ENTRANCE UH-1	20	1	30	
31	20	1	SPARE	0.0	0.2		WALKWAY UH-1	20	1	32	
33	20	1	SPARE		0.0	0.0	SPARE	20	1	34	
35	20	1	SPARE			0.0	SPARE	20	1	36	
37	20	1	SPARE	0.0	0.0		SPARE	20	1	38	
39	20	1	SPARE		0.0	0.0	SPARE	20	1	40	
41	20	1	HALLWAY LIGHTING			0.2	SPARE	20	1	42	
TOTAL				4.1	4.7	4.1	* GFCI BREAKER				

PANEL		LOCATION: GREENHOUSE CORRIDOR		LUG LOCATION: BOTTOM FEED		NEMA 3R ENCLOSURE		
1GG		VOLT: 208Y/120V, 3Ø, 4W		MAIN BUS: 60A MAIN BREAKER		PANELBOARD SCCR RATING (A):		
BUS: 100A				MOUNTING: SURFACE		10,000		
CIRCUIT NO.	BREAKER AMPS / POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS / POLES	CIRCUIT NO.
			A	B	C			
1	20 1	LIGHTING	0.3	0.3		GREENHOUSE 109 EF-1	15 3	2
3	20 1	GREENHOUSE 109 CONTROL PANEL		0.5 0.3		-	- -	4
5	20 1	SPARE			0.5 0.3	-	- -	6
7	20 1	GREENHOUSE 109 RECS.	0.5 0.3			GREENHOUSE 109 EF-1	15 3	8
9	20 1	GREENHOUSE 109 RECS.		0.5 0.3		-	- -	10
11	20 1	GREENHOUSE 109 RECS.			0.7 0.3	-	- -	12
13	20 1	GREENHOUSE 109 OVERHEAD RECS.	0.5 0.3			GREENHOUSE 109 EF-1	15 3	14
15	20 1	GREENHOUSE 109 OVERHEAD RECS.		0.5 0.3		-	- -	16
17	20 1	GREENHOUSE 109 OVERHEAD RECS.			0.5 0.3	-	- -	18
19	20 1	GREENHOUSE 109 OVERHEAD RECS.	0.5 0.2			GREENHOUSE 109 UH-1	20 1	20
21	20 1	GREENHOUSE 109 OVERHEAD RECS.		0.4 0.2		GREENHOUSE 109 UH-1	20 1	22
23	20 1	GREENHOUSE 109 OVERHEAD RECS.			0.4 0.5	GREENHOUSE 109 HF	20 1	24
25	20 1	GREENHOUSE 109 ECP SUMP PUMP	0.5 0.5			GREENHOUSE 109 HF	20 1	26
27	20 1	GREENHOUSE 109 ROOF VENT		0.2 0.0		SPARE	30 1	28
29	20 1	GREENHOUSE 109 SIDE VENT			0.2 0.0	SPARE	20 1	30
31	20 1	SPARE	0.0 0.0			SPARE	20 1	32
33	20 1	SPARE		0.0 0.0		SPARE	20 1	34
35	20 1	SPARE			0.0 0.0	SPARE	20 1	36
37	20 1	SPARE	0.0 0.0			SPARE	20 1	38
39	20 1	SPARE		0.0 0.0		SPARE	20 1	40
41	20 1	SPARE			0.0 0.0	SPARE	20 1	42
TOTAL			4.0	3.2	3.7	* GFCI BREAKER		

PANEL 1GJ		LOCATION: GREENHOUSE CORRIDOR VOLT: 208Y/120V, 3Ø, 4W BUS: 100A		LUG LOCATION: BOTTOM FEED MAIN BUS: 60A MAIN BREAKER MOUNTING: SURFACE		NEMA 3R ENCLOSURE PANEL
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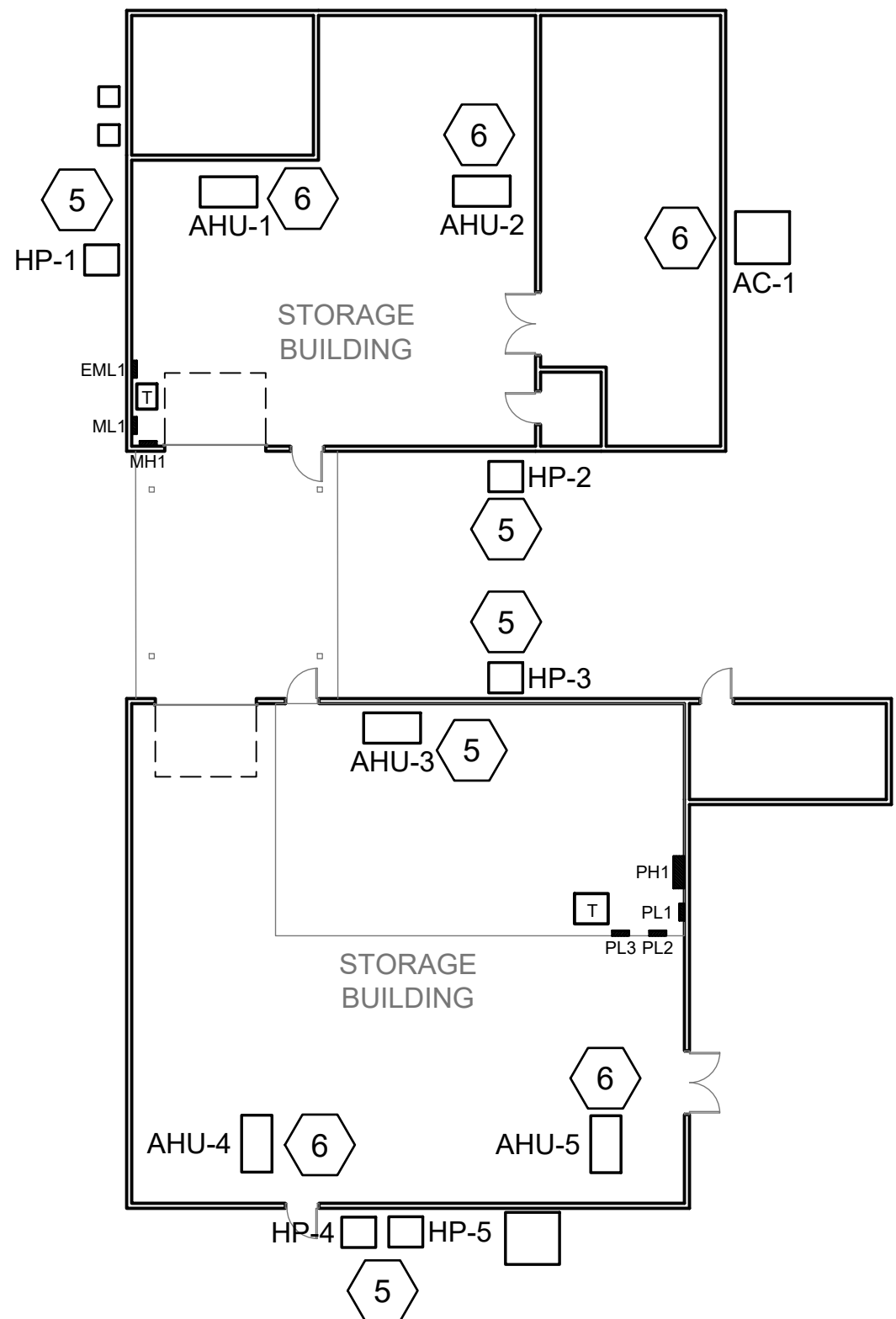
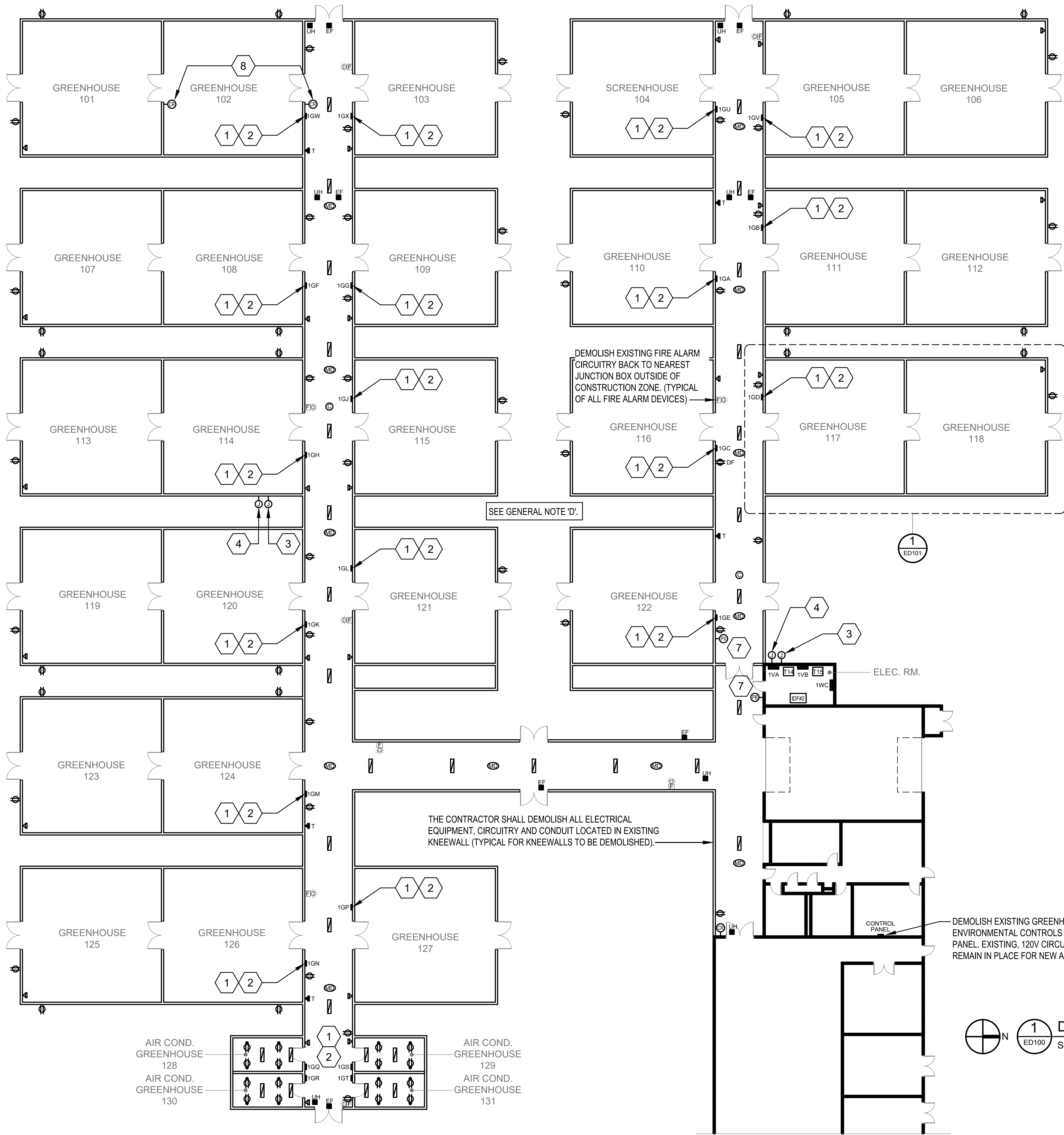
PANEL	LOCATION: GREENHOUSE CORRIDOR			LUG LOCATION: BOTTOM FEED			NEMA 3R ENCLOSURE			CIRCUIT	
1GB	VOLT: 208Y/120V, 3Ø, 4W			MAIN BUS: 100A MAIN BREAKER			PANELBOARD SCCR RATING (A):			NO.	
	BUS: 225A			MOUNTING: SURFACE							
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS	POLES	CIRCUIT NO.	
				A	B	C					
1	20	1	LIGHTING	0.5	0.3		GREENHOUSE 111 EF-1	15	3	2	
3	20	1	GREENHOUSE 111 CONTROL PANEL		0.5	0.3	-	-	-	4	
5	20	1	GREENHOUSE 112 CONTROL PANEL			0.5	0.3	-	-	6	
7	20	1	GREENHOUSE 111 RECS.	0.5	0.3		GREENHOUSE 111 EF-1	15	3	8	
9	20	1	GREENHOUSE 111 RECS.		0.5	0.3	-	-	-	10	
11	20	1	GREENHOUSE 111 RECS.			0.7	0.3	-	-	12	
13	20	1	GREENHOUSE 111 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 111 EF-1	15	3	14	
15	20	1	GREENHOUSE 111 OVERHEAD RECS.		0.5	0.3	-	-	-	16	
17	20	1	GREENHOUSE 111 OVERHEAD RECS.			0.5	0.3	-	-	18	
19	20	1	GREENHOUSE 111 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 112 EF-1	15	3	20	
21	20	1	GREENHOUSE 111 OVERHEAD RECS.		0.4	0.3	-	-	-	22	
23	20	1	GREENHOUSE 111 OVERHEAD RECS.			0.4	0.3	-	-	24	
25	20	1	GREENHOUSE 111 ECP SUMP PUMP	0.5	0.3		GREENHOUSE 112 EF-1	15	3	26	
27	20	1	GREENHOUSE 112 RECS.		0.5	0.3	-	-	-	28	
29	20	1	GREENHOUSE 112 RECS.			0.5	0.3	-	-	30	
31	20	1	GREENHOUSE 112 RECS.	0.7	0.3		GREENHOUSE 112 EF-1	15	3	32	
33	20	1	GREENHOUSE 112 OVERHEAD RECS.		0.5	0.3	-	-	-	34	
35	20	1	GREENHOUSE 112 OVERHEAD RECS.			0.5	0.3	-	-	36	
37	20	1	GREENHOUSE 112 OVERHEAD RECS.	0.5	0.4		GREENHOUSE 111 UH-1	20	1	38	
39	20	1	GREENHOUSE 112 OVERHEAD RECS.		0.5	1.0	GREENHOUSE 111 HF	20	1	40	
41	20	1	GREENHOUSE 112 OVERHEAD RECS.			0.4	0.5	GREENHOUSE 111 ROOF VENT	20	1	42
43	20	1	GREENHOUSE 112 OVERHEAD RECS.	0.4	0.5		GREENHOUSE 111 SIDE VENT	20	1	44	
45	20	1	GREENHOUSE 112 ECP SUMP PUMP		0.5	0.4	GREENHOUSE 112 UH-1	20	1	46	
47	20	1	SPARE			0.0	1.0	GREENHOUSE 112 HF	20	1	48
49	20	1	SPARE	0.0	0.5		GREENHOUSE 112 ROOF VENT	20	1	50	
51	20	1	SPARE		0.0	0.5	GREENHOUSE 112 SIDE VENT	20	1	52	
53	20	1	SPARE			0.0	0.0	SPARE	20	1	54
TOTAL				7.4	7.7	6.9	* GFCI BREAKER				

PANEL 1GD	LOCATION: GREENHOUSE CORRIDOR			LUG LOCATION: BOTTOM FEED			NEMA 3R ENCLOSURE			10,000	
	VOLT: 208Y/120V, 3Ø, 4W			MAIN BUS: 100A MAIN BREAKER			PANELBOARD SCCR RATING (A):				
	BUS: 225A			MOUNTING: SURFACE							
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS	POLES	CIRCUIT NO.	
				A	B	C					
1	20	1	LIGHTING	0.5	0.3		GREENHOUSE 117 EF-1	15	3	2	
3	20	1	GREENHOUSE 117 CONTROL PANEL		0.5	0.3	-	-	-	4	
5	20	1	GREENHOUSE 118 CONTROL PANEL			0.5	0.3	-	-	6	
7	20	1	GREENHOUSE 117 RECS.	0.5	0.3		GREENHOUSE 117 EF-1	15	3	8	
9	20	1	GREENHOUSE 117 RECS.		0.5	0.3	-	-	-	10	
11	20	1	GREENHOUSE 117 RECS.			0.7	0.3	-	-	12	
13	20	1	GREENHOUSE 117 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 117 EF-1	15	3	14	
15	20	1	GREENHOUSE 117 OVERHEAD RECS.		0.5	0.3	-	-	-	16	
17	20	1	GREENHOUSE 117 OVERHEAD RECS.			0.5	0.3	-	-	18	
19	20	1	GREENHOUSE 117 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 118 EF-1	15	3	20	
21	20	1	GREENHOUSE 117 OVERHEAD RECS.		0.4	0.3	-	-	-	22	
23	20	1	GREENHOUSE 117 OVERHEAD RECS.			0.4	0.3	-	-	24	
25	20	1	GREENHOUSE 117 ECP SUMP PUMP	0.5	0.3		GREENHOUSE 118 EF-1	15	3	26	
27	20	1	GREENHOUSE 118 RECS.		0.5	0.3	-	-	-	28	
29	20	1	GREENHOUSE 118 RECS.			0.5	0.3	-	-	30	
31	20	1	GREENHOUSE 118 RECS.	0.7	0.3		GREENHOUSE 118 EF-1	15	3	32	
33	20	1	GREENHOUSE 118 OVERHEAD RECS.		0.5	0.3	-	-	-	34	
35	20	1	GREENHOUSE 118 OVERHEAD RECS.			0.5	0.3	-	-	36	
37	20	1	GREENHOUSE 118 OVERHEAD RECS.	0.5	0.4		GREENHOUSE 117 UH-1	20	1	38	
39	20	1	GREENHOUSE 118 OVERHEAD RECS.		0.5	1.0	GREENHOUSE 117 HF	20	1	40	
41	20	1	GREENHOUSE 118 OVERHEAD RECS.			0.4	0.5	GREENHOUSE 117 ROOF VENT	20	1	42
43	20	1	GREENHOUSE 118 OVERHEAD RECS.	0.4	0.5		GREENHOUSE 117 SIDE VENT	20	1	44	
45	20	1	GREENHOUSE 118 ECP SUMP PUMP		0.5	0.4	GREENHOUSE 118 UH-1	20	1	46	
47	20	1	SPARE			0.0	1.0	GREENHOUSE 118 HF	20	1	48
49	20	1	SPARE	0.0	0.5		GREENHOUSE 118 ROOF VENT	20	1	50	
51	20	1	SPARE		0.0	0.5	GREENHOUSE 118 SIDE VENT	20	1	52	
53	20	1	SPARE			0.0	0.0	SPARE	20	1	54
TOTAL				7.4	7.7	6.9	* GFCI BREAKER				

PANEL LOCATION: GREENHOUSE CORRIDOR 1GF VOLT: 208Y/120V, 3Ø, 4W BUS: 225A			LUG LOCATION: BOTTOM FEED MAIN BUS: 100A MAIN BREAKER MOUNTING: SURFACE			NEMA 3R ENCLOSURE PANELBOARD SCCR RATING (A): 10,000					
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS	POLES	CIRCUIT NO.	
				A	B	C					
1	20	1	LIGHTING	0.5	0.3		GREENHOUSE 108 EF-1	15	3	2	
3	20	1	GREENHOUSE 108 CONTROL PANEL		0.5	0.3	-	-	-	4	
5	20	1	GREENHOUSE 107 CONTROL PANEL			0.5	0.3	-	-	6	
7	20	1	GREENHOUSE 108 RECS.	0.5	0.3		GREENHOUSE 108 EF-1	15	3	8	
9	20	1	GREENHOUSE 108 RECS.		0.5	0.3	-	-	-	10	
11	20	1	GREENHOUSE 108 RECS.			0.7	0.3	-	-	12	
13	20	1	GREENHOUSE 108 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 108 EF-1	15	3	14	
15	20	1	GREENHOUSE 108 OVERHEAD RECS.		0.5	0.3	-	-	-	16	
17	20	1	GREENHOUSE 108 OVERHEAD RECS.			0.5	0.3	-	-	18	
19	20	1	GREENHOUSE 108 OVERHEAD RECS.	0.5	0.3		GREENHOUSE 107 EF-1	15	3	20	
21	20	1	GREENHOUSE 108 OVERHEAD RECS.		0.4	0.3	-	-	-	22	
23	20	1	GREENHOUSE 108 OVERHEAD RECS.			0.4	0.3	-	-	24	
25	20	1	GREENHOUSE 108 ECP SUMP PUMP	0.5	0.3		GREENHOUSE 107 EF-1	15	3	26	
27	20	1	GREENHOUSE 107 RECS.		0.5	0.3	-	-	-	28	
29	20	1	GREENHOUSE 107 RECS.			0.5	0.3	-	-	30	
31	20	1	GREENHOUSE 107 RECS.	0.7	0.3		GREENHOUSE 107 EF-1	15	3	32	
33	20	1	GREENHOUSE 107 OVERHEAD RECS.		0.5	0.3	-	-	-	34	
35	20	1	GREENHOUSE 107 OVERHEAD RECS.			0.5	0.3	-	-	36	
37	20	1	GREENHOUSE 107 OVERHEAD RECS.	0.5	0.4		GREENHOUSE 108 UH-1	20	1	38	
39	20	1	GREENHOUSE 107 OVERHEAD RECS.		0.5	1.0	GREENHOUSE 108 HF	20	1	40	
41	20	1	GREENHOUSE 107 OVERHEAD RECS.			0.4	0.5	GREENHOUSE 108 ROOF VENT	20	1	42
43	20	1	GREENHOUSE 107 OVERHEAD RECS.	0.4	0.5		GREENHOUSE 108 SIDE VENT	20	1	44	
45	20	1	GREENHOUSE 107 ECP SUMP PUMP		0.5	0.4	GREENHOUSE 107 UH-1	20	1	46	
47	20	1	SPARE			0.0	1.0	GREENHOUSE 107 HF	20	1	48
49	20	1	SPARE	0.0	0.5		GREENHOUSE 107 ROOF VENT	20	1	50	
51	20	1	SPARE		0.0	0.5	GREENHOUSE 107 SIDE VENT	20	1	52	
53	20	1	SPARE			0.0	0.0	SPARE	20	1	54
TOTAL				7.4	7.7	6.9	* GFCI BREAKER				

PANEL 1GH			LOCATION: GREENHOUSE CORRIDOR VOLT: 208Y/120V, 3Ø, 4W BUS: 225A		LUG LOCATION: BOTTOM FEED MAIN BUS: 100A MAIN BREAKER MOUNTING: SURFACE			NEMA 3R ENCLOSURE PANELBOARD SCCR RATING (A):			10,000				
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION			PHASE LOAD (KVA)			DESCRIPTION			BREAKER AMPS	POLES	CIRCUIT NO.	
						A	B	C							
1	20	1	LIGHTING			0.5	0.3		GREENHOUSE 114 EF-1			15	3	2	
3	20	1	GREENHOUSE 114 CONTROL PANEL				0.5	0.3		-			-	-	4
5	20	1	GREENHOUSE 113 CONTROL PANEL					0.5	0.3	-			-	-	6
7	20	1	GREENHOUSE 114 RECS.			0.5	0.3			GREENHOUSE 114 EF-1			15	3	8
9	20	1	GREENHOUSE 114 RECS.				0.5	0.3		-			-	-	10
11	20	1	GREENHOUSE 114 RECS.					0.7	0.3	-			-	-	12
13	20	1	GREENHOUSE 114 OVERHEAD RECS.			0.5	0.3			GREENHOUSE 114 EF-1			15	3	14
15	20	1	GREENHOUSE 114 OVERHEAD RECS.				0.5	0.3		-			-	-	16
17	20	1	GREENHOUSE 114 OVERHEAD RECS.					0.5	0.3	-			-	-	18
19	20	1	GREENHOUSE 114 OVERHEAD RECS.			0.5	0.3			GREENHOUSE 113 EF-1			15	3	20
21	20	1	GREENHOUSE 114 OVERHEAD RECS.				0.4	0.3		-			-	-	22
23	20	1	GREENHOUSE 114 OVERHEAD RECS.					0.4	0.3	-			-	-	24
25	20	1	GREENHOUSE 114 ECP SUMP PUMP			0.5	0.3			GREENHOUSE 113 EF-1			15	3	26
27	20	1	GREENHOUSE 113 RECS.				0.5	0.3		-			-	-	28
29	20	1	GREENHOUSE 113 RECS.					0.5	0.3	-			-	-	30
31	20	1	GREENHOUSE 113 RECS.			0.7	0.3			GREENHOUSE 113 EF-1			15	3	32
33	20	1	GREENHOUSE 113 OVERHEAD RECS.				0.5	0.3		-			-	-	34
35	20	1	GREENHOUSE 113 OVERHEAD RECS.					0.5	0.3	-			-	-	36
37	20	1	GREENHOUSE 113 OVERHEAD RECS.			0.5	0.4			GREENHOUSE 114 UH-1			20	1	38
39	20	1	GREENHOUSE 113 OVERHEAD RECS.				0.5	1.0		GREENHOUSE 114 HF			20	1	40
41	20	1	GREENHOUSE 113 OVERHEAD RECS.					0.4	0.5	GREENHOUSE 114 ROOF VENT			20	1	42
43	20	1	GREENHOUSE 113 OVERHEAD RECS.			0.4	0.5			GREENHOUSE 114 SIDE VENT			20	1	44
45	20	1	GREENHOUSE 113 ECP SUMP PUMP				0.5	0.4		GREENHOUSE 113 UH-1			20	1	46
47	20	1	SPARE					0.0	1.0	GREENHOUSE 113 HF			20	1	48
49	20	1	SPARE			0.0	0.5			GREENHOUSE 113 ROOF VENT			20	1	50
51	20	1	SPARE				0.0	0.5		GREENHOUSE 113 SIDE VENT			20	1	52
53	20	1	HALLWAY LIGHTING					0.2	0.0	SPARE			20	1	54
TOTAL						7.4	7.7	7.1	* GFICI BREAKER						

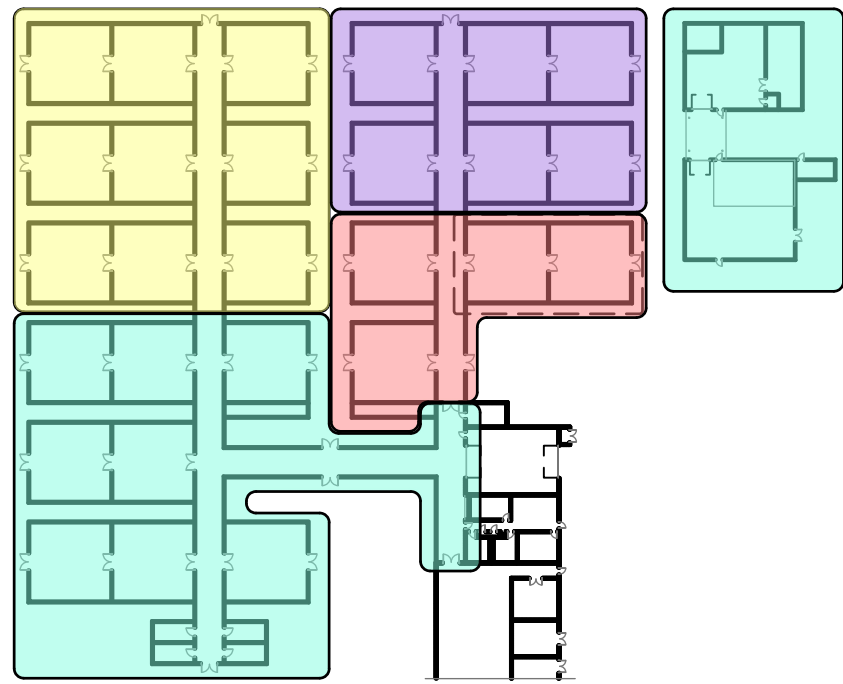
PANEL 1GK			LOCATION: GREENHOUSE CORRIDOR VOLT: 208Y/120V, 3Ø, 4W BUS: 225A		LUG LOCATION: BOTTOM FEED MAIN BUS: 100A MAIN BREAKER MOUNTING: SURFACE			NEMA 3R ENCLOSURE PANELBOARD SCCR RATING (A): 10,000		
CIRCUIT NO.	BREAKER AMPS	POLES	DESCRIPTION	PHASE LOAD (KVA)			DESCRIPTION	BREAKER AMPS	POLES	CIRCUIT NO.
				A	B	C				
1	20	1	LIGHTING	0.5	0.3					
3	20	1	GREENHOUSE 120 CONTROL PANEL		0.5	0.3		GREENHOUSE 120 EF-1	15	3
5	20	1	GREENHOUSE 110 CONTROL PANEL				0.5	-	-	4
7	20	1	GREENHOUSE 120 RECS.	0.5	0.3			GREENHOUSE 120 EF-1	15	3
9	20	1	GREENHOUSE 120 RECS.		0.5	0.3		-	-	10
11	20	1	GREENHOUSE 120 RECS.			0.7	0.3	-	-	12
13	20	1	GREENHOUSE 120 OVERHEAD RECS.	0.5	0.3			GREENHOUSE 120 EF-1	15	3
15	20	1	GREENHOUSE 120 OVERHEAD RECS.		0.5	0.3		-	-	16
17	20	1	GREENHOUSE 120 OVERHEAD RECS.			0.5	0.3	-	-	18
19	20	1	GREENHOUSE 120 OVERHEAD RECS.	0.5	0.3			GREENHOUSE 119 EF-1	15	3
21	20	1	GREENHOUSE 120 OVERHEAD RECS.		0.4	0.3		-	-	22
23	20	1	GREENHOUSE 120 OVERHEAD RECS.			0.4	0.3	-	-	24
25	20	1	GREENHOUSE 120 ECP SUMP PUMP	0.5	0.3			GREENHOUSE 119 EF-1	15	3
27	20	1	GREENHOUSE 119 RECS.		0.5	0.3		-	-	28
29	20	1	GREENHOUSE 119 RECS.			0.5	0.3	-	-	30
31	20	1	GREENHOUSE 119 RECS.	0.7	0.3			GREENHOUSE 119 EF-1	15	3
33	20	1	GREENHOUSE 119 OVERHEAD RECS.		0.5	0.3		-	-	34
35	20	1	GREENHOUSE 119 OVERHEAD RECS.			0.5	0.3	-	-	36
37	20	1	GREENHOUSE 119 OVERHEAD RECS.	0.5	0.4			GREENHOUSE 120 UH-1	20	1
39	20	1	GREENHOUSE 119 OVERHEAD RECS.		0.5	1.0		GREENHOUSE 120 HF	20	1
41	20	1	GREENHOUSE 119 OVERHEAD RECS.			0.4	0.5	GREENHOUSE 120 ROOF VENT	20	1
43	20	1	GREENHOUSE 119 OVERHEAD RECS.	0.4	0.5			GREENHOUSE 120 SIDE VENT	20	1
45	20	1	GREENHOUSE 119 ECP SUMP PUMP		0.5	0.4		GREENHOUSE 119 UH-1	20	1
47	20	1	SPARE			0.0	1.0	GREENHOUSE 119 HF	20	1
49	20	1	SPARE	0.0	0.5			GREENHOUSE 119 ROOF VENT	20	1
51	20	1	SPARE		0.0	0.5		GREENHOUSE 119 SIDE VENT	20	1
53	20	1	SPARE			0.0	0.0	SPARE	20	1
TOTAL				7.4	7.7	6.9		* GFCI BREAKER		



NOTE: WORK IN THIS AREA CAN BE DONE DURING ANY BID OPTION.

BID OPTION LEGEND

	BASE BID		BID OPTION #2
	BID OPTION #1		BID OPTION #3



2 KEY PLAN
ED100 SCALE: N.T.S

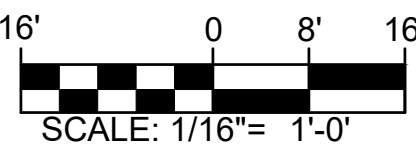
GENERAL DEMOLITION NOTES

- PRIOR TO BEGINNING WORK, SURVEY EXISTING ELECTRICAL SYSTEMS. DOCUMENT, IN WRITING, SIGNED BY THE CONTRACTING OFFICER, ANY PORTIONS OF THE EXISTING SYSTEMS THAT ARE NOT OPERATING PROPERLY BEFORE CONSTRUCTION BEGINS. ANY ELECTRICAL SYSTEMS FOUND INOPERABLE AT THE END OF THE CONSTRUCTION PROCESS THAT HAVE NOT BEEN DOCUMENTED SHALL BE REPAIRED BY THE CONTRACTOR AT HIS OWN EXPENSE.
- PRIOR TO BEGINNING THE DEMOLITION PROCESS, CONDUCT A COMPREHENSIVE SITE SURVEY WITH THE ASSISTANCE OF THE CONTRACTING OFFICER. IDENTIFY AND MARK ALL EXISTING EQUIPMENT TO REMAIN AND ALL EXISTING RACEWAYS AND CIRCUITRY THAT ARE PRESENTLY PASSING THROUGH THE CONSTRUCTION AREA THAT ARE FEEDING EXISTING EQUIPMENT TO REMAIN.
- REMOVE ELECTRICAL EQUIPMENT IN AREAS BEING DEMOLISHED, AND ELECTRICAL EQUIPMENT FEEDING OTHER EQUIPMENT BEING DEMOLISHED. REMOVE RACEWAYS AND CIRCUITRY BACK TO THE PANEL OF ORIGINATION. WHERE CIRCUITS ARE NOT BEING COMPLETELY DEMOLISHED, REMOVE RACEWAYS AND CIRCUITRY BACK TO A JUNCTION BOX OR OTHER COMMON CONNECTION POINT AND RE-CIRCUIT EXISTING ELECTRICAL EQUIPMENT TO REMAIN AS REQUIRED. WHERE NECESSARY, COMPLETELY REFEED EXISTING ELECTRICAL EQUIPMENT THAT IS TO REMAIN IF REQUIRED. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL EXISTING EQUIPMENT TO REMAIN BE KEPT COMPLETELY OPERABLE DURING CONSTRUCTION AND LEFT COMPLETELY OPERABLE AT THE END OF THE CONSTRUCTION PROCESS.
- THE ELECTRICAL EQUIPMENT SHOWN IS PROVIDED TO CONVEY THE APPROXIMATE SCOPE OF THE WORK. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL ELECTRICAL EQUIPMENT, RACEWAYS, AND CIRCUITRY IN THE AREA OF WORK BE DEMOLISHED UNLESS FEEDING EXISTING EQUIPMENT TO REMAIN.
- PERFORM DEMOLITION WORK AS FOLLOWS:
 - DEMOLISH DEVICE OR END OF LINE EQUIPMENT.
 - DEMOLISH RACEWAYS AND CIRCUITRY FEEDING END OF LINE EQUIPMENT (INCLUDING ALL RACEWAYS AND CIRCUITRY ASSOCIATED WITH CONTROLS).
 - LEAVE ALL RACEWAYS AND CIRCUITRY THAT COMPLETELY PASS-THROUGH THE SPACE COMPLETELY OPERABLE BY REROUTING THE CIRCUITS AS REQUIRED.

KEYED NOTES

- THE EXISTING FEEDER CONDUIT AND CIRCUITRY FOR ALL GREENHOUSE PANELS IS EXISTING TO REMAIN AND SHALL BE PROTECTED THROUGHOUT DEMOLITION. SEE 2/ED101 FOR TYPICAL ELEVATION VIEW OF GREENHOUSE PANELS.
- THE EXISTING UNDERGROUND BRANCH CIRCUIT CONDUITS AT EACH GREENHOUSE SHALL BE CUT LEVEL WITH THE SLAB AND FILLED. SEE 2/ED101 FOR TYPICAL ELEVATION VIEW OF GREENHOUSE PANELS.
- EXISTING 3" CONDUIT ROUTED BETWEEN ELECTRICAL ROOM AND EXTERIOR OF GREENHOUSE 114 IS EXISTING TO REMAIN FOR FUTURE USE. REMOVE EXISTING CABLE BACK TO IDF #2.
- EXISTING JUNCTION BOX, UNDERGROUND 1", AND CABLE ROUTED TO TELEPHONE J-BOX SHALL BE DEMOLISHED IN ITS ENTIRETY.
- DEMOLISH EXISTING DISCONNECT AND CIRCUITRY BETWEEN DISCONNECT AND UNIT. EXISTING CIRCUITRY BETWEEN PANEL AND DISCONNECT SHALL REMAIN.
- EXISTING CONDUIT & CIRCUITRY SHALL BE DEMOLISHED BACK TO PANEL. TURN BREAKER OFF & LABEL AS SPARE.
- REMOVE EXISTING ACCESSIBLE DOOR OPENER AND STORE DURING DEMOLITION. REINSTALL ONCE NEW DOOR HARDWARE IS IN PLACE.
- EXISTING ACCESS CONTROL SYSTEM SHALL BE REPLACED. SEE E-002 FOR DETAILS ON NEW SYSTEM. COORDINATE DEMO/REPLACEMENT WITH PHASING PLAN.

GRAPHIC SCALE



THE JOHNSON & McADAMS, INC.
ELECTRICAL ENGINEERS • PLANNING • SURVEYING • MANAGEMENT
P.O. BOX 513, GREENWOOD, MISSISSIPPI 38902-0513 • (662) 455-4943

A-E FIRM	USDA
PROJECT MANAGER	EPM
BP	XXX
DESIGNER	PPM
EH	XXX
CHECKED BY	SAFETY & HEALTH
BTD	XXX
JOB #	REAL PROPERTY
4825	XXX

USDA

USDA Agricultural Research Service

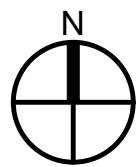
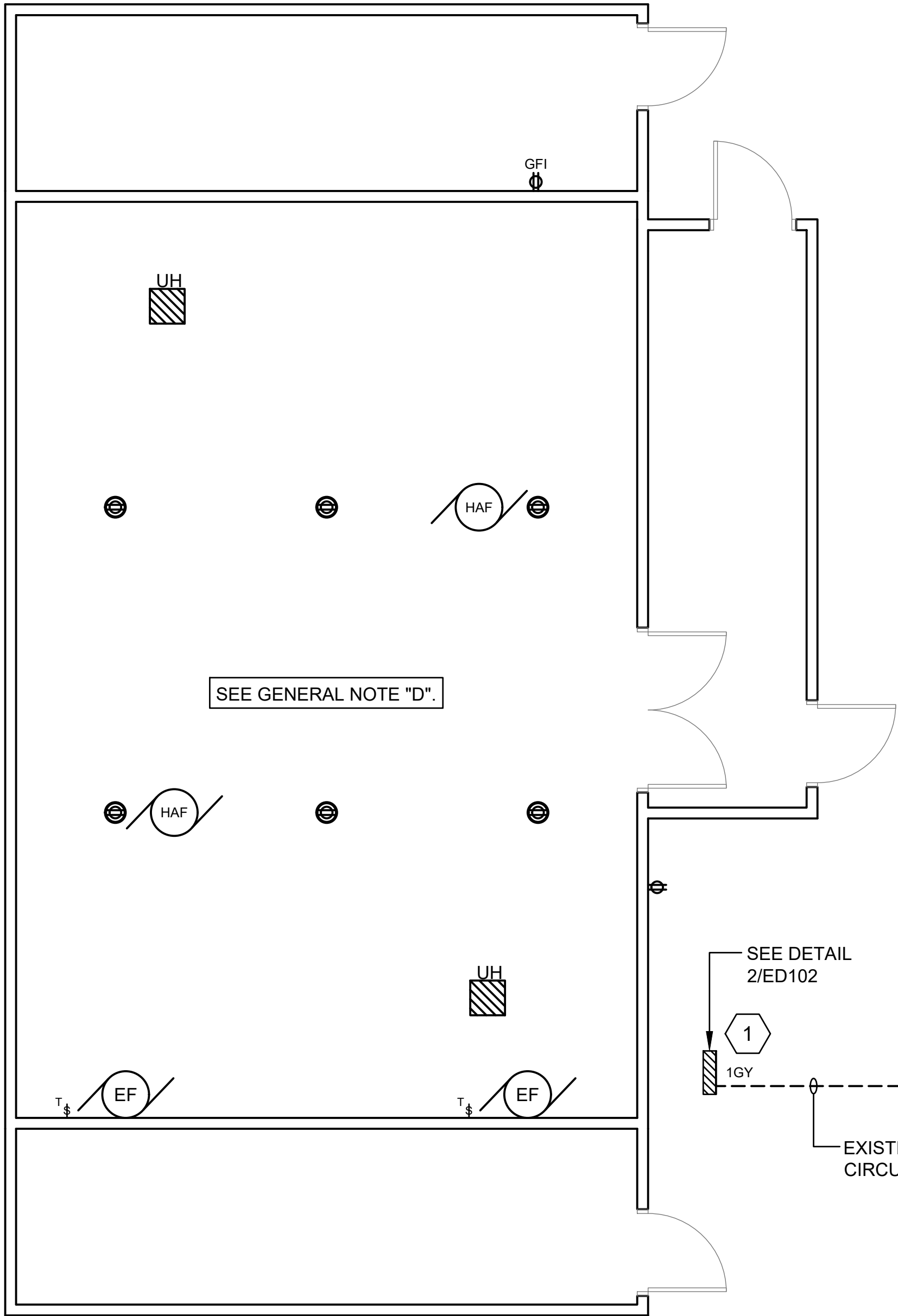
USDA AGRICULTURAL RESEARCH SERVICE
US HORTICULTURAL RESEARCH LABORATORY
REPAIR TORNADO DAMAGE
FT. PIERCE, FL

OVERALL ELECTRICAL DEMOLITION PLAN

PROJECT NO.	AGRICULTURAL RESEARCH SERVICE	SHEET
XXX		ED100
DATE	FT. PIERCE, FL	92 OF 108
04/27/2026		



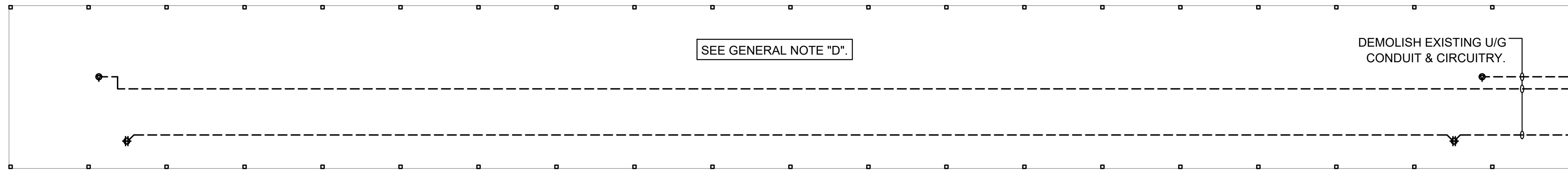
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1
ED102

BID OPTION #4 - PICO'S FARM ENLARGED GREENHOUSE DEMOLITION PLAN

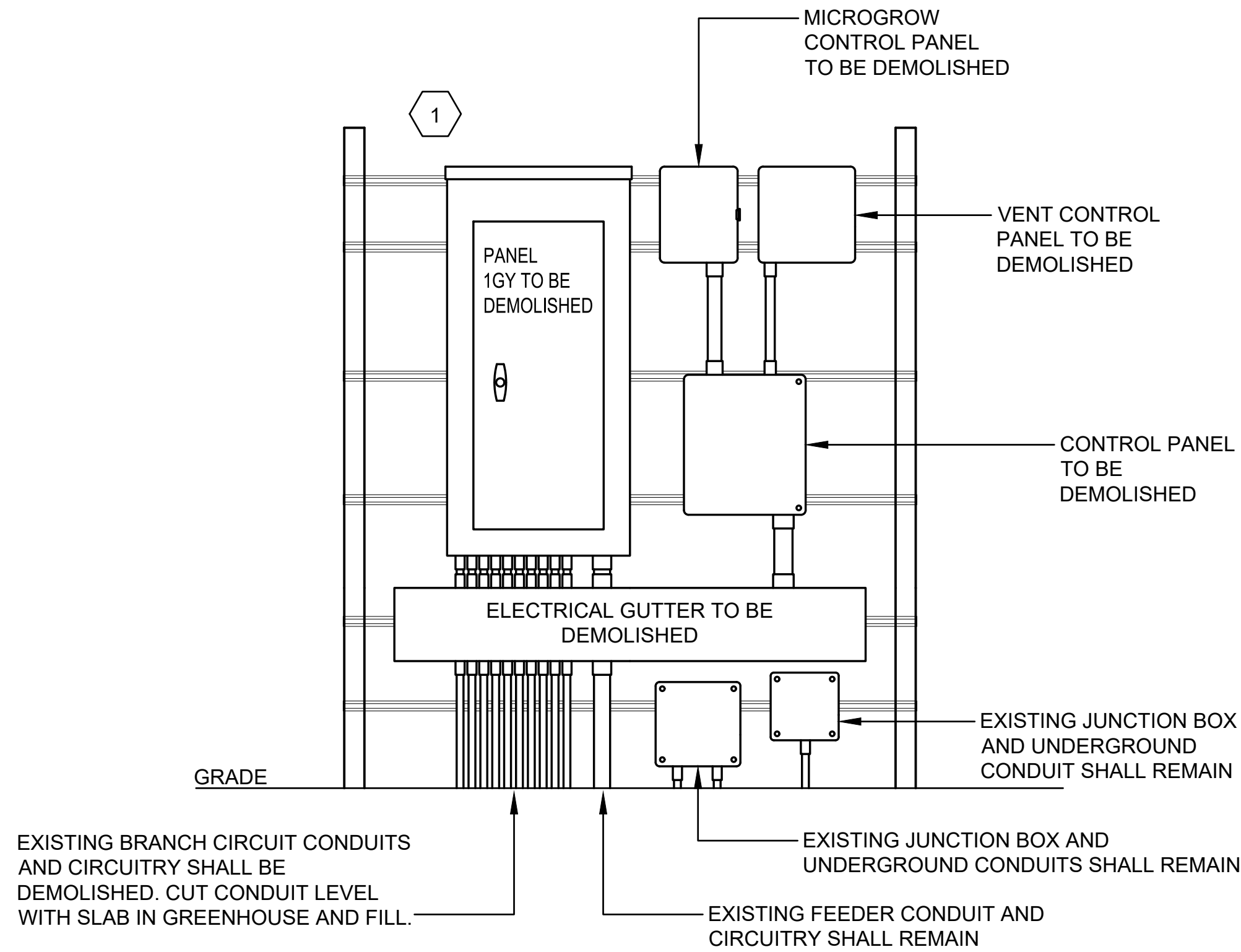
SCALE: 1/4" = 1'-0"



3
ED102

BID OPTION #4 - PICO'S FARM HOOPHOUSE DEMOLITION PLAN

SCALE: 3/16" = 1'-0"



2
ED102

GREENHOUSE PANEL DETAIL

SCALE: NONE



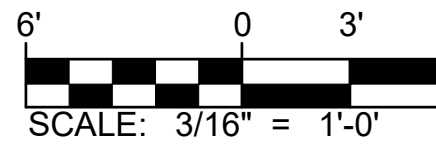
KEYED NOTES

- EXISTING PANEL, ELECTRICAL GUTTER, AND ALL CONTROL PANELS SHALL BE DEMOLISHED. THE EXISTING FEEDER CONDUIT AND CIRCUITRY SHALL REMAIN AND BE PROTECTED DURING DEMOLITION. FEEDER BREAKER IS LOCATED IN PANEL "1VC" (13,15,17) IN THE MAIN ELECTRICAL ROOM.
- EXISTING DISCONNECT, TRANSFORMER, PANEL, UNDERGROUND CONDUIT, BRANCH CIRCUITRY, AND RECEPTACLES SHALL BE DEMOLISHED. EXISTING 480V FEEDER CONDUIT AND CONDUCTORS SHALL REMAIN AND BE PROTECTED DURING DEMOLITION. FEEDER BREAKER IS LOCATED IN PANEL "1WD" IN MAIN ELECTRICAL ROOM.

GRAPHIC SCALE



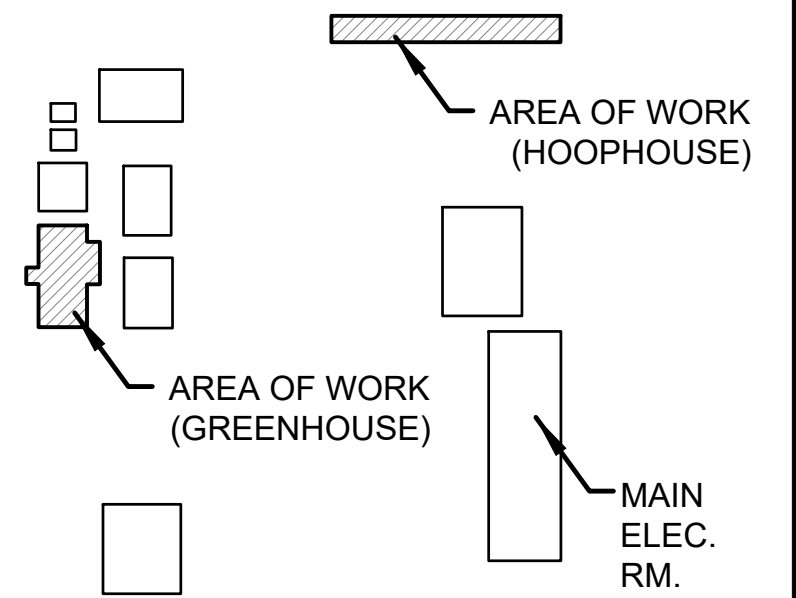
SCALE: 1/4"=1'-0"



SCALE: 3/16" = 1'-0"

GENERAL NOTES

- A. PRIOR TO BEGINNING WORK, SURVEY EXISTING ELECTRICAL SYSTEMS. DOCUMENT, IN WRITING, SIGNED BY THE CONTRACTING OFFICER, ANY PORTIONS OF THE EXISTING SYSTEMS THAT ARE NOT OPERATING PROPERLY BEFORE CONSTRUCTION BEGINS. ANY ELECTRICAL SYSTEMS FOUND INOPERABLE AT THE END OF THE CONSTRUCTION PROCESS THAT HAVE NOT BEEN SO DOCUMENTED SHALL BE REPAIRED BY THE CONTRACTOR AT HIS OWN EXPENSE.
- B. PRIOR TO BEGINNING THE DEMOLITION PROCESS, CONDUCT A COMPREHENSIVE SITE SURVEY WITH THE ASSISTANCE OF THE CONTRACTING OFFICER. IDENTIFY AND MARK ALL EXISTING EQUIPMENT TO REMAIN AND ALL EXISTING RACEWAYS AND CIRCUITRY THAT ARE PRESENTLY PASSING THROUGH THE CONSTRUCTION AREA THAT ARE FEEDING EXISTING EQUIPMENT TO REMAIN.
- C. REMOVE ELECTRICAL EQUIPMENT IN AREAS BEING DEMOLISHED, AND ELECTRICAL EQUIPMENT FEEDING OTHER EQUIPMENT BEING DEMOLISHED. REMOVE RACEWAYS AND CIRCUITRY BACK TO THE PANEL OF ORIGIN. WHERE CIRCUITS ARE NOT BEING COMPLETELY DEMOLISHED, REMOVE RACEWAYS AND CIRCUITRY BACK TO A JUNCTION BOX OR OTHER COMMON CONNECTION POINT AND RECIRCUIT EXISTING ELECTRICAL EQUIPMENT TO REMAIN AS REQUIRED. WHERE NECESSARY, COMPLETELY REFEED EXISTING ELECTRICAL EQUIPMENT THAT IS TO REMAIN IF REQUIRED. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL EXISTING EQUIPMENT TO REMAIN BE KEPT COMPLETELY OPERABLE DURING CONSTRUCTION AND LEFT COMPLETELY OPERABLE AT THE END OF THE CONSTRUCTION PROCESS.
- D. THE ELECTRICAL EQUIPMENT SHOWN IS PROVIDED TO CONVEY THE APPROXIMATE SCOPE OF THE WORK. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL ELECTRICAL EQUIPMENT, RACEWAYS, AND CIRCUITRY IN THE AREA OF WORK BE DEMOLISHED UNLESS FEEDING EXISTING EQUIPMENT TO REMAIN.
- E. PERFORM DEMOLITION WORK AS FOLLOWS:
- DEMOLISH DEVICE OR END OF LINE EQUIPMENT.
 - DEMOLISH RACEWAYS AND CIRCUITRY FEEDING END OF LINE EQUIPMENT (INCLUDING ALL RACEWAYS AND CIRCUITRY ASSOCIATED WITH CONTROLS).
 - LEAVE ALL RACEWAYS AND CIRCUITRY THAT COMPLETELY PASS-THROUGH THE SPACE COMPLETELY OPERABLE BY REROUTING THE CIRCUITS AS REQUIRED.



PICOS FARM KEYPLAN

SCALE: N.T.S

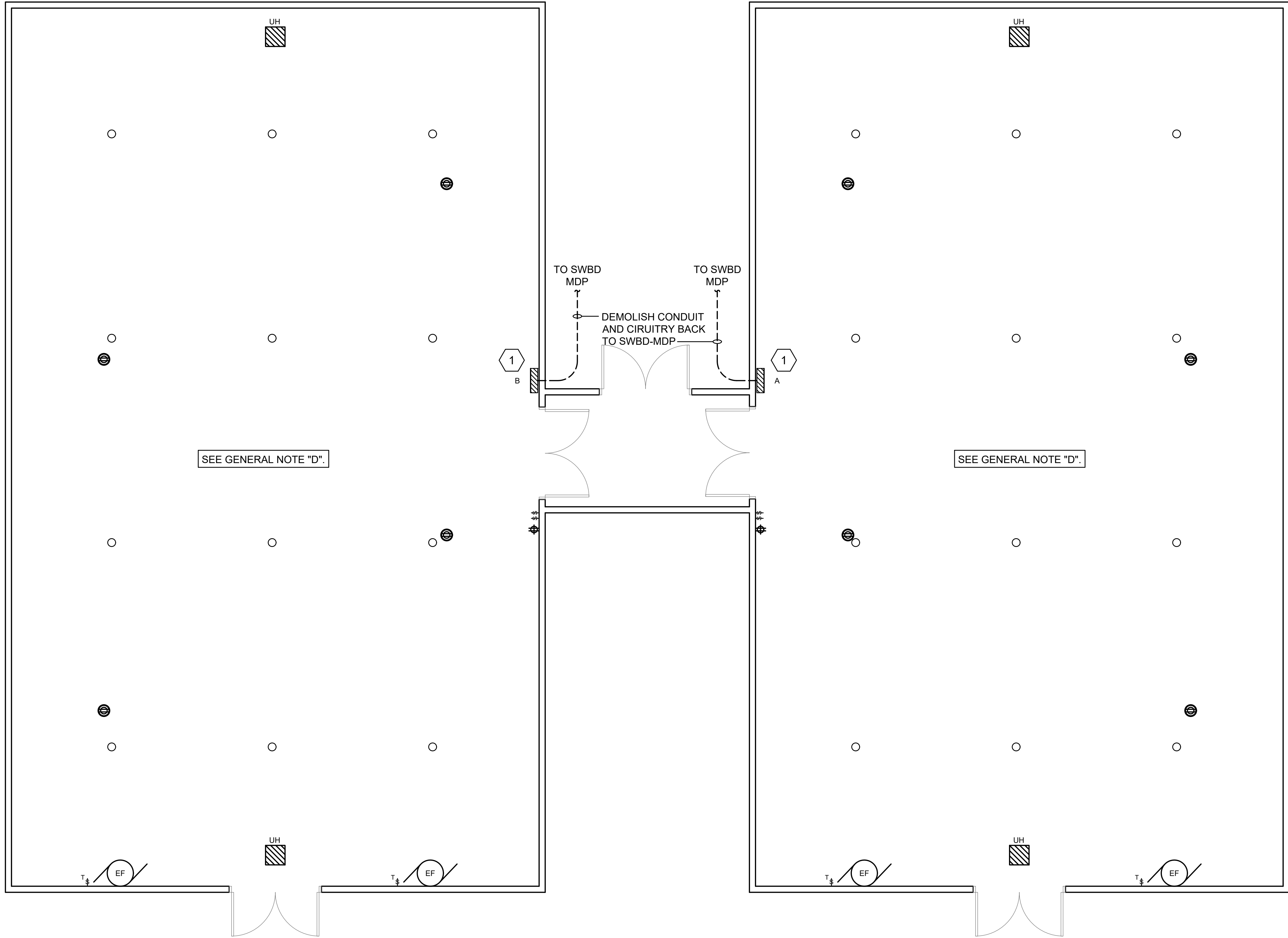
Johnson & Adams
P.O. BOX 513, GREENWOOD, MISSISSIPPI 38921-0513
ARCHITECTS/ENGINEERS • PLANNING • SURVEYING • MANAGEMENT

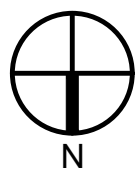
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PROJECT MANAGER	EPM
BP	XXX
DESIGNER	PPM
EH	XXX
CHECKED BY	SAFETY & HEALTH
BTD	XXX
JOB #	REAL PROPERTY
4825	XXX

NO.	DATE	BY	DESCRIPTION
REVISIONS			
USDA Agricultural Research Service			
USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL			
BID OPTION #4 ELECTRICAL DEMOLITION PLAN			
PROJECT NO.	XXX	AGRICULTURAL RESEARCH SERVICE	SHEET
DATE	04/27/2026	FT. PIERCE, FL	ED102
			94 OF 108



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Apr 27, 2026 9:55am

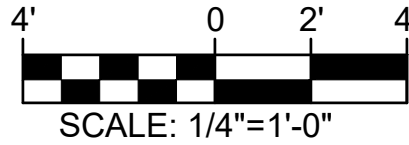


 **1** **BID OPTION #5 - LEESBURG FARM ENLARGED GREENHOUSE DEMOLITION PLAN**
ED103 SCALE: 1/4" = 1'-0"

KEYED NOTES

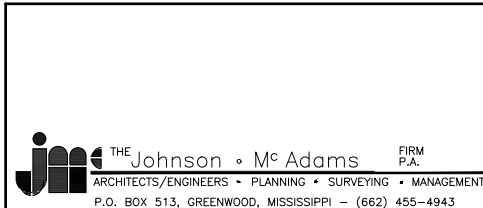
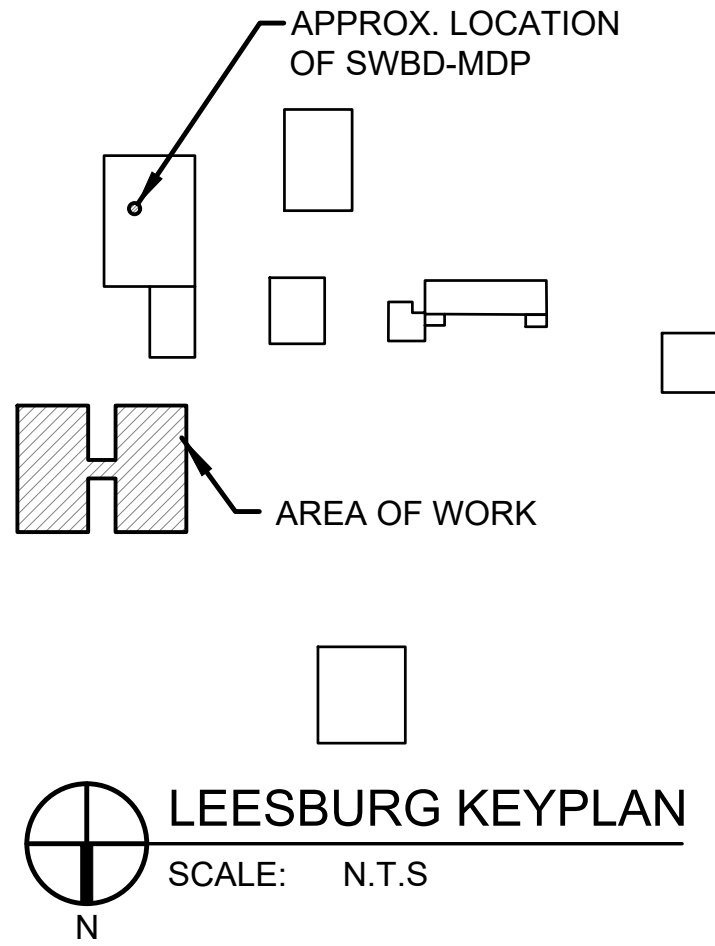
1. DEMOLISH EXISTING PANEL. EXISTING UNDERGROUND FEEDER CONDUIT AND CIRCUITRY SHALL BE DEMOLISHED BACK TO SWBD MDP APPROX 150' NORTH OF GREENHOUSE. TURN BREAKER OFF AND LABEL AS SPARE.

GRAPHIC SCALE



GENERAL DEMOLITION NOTES

- A. PRIOR TO BEGINNING WORK, SURVEY EXISTING ELECTRICAL SYSTEMS. DOCUMENT, IN WRITING, SIGNED BY THE CONTRACTING OFFICER, ANY PORTIONS OF THE EXISTING SYSTEMS THAT ARE NOT OPERATING PROPERLY BEFORE CONSTRUCTION BEGINS. ANY ELECTRICAL SYSTEMS FOUND INOPERABLE AT THE END OF THE CONSTRUCTION PROCESS THAT HAVE NOT BEEN SO DOCUMENTED SHALL BE REPAIRED BY THE CONTRACTOR AT HIS OWN EXPENSE.
- B. PRIOR TO BEGINNING THE DEMOLITION PROCESS, CONDUCT A COMPREHENSIVE SITE SURVEY WITH THE ASSISTANCE OF THE CONTRACTING OFFICER. IDENTIFY AND MARK ALL EXISTING EQUIPMENT TO REMAIN AND ALL EXISTING RACEWAYS AND CIRCUITRY THAT ARE PRESENTLY PASSING THROUGH THE CONSTRUCTION AREA THAT ARE FEEDING EXISTING EQUIPMENT TO REMAIN.
- C. REMOVE ELECTRICAL EQUIPMENT IN AREAS BEING DEMOLISHED, AND ELECTRICAL EQUIPMENT FEEDING OTHER EQUIPMENT BEING DEMOLISHED. REMOVE RACEWAYS AND CIRCUITRY BACK TO THE PANEL OF ORIGINATION. WHERE CIRCUITS ARE NOT BEING COMPLETELY DEMOLISHED, REMOVE RACEWAYS AND CIRCUITRY BACK TO A JUNCTION BOX OR OTHER COMMON CONNECTION POINT AND RECIRCUIT EXISTING ELECTRICAL EQUIPMENT TO REMAIN AS REQUIRED. WHERE NECESSARY, COMPLETELY REFEED EXISTING ELECTRICAL EQUIPMENT THAT IS TO REMAIN IF REQUIRED. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL EXISTING EQUIPMENT TO REMAIN BE KEPT COMPLETELY OPERABLE DURING CONSTRUCTION AND LEFT COMPLETELY OPERABLE AT THE END OF THE CONSTRUCTION PROCESS.
- D. THE ELECTRICAL EQUIPMENT SHOWN IS PROVIDED TO CONVEY THE APPROXIMATE SCOPE OF THE WORK. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL ELECTRICAL EQUIPMENT, RACEWAYS, AND CIRCUITRY IN THE AREA OF WORK BE DEMOLISHED UNLESS FEEDING EXISTING EQUIPMENT TO REMAIN.
- E. PERFORM DEMOLITION WORK AS FOLLOWS:
 1. DEMOLISH DEVICE OR END OF LINE EQUIPMENT.
 2. DEMOLISH RACEWAYS AND CIRCUITRY FEEDING END OF LINE EQUIPMENT (INCLUDING ALL RACEWAYS AND CIRCUITRY ASSOCIATED WITH CONTROLS).
 3. LEAVE ALL RACEWAYS AND CIRCUITRY THAT COMPLETELY PASS-THROUGH THE SPACE COMPLETELY OPERABLE BY REROUTING THE CIRCUITS AS REQUIRED.

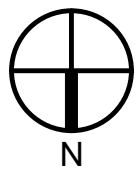


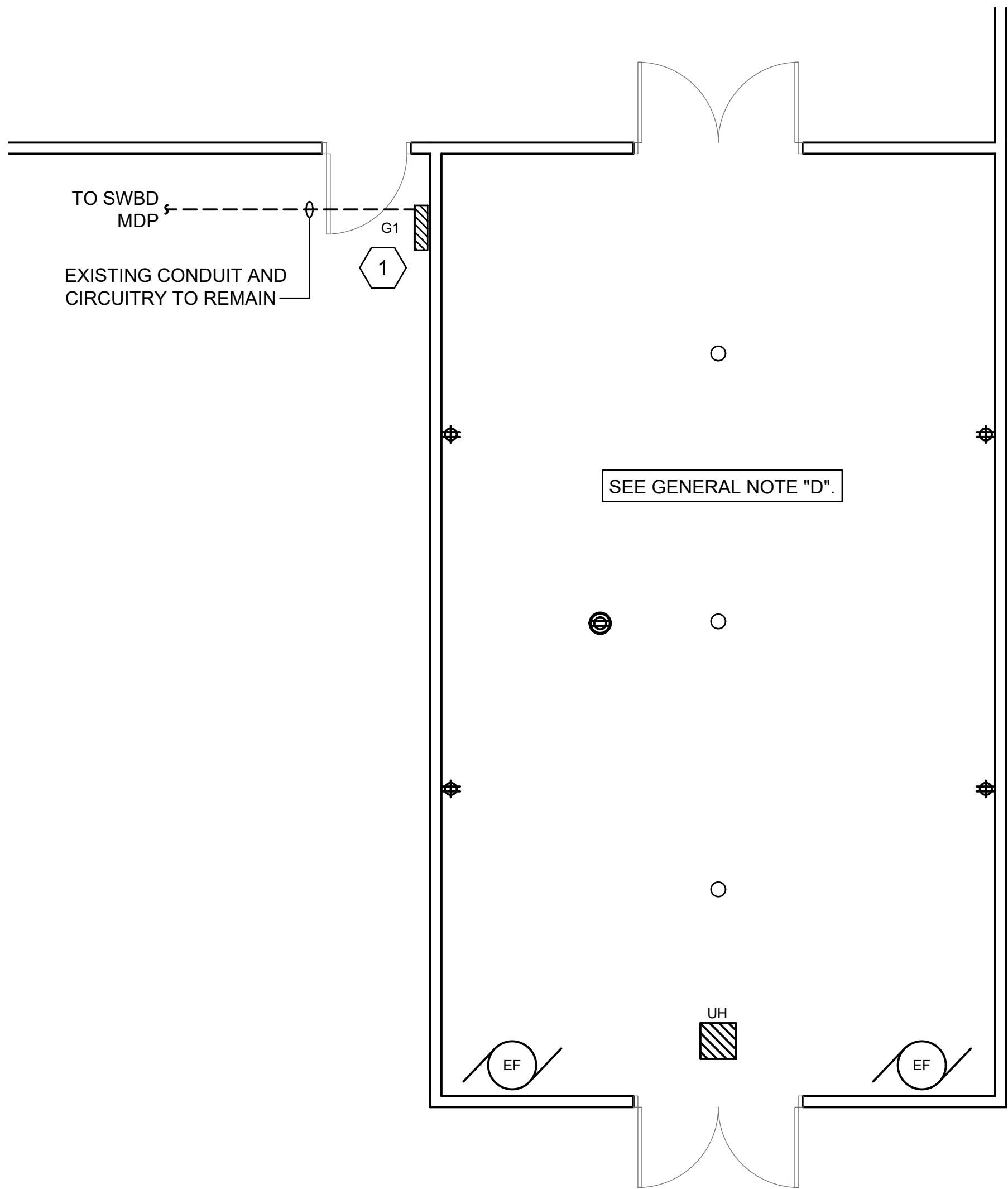
A-E FIRM		USDA
PROJECT MANAGER	EPM	XXX
DESIGNER	EH	XXX
CHECKED BY	BTB	XXX
JOB #	4825	XXX

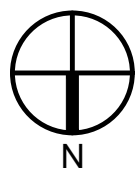
NO.	DATE	BY	DESCRIPTION
REVISIONS			
USDA Agricultural Research Service			
USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL			
BID OPTION #5 ELECTRICAL DEMOLITION PLAN			
PROJECT NO.	XXX	AGRICULTURAL RESEARCH SERVICE	SHEET
DATE	04/27/2026	FT. PIERCE, FL	ED103
		95 OF 108	

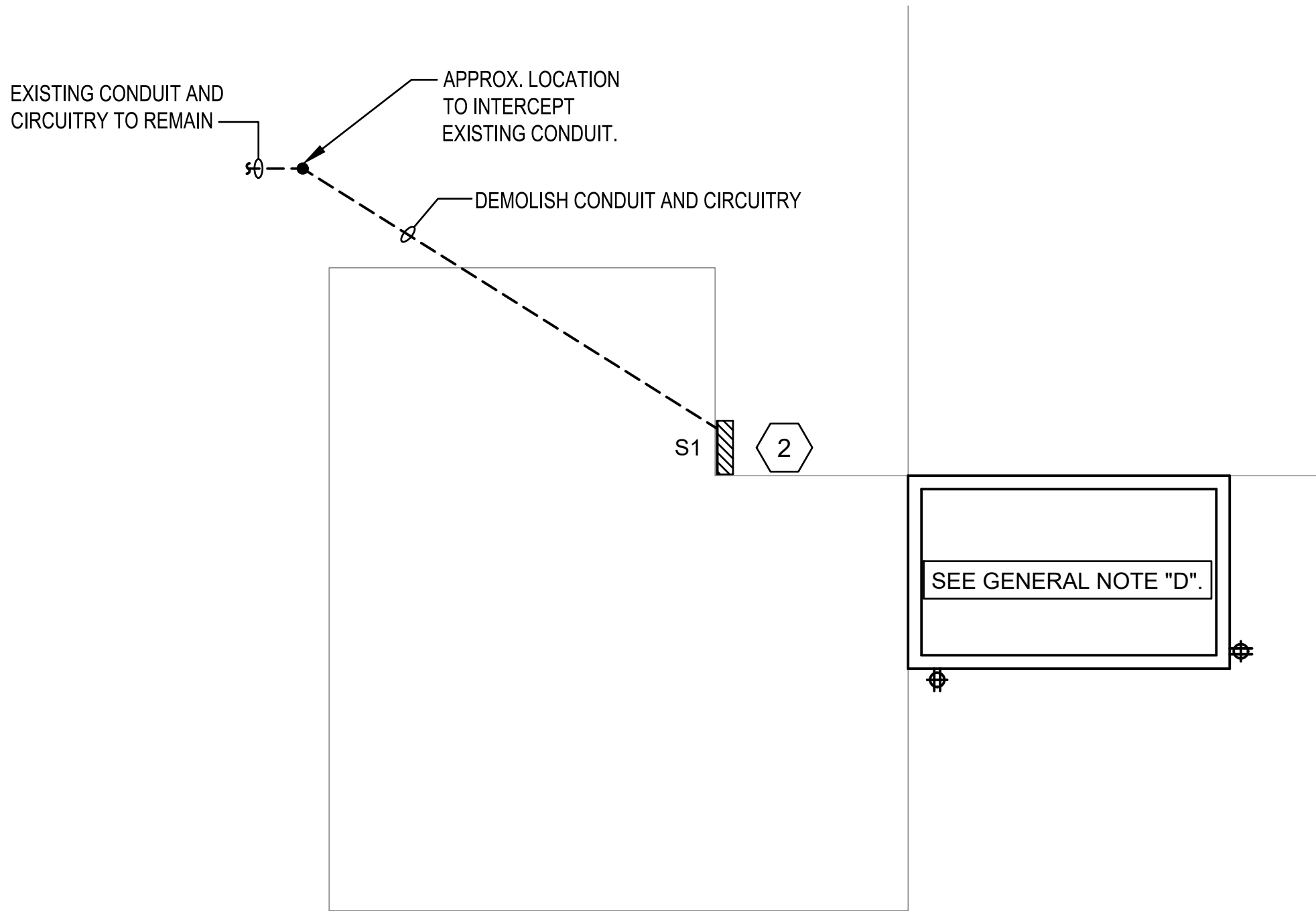


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 **1** **BID OPTION #5 - LEESBURG FARM ENLARGED GREENHOUSE DEMOLITION PLAN**
ED104 SCALE: 1/4" = 1'-0"



 **2** **BID OPTION #5 - LEESBURG FARM ENLARGED PUMP STORAGE DEMOLITION PLAN**
ED104 SCALE: 1/4" = 1'-0"



KEYED NOTES

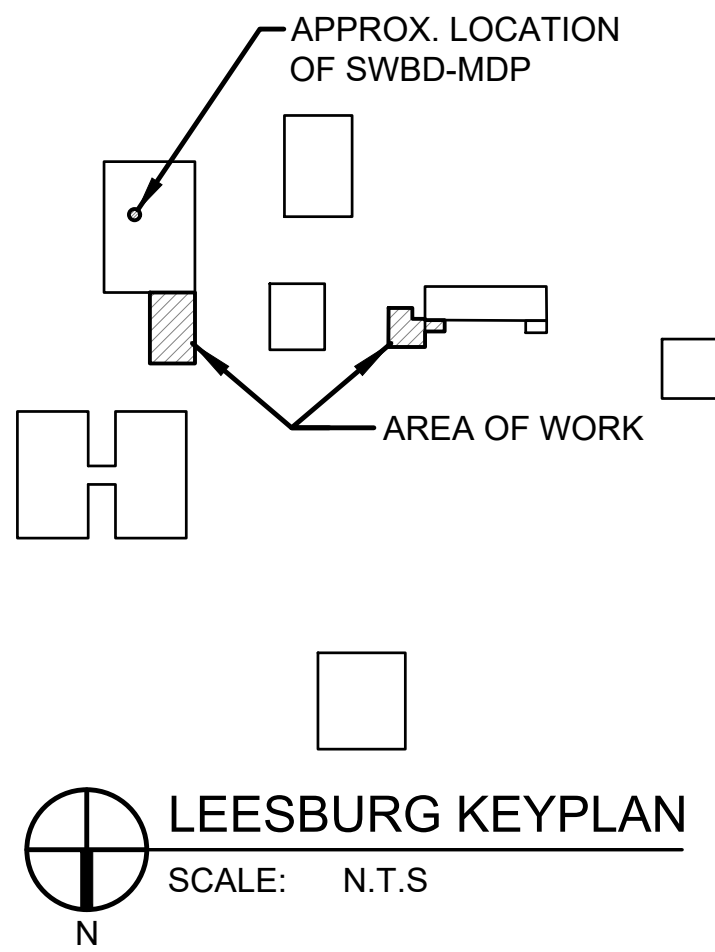
1. DEMOLISH EXISTING PANEL AND UNDERGROUND BRANCH CIRCUITS. BRANCH CIRCUIT CONDUITS SHALL BE CUT LEVEL WITH THE SLAB AND FILLED. THE EXISTING FEEDER CONDUIT AND CIRCUITRY SHALL REMAIN AND BE PROTECTED DURING DEMOLITION.
2. DEMOLISH EXISTING PANEL AND UNDERGROUND BRANCH CIRCUITS. BRANCH CIRCUIT CONDUITS SHALL BE CUT LEVEL WITH THE SLAB AND FILLED. THE EXISTING FEEDER CONDUIT SHALL BE INTERCEPTED OUTSIDE THE CONSTRUCTION ZONE AND REROUTED TO NEW PANEL LOCATION. SEE EP104.

GRAPHIC SCALE



GENERAL DEMOLITION NOTES

- A. PRIOR TO BEGINNING WORK, SURVEY EXISTING ELECTRICAL SYSTEMS. DOCUMENT, IN WRITING, SIGNED BY THE CONTRACTING OFFICER, ANY PORTIONS OF THE EXISTING SYSTEMS THAT ARE NOT OPERATING PROPERLY BEFORE CONSTRUCTION BEGINS. ANY ELECTRICAL SYSTEMS FOUND INOPERABLE AT THE END OF THE CONSTRUCTION PROCESS THAT HAVE NOT BEEN SO DOCUMENTED SHALL BE REPAIRED BY THE CONTRACTOR AT HIS OWN EXPENSE.
- B. PRIOR TO BEGINNING THE DEMOLITION PROCESS, CONDUCT A COMPREHENSIVE SITE SURVEY WITH THE ASSISTANCE OF THE CONTRACTING OFFICER. IDENTIFY AND MARK ALL EXISTING EQUIPMENT TO REMAIN AND ALL EXISTING RACEWAYS AND CIRCUITRY THAT ARE PRESENTLY PASSING THROUGH THE CONSTRUCTION AREA THAT ARE FEEDING EXISTING EQUIPMENT TO REMAIN.
- C. REMOVE ELECTRICAL EQUIPMENT IN AREAS BEING DEMOLISHED, AND ELECTRICAL EQUIPMENT FEEDING OTHER EQUIPMENT BEING DEMOLISHED. REMOVE RACEWAYS AND CIRCUITRY BACK TO THE PANEL OF ORIGINATION. WHERE CIRCUITS ARE NOT BEING COMPLETELY DEMOLISHED, REMOVE RACEWAYS AND CIRCUITRY BACK TO A JUNCTION BOX OR OTHER COMMON CONNECTION POINT AND RECIRCUIT EXISTING ELECTRICAL EQUIPMENT TO REMAIN AS REQUIRED. WHERE NECESSARY, COMPLETELY REFEED EXISTING ELECTRICAL EQUIPMENT THAT IS TO REMAIN IF REQUIRED. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL EXISTING EQUIPMENT TO REMAIN BE KEPT COMPLETELY OPERABLE DURING CONSTRUCTION AND LEFT COMPLETELY OPERABLE AT THE END OF THE CONSTRUCTION PROCESS.
- D. THE ELECTRICAL EQUIPMENT SHOWN IS PROVIDED TO CONVEY THE APPROXIMATE SCOPE OF THE WORK. IT IS THE INTENT OF THESE DOCUMENTS THAT ALL ELECTRICAL EQUIPMENT, RACEWAYS, AND CIRCUITRY IN THE AREA OF WORK BE DEMOLISHED UNLESS FEEDING EXISTING EQUIPMENT TO REMAIN.
- E. PERFORM DEMOLITION WORK AS FOLLOWS:
 1. DEMOLISH DEVICE OR END OF LINE EQUIPMENT.
 2. DEMOLISH RACEWAYS AND CIRCUITRY FEEDING END OF LINE EQUIPMENT (INCLUDING ALL RACEWAYS AND CIRCUITRY ASSOCIATED WITH CONTROLS).
 3. LEAVE ALL RACEWAYS AND CIRCUITRY THAT COMPLETELY PASS-THROUGH THE SPACE COMPLETELY OPERABLE BY REROUTING THE CIRCUITS AS REQUIRED.



THE JOHNSON & Mc ADAMS, INC.
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P.O. BOX 513, GREENWOOD, MISSISSIPPI 38902-0513

A-E FIRM
BP
DESIGNER
EH
CHECKED BY
BTD
JOB #
4825

USDA

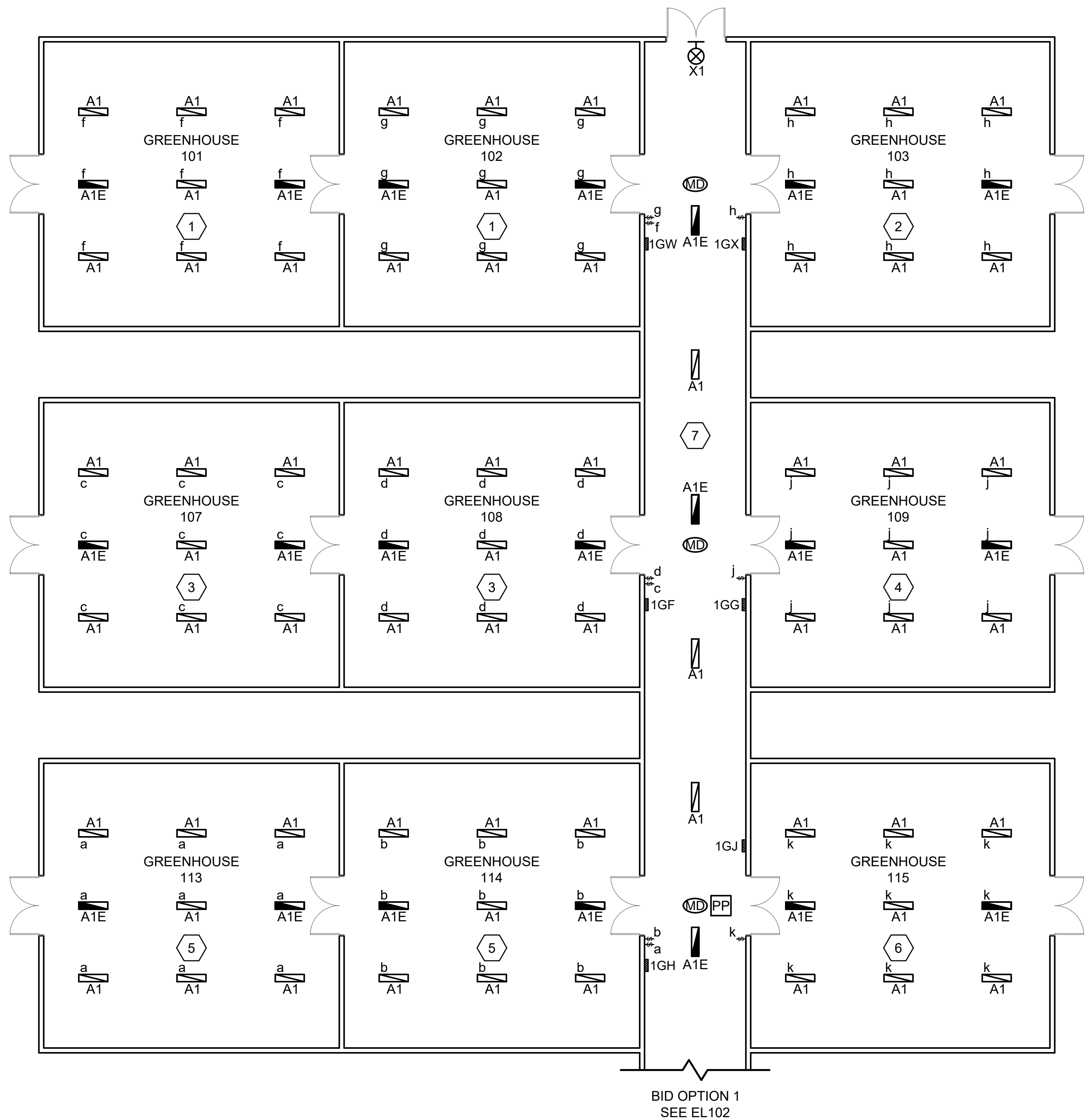
EPM
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PPM
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SAFETY & HEALTH
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REAL PROPERTY
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USDA

USDA Agricultural Research Service
USDA AGRICULTURAL RESEARCH SERVICE
US HORTICULTURAL RESEARCH LABORATORY
REPAIR TORNADO DAMAGE
FT. PIERCE, FL
BID OPTION #5 ELECTRICAL DEMOLITION PLAN
PROJECT NO. XXX
DATE 04/27/2026
AGRICULTURAL RESEARCH SERVICE
FT. PIERCE, FL
SHEET
ED104
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Apr 27, 2026 - 9:55am



1 BASE BID LIGHTING PLAN
EL101 SCALE: 3/32" = 1'-0"

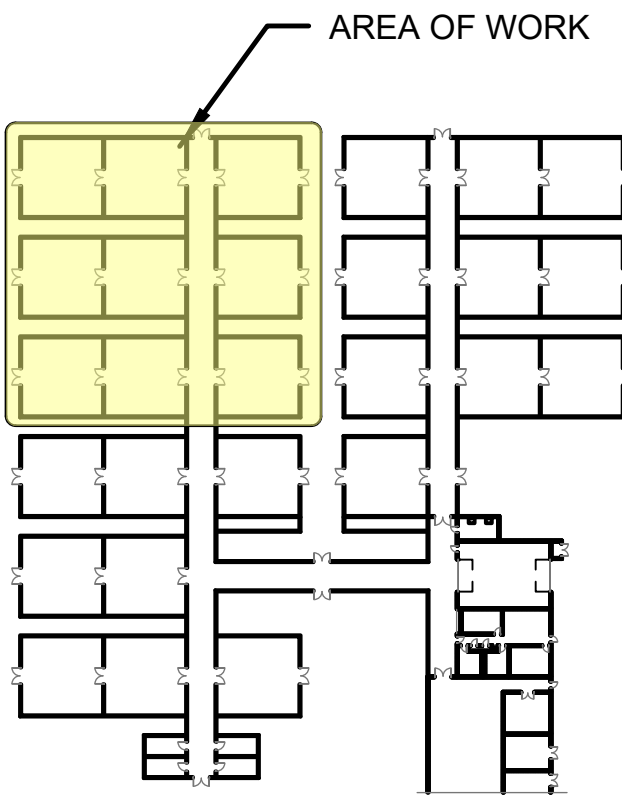
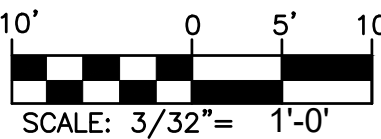
GENERAL NOTES

- A. ALL FIXTURES THROUGHOUT THE GREENHOUSE SHALL BE VAPOR-TIGHT GASKETED 4' LINEAR FIXTURES.
- B. ALL EMERGENCY FIXTURES SHALL BE PROVIDED WITH AN INTEGRAL 10W, 120V EMERGENCY BATTERY PACK.
- C. ALL FIXTURES LOCATED IN THE GREENHOUSE CORRIDORS SHALL BE AUTOMATICALLY CONTROLLED BY MOTION SENSORS. MOTION SENSORS SHALL AUTOMATICALLY TURN LIGHTS OFF AFTER 20 MINUTES OF VACANCY.
- D. ALL FIXTURES LOCATED IN THE GREENHOUSES SHALL BE MANUALLY CONTROLLED WITH WEATHERPROOF SWITCHES LOCATED OUTSIDE EACH GREENHOUSE.
- E. ALL GREENHOUSE LIGHT SWITCHES SHALL BE PROVIDED WITH WEATHERPROOF COVERS.

KEYED NOTES

- 1. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GW-1".
- 2. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GX-1".
- 3. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GF-1".
- 4. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GG-1".
- 5. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GH-1".
- 6. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GJ-1".
- 7. LIGHTING FIXTURES SHOWN IN THIS SECTION OF HALLWAY SHALL BE FED FROM CIRCUIT "1GH-53".

GRAPHIC SCALE



2 KEY PLAN
EL101 SCALE: N.T.S

Johnson & Adams, Inc.
ARCHITECTS/ENGINEERS • PLANNING • SURVEYING • MANAGEMENT
P.O. BOX 513, GREENWOOD, MISSISSIPPI 38943 • (662) 455-4943

A-E FIRM	USDA
PROJECT MANAGER	EPM
BP	XXX
DESIGNER	PPM
EH	XXX
CHECKED BY	SAFETY & HEALTH
BTD	XXX
JOB #	REAL PROPERTY
4825	XXX

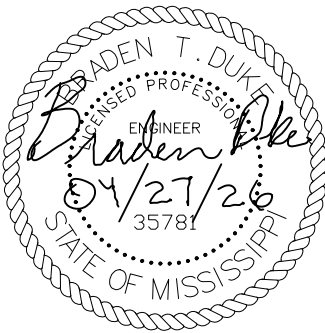
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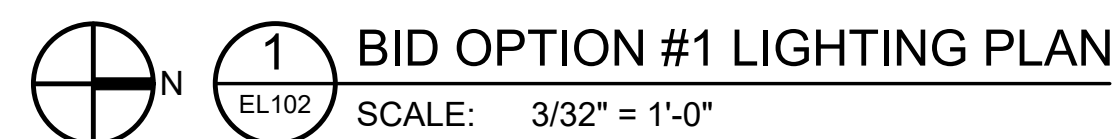
USDA Agricultural Research Service

USDA AGRICULTURAL RESEARCH SERVICE
US HORTICULTURAL RESEARCH LABORATORY
REPAIR TORNADO DAMAGE
FT. PIERCE, FL

BASE BID LIGHTING PLAN

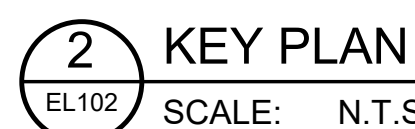
PROJECT NO.	AGRICULTURAL RESEARCH SERVICE	SHEET
DATE	FT. PIERCE, FL	EL101
04/27/2026		97 OF 108





- A. ALL FIXTURES THROUGHOUT THE GREENHOUSE SHALL BE VAPOR-TIGHT GASKETED 4" LINEAR FIXTURES.
- B. ALL EMERGENCY FIXTURES SHALL BE PROVIDED WITH AN INTEGRAL 10W, 120V EMERGENCY BATTERY PACK.
- C. ALL FIXTURES LOCATED IN THE GREENHOUSE CORRIDORS SHALL BE AUTOMATICALLY CONTROLLED BY MOTION SENSORS. MOTION SENSORS SHALL AUTOMATICALLY TURN LIGHTS OFF AFTER 20 MINUTES OF VACANCY.
- D. ALL FIXTURES LOCATED IN THE GREENHOUSES SHALL BE MANUALLY CONTROLLED WITH WEATHERPROOF SWITCHES LOCATED OUTSIDE EACH GREENHOUSE.
- E. ALL GREENHOUSE LIGHT SWITCHES SHALL BE PROVIDED WITH WEATHERPROOF COVERS.

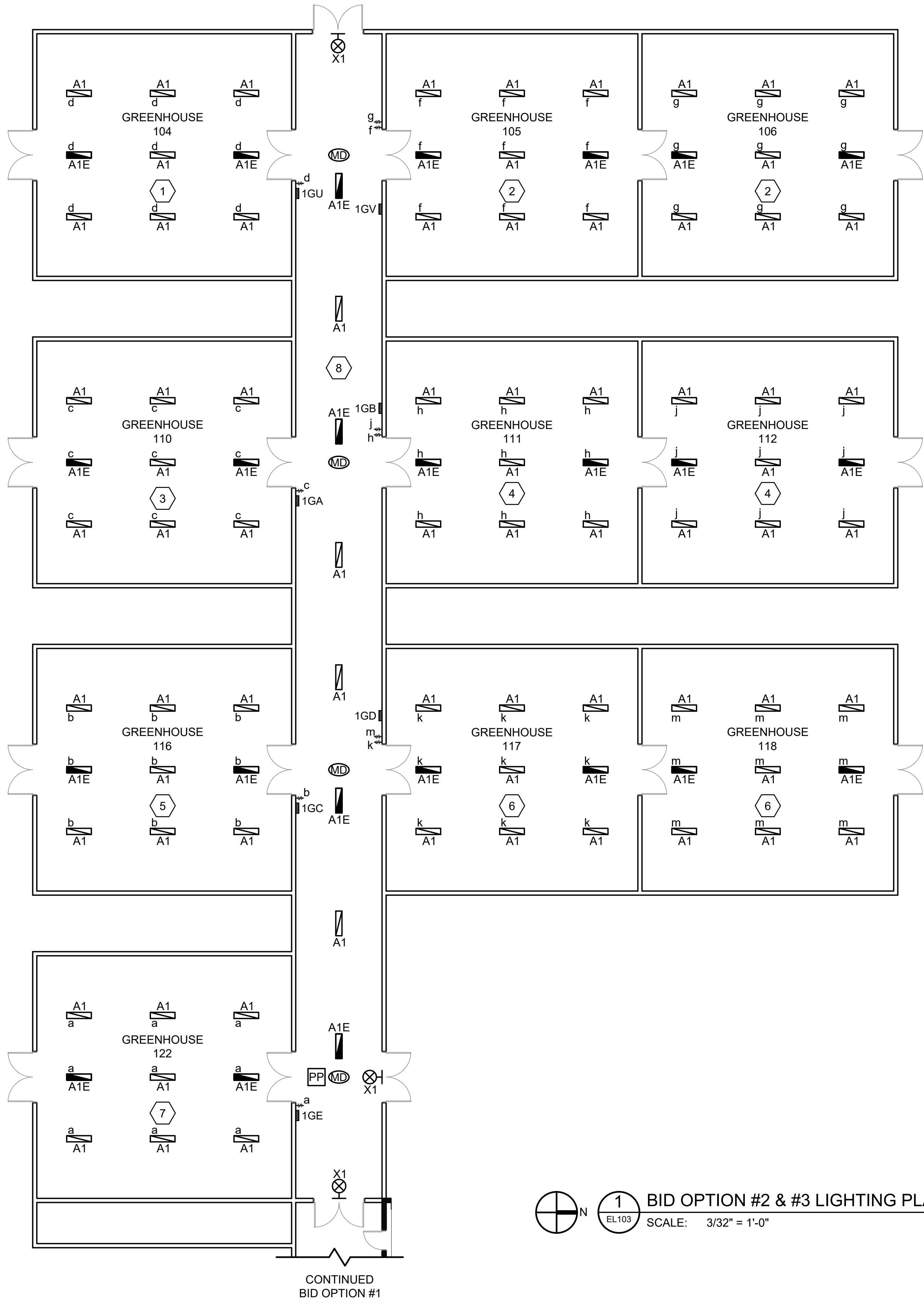
1. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GK-1".
2. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GL-1".
3. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GM-1".
4. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GN-1".
5. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GP-1".
6. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GQ-1".
7. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GS-1".
8. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GR-1".
9. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GT-1".
10. LIGHTING FIXTURES SHOWN IN THIS SECTION OF HALLWAY SHALL BE FED FROM CIRCUIT "1GM-53".



U S D A	 Agricultural Research Service	
	USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL	
	BID OPTION #1 LIGHTING PLAN	
PROJECT NO. XXX	AGRICULTURAL RESEARCH SERVICE	SHEET EL102
DATE 04/27/2026	FT. PIERCE, FL	98 OF 108



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1 BID OPTION #2 & #3 LIGHTING PLAN
SCALE: 3/32" = 1'-0"

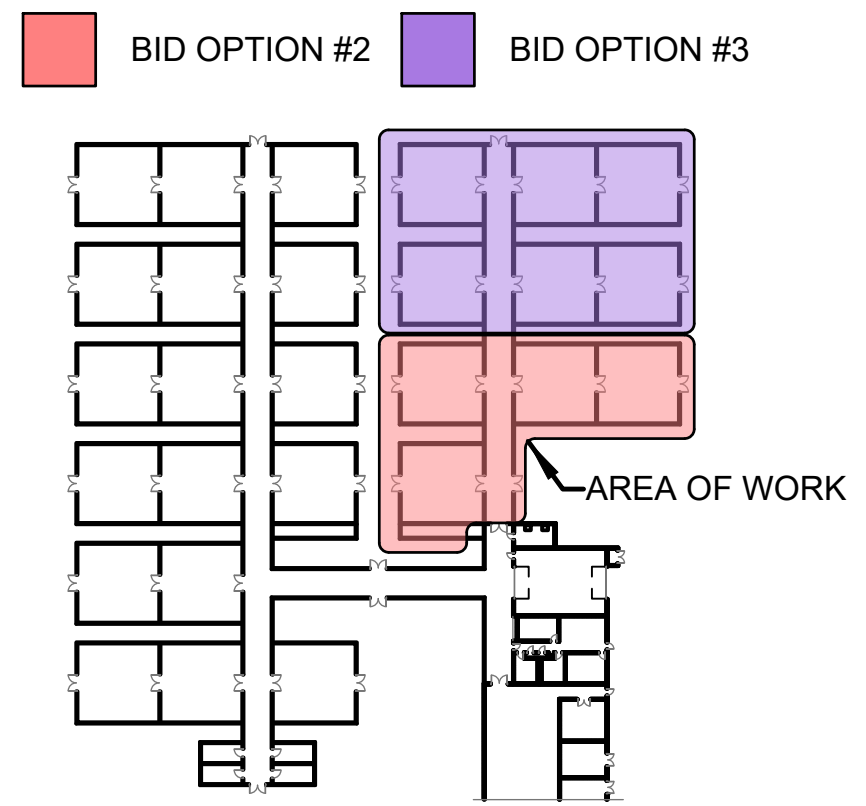
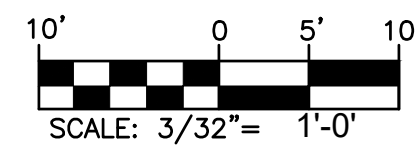
GENERAL NOTES

- A. ALL FIXTURES THROUGHOUT THE GREENHOUSE SHALL BE VAPOR-TIGHT GASKETED 4' LINEAR FIXTURES.
- B. ALL EMERGENCY FIXTURES SHALL BE PROVIDED WITH AN INTEGRAL 10W, 120V EMERGENCY BATTERY PACK.
- C. ALL FIXTURES LOCATED IN THE GREENHOUSE CORRIDORS SHALL BE AUTOMATICALLY CONTROLLED BY MOTION SENSORS. MOTION SENSORS SHALL AUTOMATICALLY TURN LIGHTS OFF AFTER 20 MINUTES OF VACANCY.
- D. ALL FIXTURES LOCATED IN THE GREENHOUSES SHALL BE MANUALLY CONTROLLED WITH WEATHERPROOF SWITCHES LOCATED OUTSIDE EACH GREENHOUSE.
- E. ALL GREENHOUSE LIGHT SWITCHES SHALL BE PROVIDED WITH WEATHERPROOF COVERS.

KEYED NOTES

1. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GU-1".
2. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GV-1".
3. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GA-1".
4. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GB-1".
5. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GC-1".
6. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GD-1".
7. LIGHTING FIXTURES IN THIS ROOM SHALL BE FED FROM CIRCUIT "1GE-1".
8. LIGHTING FIXTURES SHOWN IN THIS SECTION OF HALLWAY SHALL BE FED FROM CIRCUIT "1GE-41".

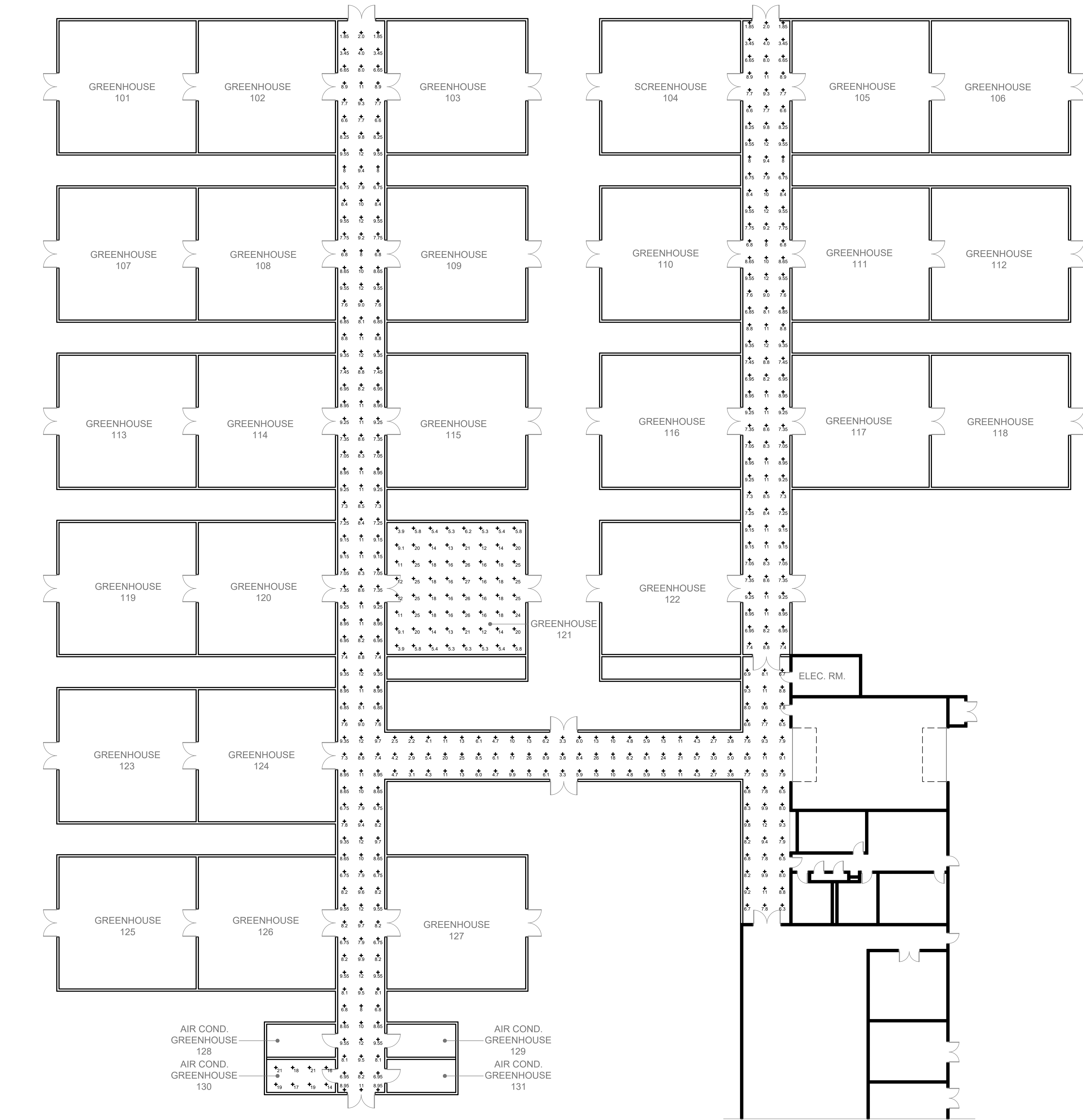
GRAPHIC SCALE

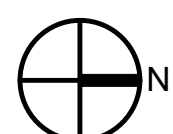


2 KEY PLAN
SCALE: N.T.S.

 JJA THE JOHNSON & ADAMS FIRM ARCHITECTS/ENGINEERS • PLANNING • INTERIORS • MANAGEMENT P.O. BOX 513, GREENWOOD, MISSISSIPPI 38902-0513 • (662) 455-4943		NO.	DATE	BY	DESCRIPTION
REVISIONS					
USDA	A-E FIRM	USDA			
	PROJECT MANAGER	EPM			
	DESIGNER	PPM			
	CHECKED BY	SAFETY & HEALTH			
USDA	BP	XXX			
	EH	XXX			
	BTD	XXX			
	JOB #	REAL PROPERTY			
		PROJECT NO.	AGRICULTURAL RESEARCH SERVICE		SHEET
		DATE	FT. PIERCE, FL		EL103
		04/27/2026			99 OF 108

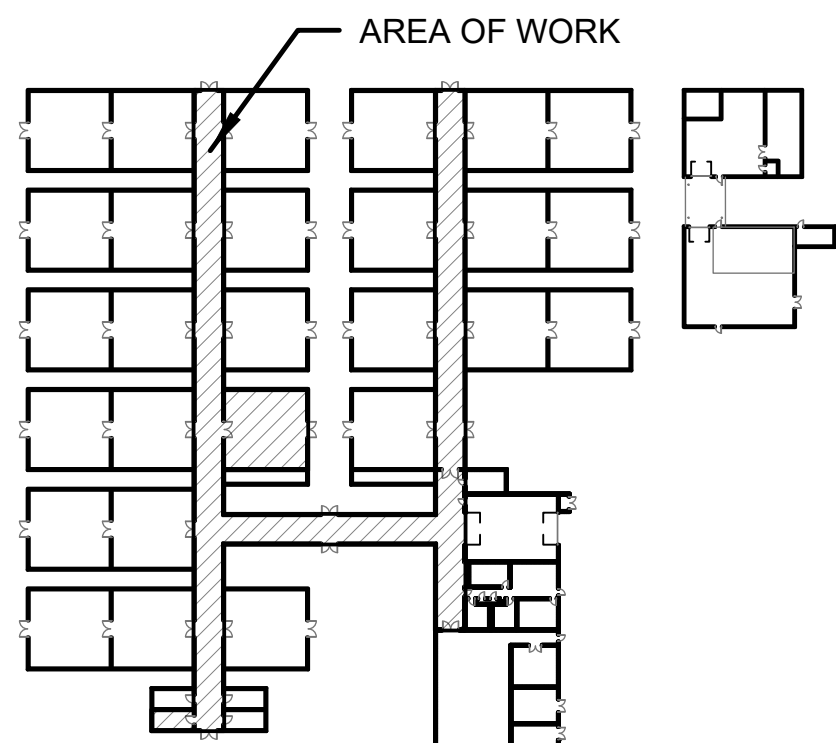
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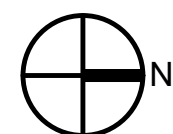


 **1** NORMAL LIGHTING PHOTOMETRIC PLAN (FOR REFERENCE ONLY)
SCALE: 1/16" = 1'-0"

PHOTOMETRIC CALCULATIONS

ROOM NUMBER	ROOM NAME	AVERAGE	MAXIMUM	MINIMUM	AVG/MIN	MAX/MIN
121	GREENHOUSE BAY	14	27	4	3.7	6.8
X	HALLWAY	9	26	2	4.5	13
130	SMALL GH BAY	18	21	14	1.3	1.5

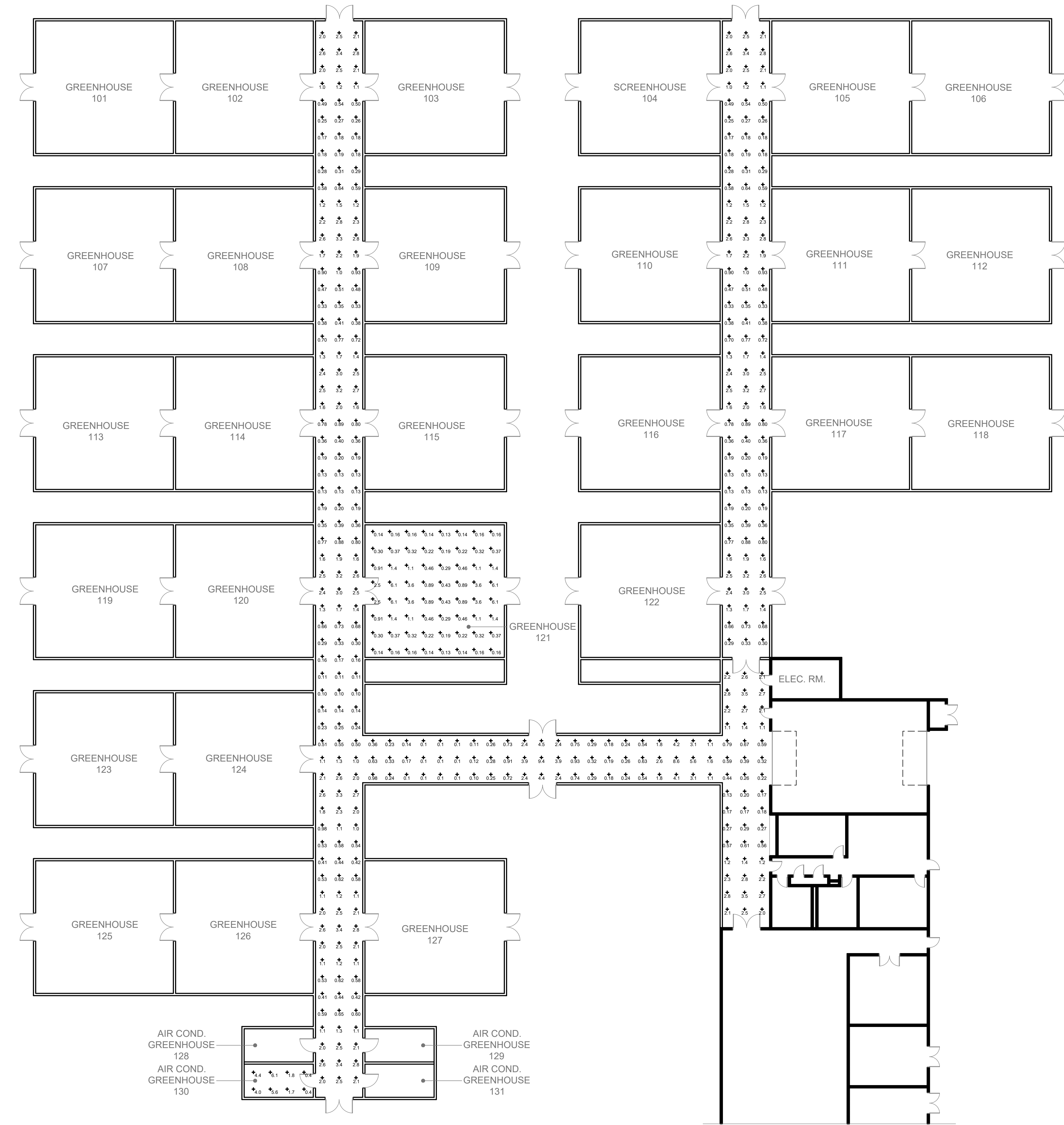


 **2** KEY PLAN
SCALE: N.T.S.

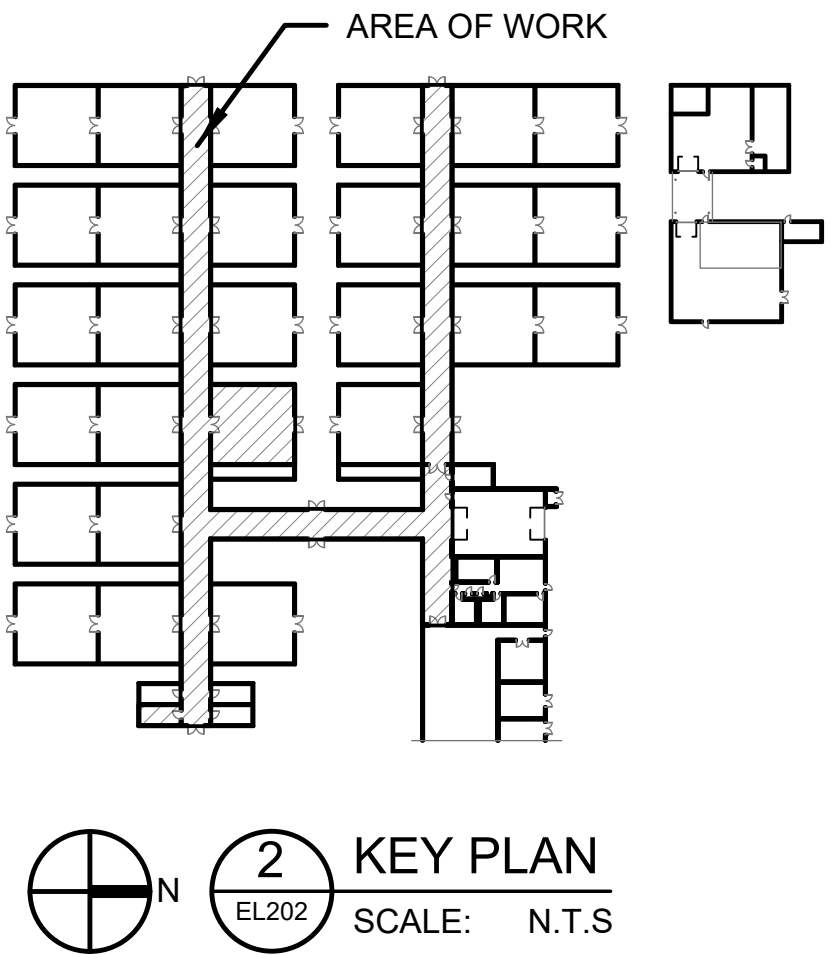
 THE JOHNSON & MCADAMS FIRM, INC. ARCHITECTS/ENGINEERS • PLANNING • SURVEYING • MANAGEMENT P.O. BOX 513, GREENWOOD, MISSISSIPPI 38902-0513 • (662) 455-4943		NO.		DATE	BY	DESCRIPTION
A-E FIRM		USDA				
PROJECT MANAGER	EPM					
BP	XXX					
DESIGNER	PPM					
EH	XXX					
CHECKED BY	SAFETY & HEALTH					
BTD	XXX					
JOB #	REAL PROPERTY					
4825	XXX					
		PROJECT NO.	AGRICULTURAL RESEARCH SERVICE		SHEET	
		DATE	FT. PIERCE, FL		EL201	
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PHOTOMETRIC CALCULATIONS						
ROOM NUMBER	ROOM NAME	AVERAGE	MAXIMUM	MINIMUM	AVG/MIN	MAX/MIN
121	GREENHOUSE BAY	1	6	0.14	8.3	40
X	HALLWAY	1	9	0.10	47	350
130	SMALL GH BAY	3	6	0.4	6.8	13.5



1
EL202
EMERGENCY LIGHTING PHOTOMETRIC PLAN (FOR REFERENCE ONLY)
SCALE: 1/16" = 1'-0"



		NO.		DATE	BY	DESCRIPTION
A-E FIRM		USDA				
PROJECT MANAGER	EPM					
BP	XXX					
DESIGNER	PPM					
EH	XXX					
CHECKED BY	SAFETY & HEALTH					
BTD	XXX					
JOB #	REAL PROPERTY					
4825	XXX					
		PROJECT NO.	AGRICULTURAL RESEARCH SERVICE		SHEET	
		DATE	FT. PIERCE, FL		EL202	
		04/27/2026			101 OF 108	



LUMINAIRE REQUIREMENTS:

1. HOUSING - ONE-PIECE 5VA FIBERGLASS HOUSING WITH INTERGRAL PERIMETER CHANNEL UTILIZING CONTINUOUS POURED-IN-PLACE NEMA 4X GASKET. APPROVED FOR THROUGH WIRING. CAPTIVE POLYMERIC LATCHES. FROSTED, LOW PROFILE, POLYCARBONATE LENSE.
2. LIGHT SOURCE - 4000K COLOR TEMPERATURE, MINIMUM EFFICACY OF 100 LUMENS/WATT, WITH A MINIMUM CRI OF 80.

TYPE "A1 & A1E" FIXTURE WITH A 4,000 LUMEN PACKAGE.
3. DRIVER - OPERATING VOLTAGE OF 120-277V. DIMMABLE TO 10%. (0-10 VOLT DIMMING)
4. CERTIFICATION - CSA CERTIFIED TO UL STANDARDS. SUITABLE FOR WET LOCATION. IP65, IP66, IP67 RATED. NSF SPLASH ZONE 2 AND NON-FOOD RATED. NEMA 4X RATED.
5. MOUNTING - SURFACE MOUNT.
6. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.
7. EMERGENCY PACK - FIXTURES "A1E" SHALL HAVE AN EMERGENCY SELF-DIAGNOSTIC BATTERY PACK, 10W CONSTANT POWER, CERTIFIED IN CA TITLE 20.

4' LED GASKETED STRIP FIXTURE

LEGEND:   FIXTURE TYPE: A1, A1E



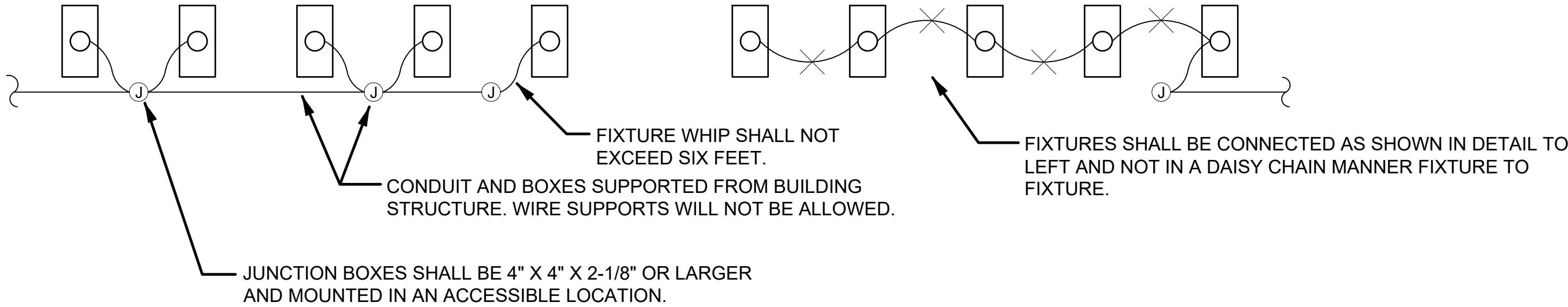
LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-CAST ALUMINUM CONSTRUCTION - COMPACT HOUSING. FULLY OVERLAPPING LIGHT SEAL TO PREVENT LIGHT LEAKS. UNIVERSAL DIRECTIONAL CHEVRON KNOCKOUTS.

X1 - SINGLE SIDED CHEVRON SIGN.
2. LIGHT SOURCE - LED'S MOUNTED ON PRINTED CIRCUIT BOARD. 10 YEARS WARRANTY.
3. LETTERS/CHEVRONS - MINIMUM 6" HIGH WITH 3/4" STROKE. RED LETTERS AS INDICATED. PROVIDE CHEVRONS AS INDICATED EITHER LEFT, RIGHT OR BOTH DIRECTIONS AS INDICATED. CHEVRONS PUNCHED OUT THROUGH HOUSING AS REQUIRED.
4. EMERGENCY PACK - SOLID-STATE, CONSTANT-CURRENT TYPE BATTERY CHARGER WITH MAINTENANCE-FREE, NICKEL-CADMIUM BATTERY, AC-ON INDICATOR LAMP AND TEST SWITCH.
5. MOUNTING - UNIVERSAL MOUNTING KIT FOR CEILING, WALL OR END-OF-FIXTURE MOUNTING.
6. ILLUMINATION - PROVIDED BY RED HIGH-OUTPUT LEDS INSIDE OF FIXTURE HOUSING, PROVIDE POLYSTYRENE DIFFUSER IN COLOR INDICATED WITH FREQUENCY-MATCHED SILKSCREEN COATING FOR MAXIMUM LED LIGHT OUTPUT.
7. VOLTAGE - OPERATING VOLTAGE OF 120-277V.
8. NOM. DIMENSIONS: (L=11.75" X W=8.25" X D=2").

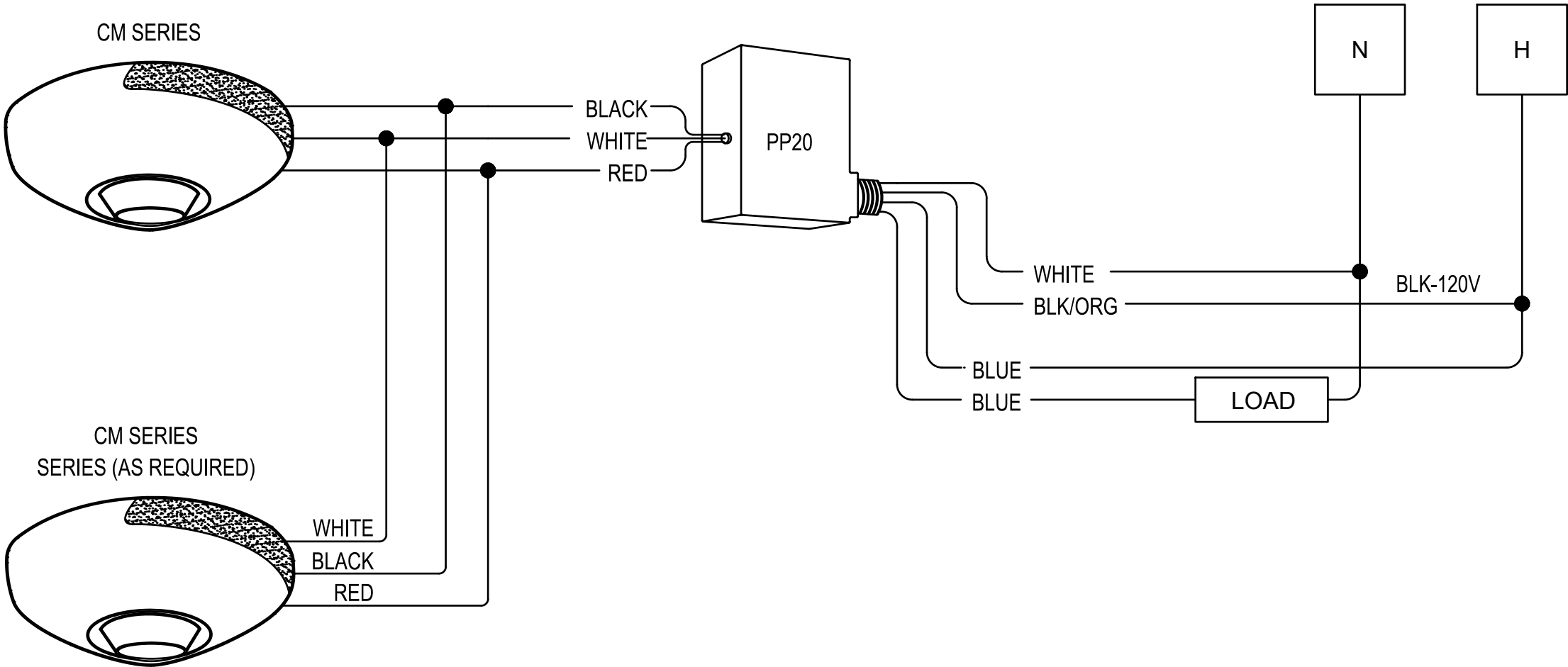
LEGEND:  FIXTURE TYPE: X1

NOTE: SUPPORT THE LIGHT FIXTURES AT ALL FOUR CORNERS OF THE FIXTURE.



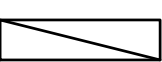
1 FIXTURE CONNECTION DETAIL
SCALE: NONE

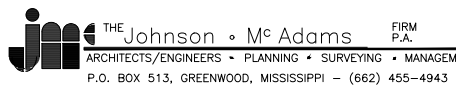
NOTE: ALL CEILING MOUNTED OCCUPANCY SENSORS LOCATED IN THE GREENHOUSE CORRIDORS SHALL BE WET/DAMP LOCATION RATED



2 TYPICAL CEILING MOUNTED OCCUPANCY SENSOR ON/OFF
SCALE: NONE

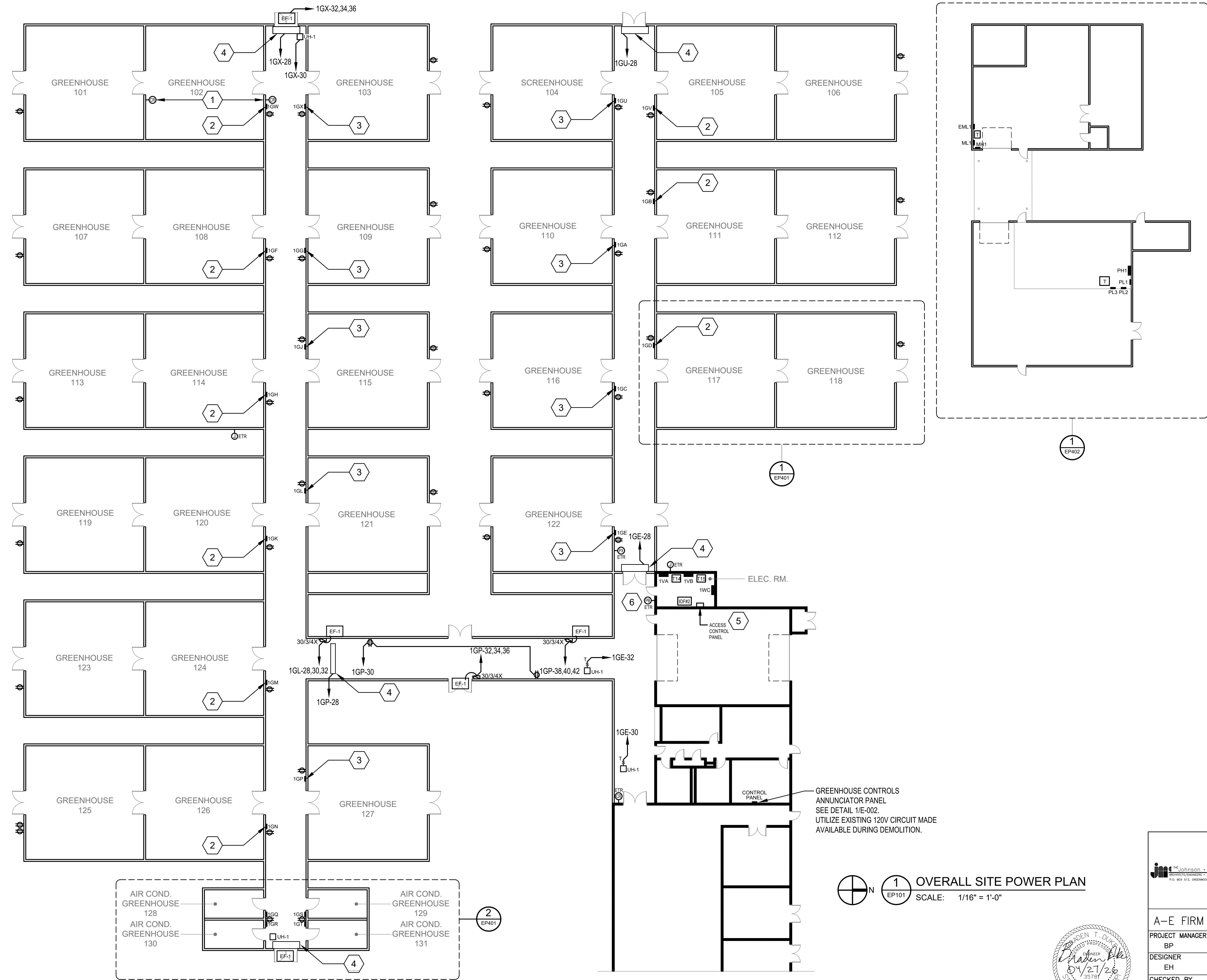
LUMINAIRE SCHEDULE

FIXTURE NAME	SYMBOL	MODEL	DESCRIPTION	VOLTS	INPUT VA	TOTAL LUMENS	NOTE 1
A1		4' LINEAR LED STRIP	4000 K, MEDIUM DISTRIBUTION, WHITE, SURFACE	120V 1P 2W	28.2	4000	4' VAPOR TIGHT FIXTURE.
A1E		4' LINEAR LED STRIP EMERGENCY	4000 K, MEDIUM DISTRIBUTION, WHITE, SURFACE	120V 1P 2W	28.2	4000	4' VAPOR TIGHT FIXTURE. PROVIDE WITH INTEGRAL EMERGENCY BATTERY.
X1E		EXIT SIGN	WET LOCATION RATED EXIT SIGN	120V 1P 2W			SELF DIAGNOSTICS AND INTEGRAL EMERGENCY BATTERY.

		NO.	DATE	BY	DESCRIPTION
		REVISIONS			
A-E FIRM		USDA			
PROJECT MANAGER	BP	EPM	XXX	 USDA Agricultural Research Service	
DESIGNER	EH	PPM	XXX		
CHECKED BY	BTD	SAFETY & HEALTH	XXX	USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL	
JOB #	4825	REAL PROPERTY	XXX		
		PROJECT NO.	XXX	AGRICULTURAL RESEARCH SERVICE	
		DATE	04/27/2026		
		FT. PIERCE, FL			SHEET EL501 102 OF 108



\\jagnework\2025_jah\4825 USDA Tornado Damage Repair - Ft. Pierce FL\07 Working Folder\Elec\CAD\EP101.dwg
4/27/2026 9:55am



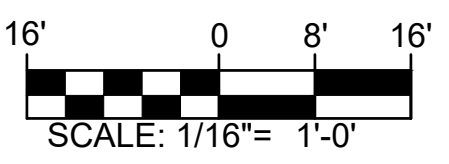
GENERAL NOTES

- A. ALL CONDUIT ASSOCIATED WITH THE GREENHOUSES SHALL BE UV RESISTANT, SCHEDULE 80 PVC.
- B. ALL CIRCUITRY SHALL BE XHHW-2.
- C. ALL JUNCTION BOXES ASSOCIATED WITH THE GREENHOUSES SHALL BE WATER-TIGHT POLYCARBONATE BOXES WITH WEATHERPROOF GASKETS AND FITTINGS.
- D. ALL RECEPTACLES SHALL HAVE A WEATHERPROOF COVER AND BE GFI TYPE OR FED WITH A GFI BREAKER.
- E. REFER TO ARCHITECTURAL PLANS FOR DEMOLITION AND NEW CONSTRUCTION PHASING PLAN. ALL EQUIPMENT NOT UNDER CONSTRUCTION SHALL REMAIN IN OPERATION THROUGHOUT THE ENTIRETY OF THE PROJECT.
- F. ALL BREAKERS FOR EQUIPMENT SHALL BE LOCKABLE IN THE OPEN POSITION.

KEYED NOTES

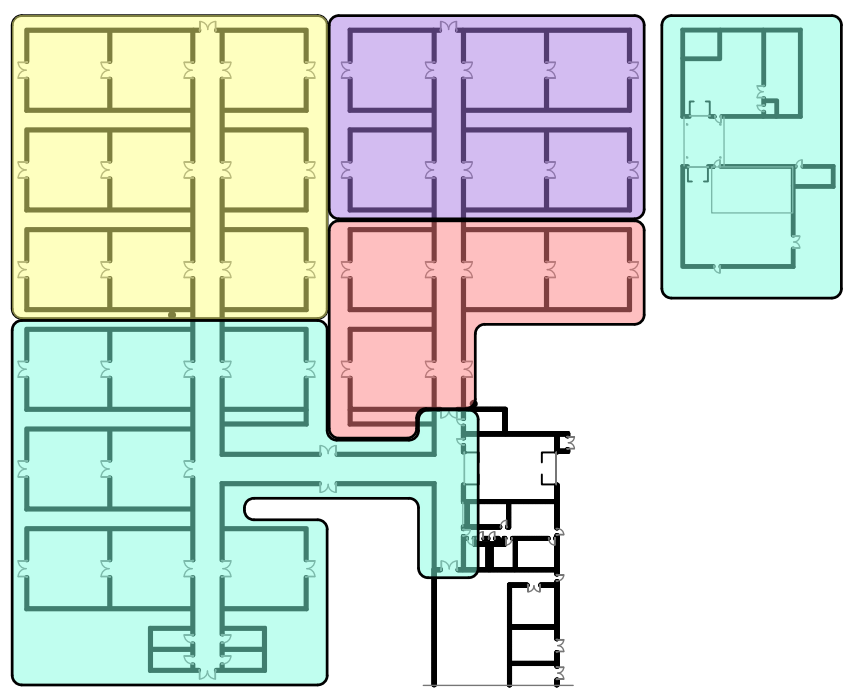
- 1. SEE ACCESS CONTROL DETAIL ON SHEET E-002 FOR REQUIRED DEVICES AND MORE INFORMATION.
- 2. SEE TYPICAL DOUBLE GREENHOUSE PANEL AND MICROGROW CONTROLLER LAYOUT DETAIL 4/E-002 FOR MORE INFORMATION.
- 3. SEE TYPICAL SINGLE GREENHOUSE PANEL AND MICROGROW CONTROLLER LAYOUT DETAIL 5/E-002 FOR MORE INFORMATION.
- 4. DAC-1: AIR CURTAIN, 1500W, 120V, 1Ø, PROVIDE 3Ø/1 BREAKER IN PANEL AS INDICATED.
- 5. ACCESS CONTROL PANEL. UTILIZE SPARE 2Ø/1 BREAKER IN PANEL 1GE FOR 120V POWER.
- 6. RECONNECT DOOR CONTROL PUSH BUTTON CIRCUITRY AFTER NEW DOOR INSTALLATION.

GRAPHIC SCALE



BID OPTION LEGEND

- BASE BID
- BID OPTION #1
- BID OPTION #2
- BID OPTION #3



2 KEY PLAN
EP101 SCALE: N.T.S

1 OVERALL SITE POWER PLAN
EP101 SCALE: 1/16" = 1'-0"

NO.		DATE	BY	DESCRIPTION
REVISIONS				
A-E FIRM		USDA		
PROJECT MANAGER	BP	EPM	XXX	
DESIGNER	EH	PPM	XXX	
CHECKED BY	BTD	SAFETY & HEALTH	XXX	
JOB #	4825	REAL PROPERTY	XXX	
PROJECT NO.	XXX	AGRICULTURAL RESEARCH SERVICE	SHEET	
DATE	04/27/2026	FT. PIERCE, FL	EP101	



PANEL L1			LOCATION: EXTERIOR BUS: 240/120V, 1Ø, 3W 100A		LUG LOCATION: BOTTOM FEED MAIN BUS: 70A MAIN BREAKER MOUNTING: SURFACE						NEMA 3R ENCLOSURE PANELBOARD AIC RATING (A): 10,000	
CIRCUIT NO.	BREAKER		DESCRIPTION	PHASE LOAD (KVA)				DESCRIPTION	BREAKER		CIRCUIT NO.	
	AMPS	POLES		L1	L2				AMPS	POLES		
1	20	1	LIGHTING	0.3	0.3			EF-2 (1/2 HP)	15	2	2	
3	20	1	GENERAL USE RECS.	-	0.3	0.4	0.3	-	-	-	4	
7	30	1	SPECIAL USE RECS.	1.8	0.3			EF-2 (1/2 HP)	15	2	6	
9	30	1	SPECIAL USE RECS.	-		1.8	0.3	-	-	-	8	
11	20	1	SPARE	0.0	0.5			SPARE	20	1	10	
13	20	1	SPARE	0.0		0.0	0.5	SPARE	15	1	12	
15	20	1	SPARE	0.0	0.0			SPARE	20	1	14	
17	20	1	SPARE	0.0		0.0	0.0	SPARE	20	1	16	
19	20	1	SPARE	0.0	0.0		0.0	SPARE	20	1	18	
21	20	1	SPARE	-		0.0	0.0	SPARE	20	1	20	
23	20	1	SPARE	0.0	0.0			SPARE	20	1	22	
25	20	1	SPARE			0.0	0.0	SPARE	20	1	24	
TOTAL				3.1		3.2		* GFCI BREAKER				

- A. ALL CONDUIT MOUNTED OVERHEAD SHALL BE UV RESISTANT, SCHEDULE 80 PVC CONDUIT.
- B. ALL BRANCH CIRCUITS SHALL BE XHHW-2 CONDUCTORS.
- C. ALL RECEPTACLES SHALL BE MOUNTED TO UNISTRUT IN THE HOOP STYLE GREENHOUSE.
- D. ALL JUNCTION BOXES ASSOCIATED WITH THE GREENHOUSES SHALL BE WATER-TIGHT POLYCARBONATE BOXES WITH WEATHERPROOF GASKETS AND FITTINGS.
- E. ALL RECEPTACLES SHALL HAVE A WEATHERPROOF COVER AND BE GFI TYPE OR FED WITH A GFI BREAKER.
- F. THE CONTRACTOR SHALL REPAIR THE EXISTING SAGGING CONDUIT AT THE PUMP FILTRATION STATION. SEE ARCHITECTURAL PLANS FOR LOCATION.
- G. RECEPTACLES ARE TO BE MOUNTED ABOVE TABLES, APPROX. 48" A.F.F.

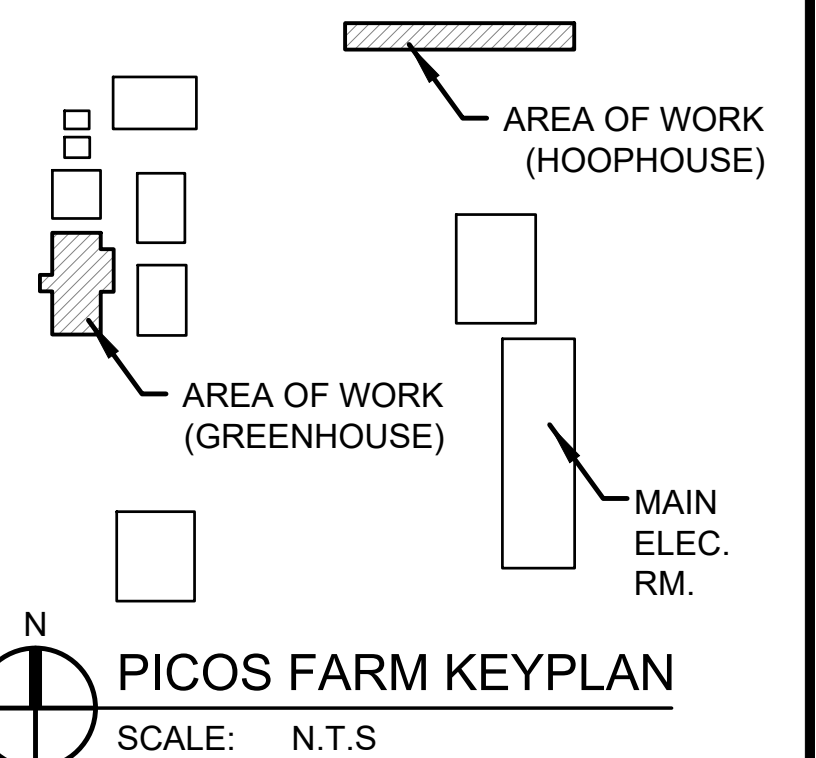
1. THE CONTRACTOR SHALL PROVIDE A FUSED DISCONNECT (60/23R, F-40), 15kVA NEMA 3R DRY TYPE TRANSFORMER (480V/240/120V) MOUNTED ON A CONCRETE PAD, AND PANEL 'L1'. MOUNT EQUIPMENT ON GALVANIZED STRUT RACK. SEE PANEL SCHEDULE FOR MORE INFORMATION.
2. NEMA 5-20R RECEPTACLE MOUNTED IN WEATHERPROOF BOX W/ A WEATHERPROOF WHILE-IN-USE COVERPLATE.
3. NEMA 5-30R RECEPTACLE MOUNTED IN WEATHERPROOF CAST ALUMINUM BOX W/ A WEATHERPROOF WHILE-IN-USE COVERPLATE.
4. THE CONTRACTOR SHALL PROVIDE (2) 3/4" x 10" COPPER CLAD DRIVEN GROUND RODS DRIVEN. BOND TO GROUND BAR OF PANEL WITH #4 CU.
5. DAC-1: AIR CURTAIN, 1500W, 120V, 1Ø, PROVIDE 30/1 BREAKER IN PANEL AS INDICATED.
6. LIGHTING FIXTURES WITHIN THIS ROOM SHALL BE SERVED FROM "1GY-1".
7. LIGHTING FIXTURES WITHIN THIS ROOM SHALL BE SERVED FROM "L1-1".

4' 0 2' 4'

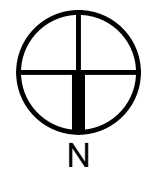
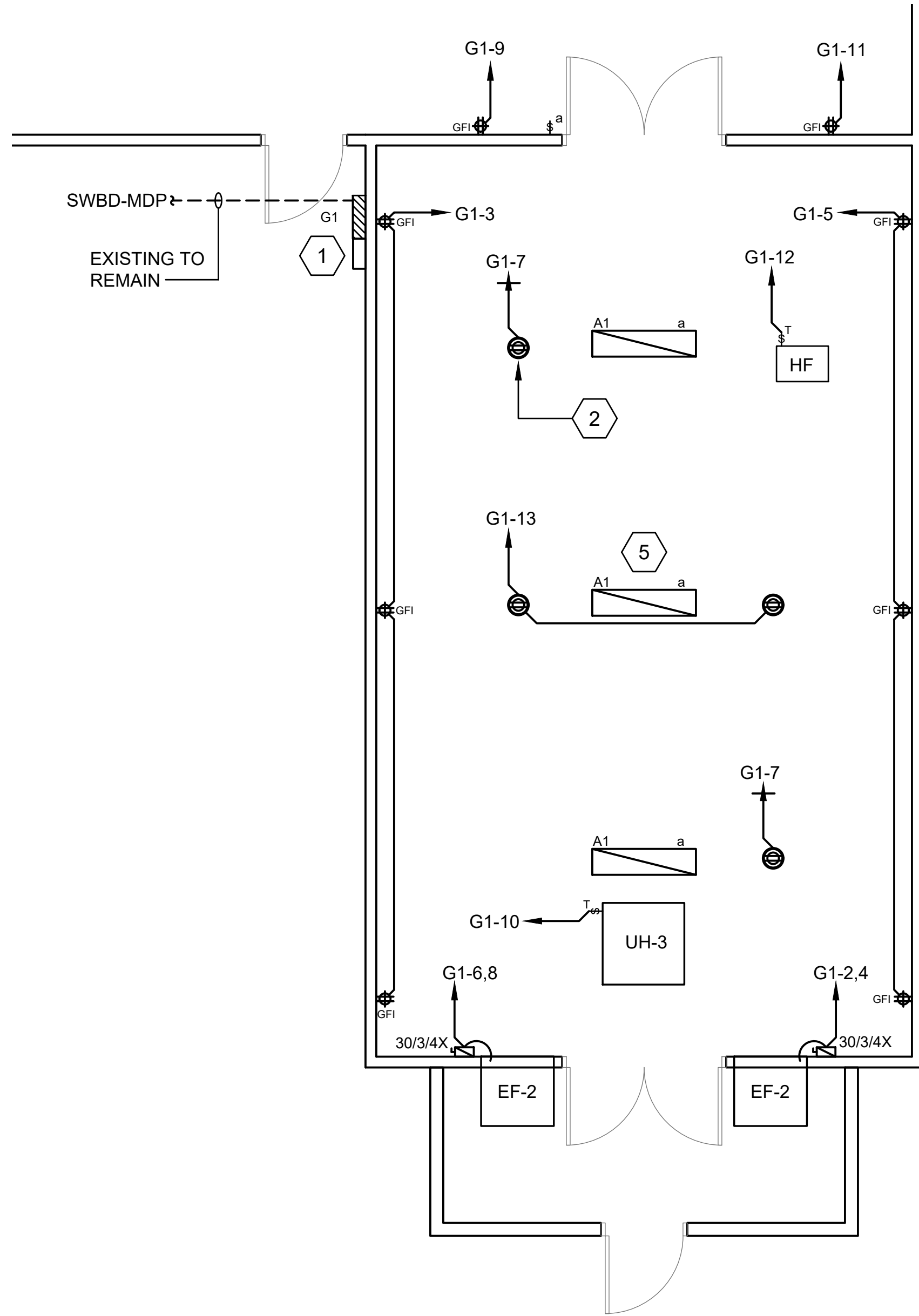
SCALE: $\frac{1}{4}" = 1'-0"$

6' 0 3' 6'

SCALE: $\frac{3}{16}" = 1'-0"$



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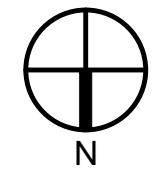
1
EP103

BID OPTION #5 - LEESBURG FARM ENLARGED GREENHOUSE ELECTRICAL PLAN

SCALE: 1/4" = 1'-0"

PANEL		LOCATION:		EXTERIOR		LUG LOCATION:		BOTTOM FEED						NEMA 3R ENCLOSURE	
G1		VOLT: BUS:		208/120V, 1Ø, 3W 100A		MAIN BUS: MOUNTING:		100A MAIN BREAKER SURFACE						PANELBOARD AIC RATING (A): 10,000	
CIRCUIT NO.	BREAKER		DESCRIPTION	PHASE LOAD (KVA)				DESCRIPTION	BREAKER		CIRCUIT NO.				
	AMPS	POLES		L1	L2				AMPS	POLES					
1	20	1	LIGHTING	0.1	0.3			EF-2 (1/2 HP)	15	2	2				
3	20	1	GENERAL USE RECS.		0.5	0.3		-			4				
5	20	1	GENERAL USE RECS.	0.5	0.3			EF-2 (1/2 HP)	15	2	6				
7	20	1	OVERHEAD RECS.		0.7	0.3		-			8				
9	20	1	ECP SUMP PUMP REC.	0.5	0.5			UH-3	20	1	10				
11	20	1	ECP SUMP PUMP REC.		0.5	0.5		HF FAN	15	1	12				
13	20	1	OVERHEAD RECS.	0.7	0.0			SPARE	15	2	14				
15	20	1	SPARE		0.0	0.0		-	-	-	16				
17	20	1	SPARE	0.0	0.0			SPARE	20	1	18				
19	20	1	SPARE		0.0	0.0		SPARE	20	1	20				
21	20	1	SPARE	0.0	0.0			SPARE	20	1	22				
23	20	1	SPARE		0.0	0.0		SPARE	20	1	24				
TOTAL				2.9	2.8	* GFCI BREAKER									

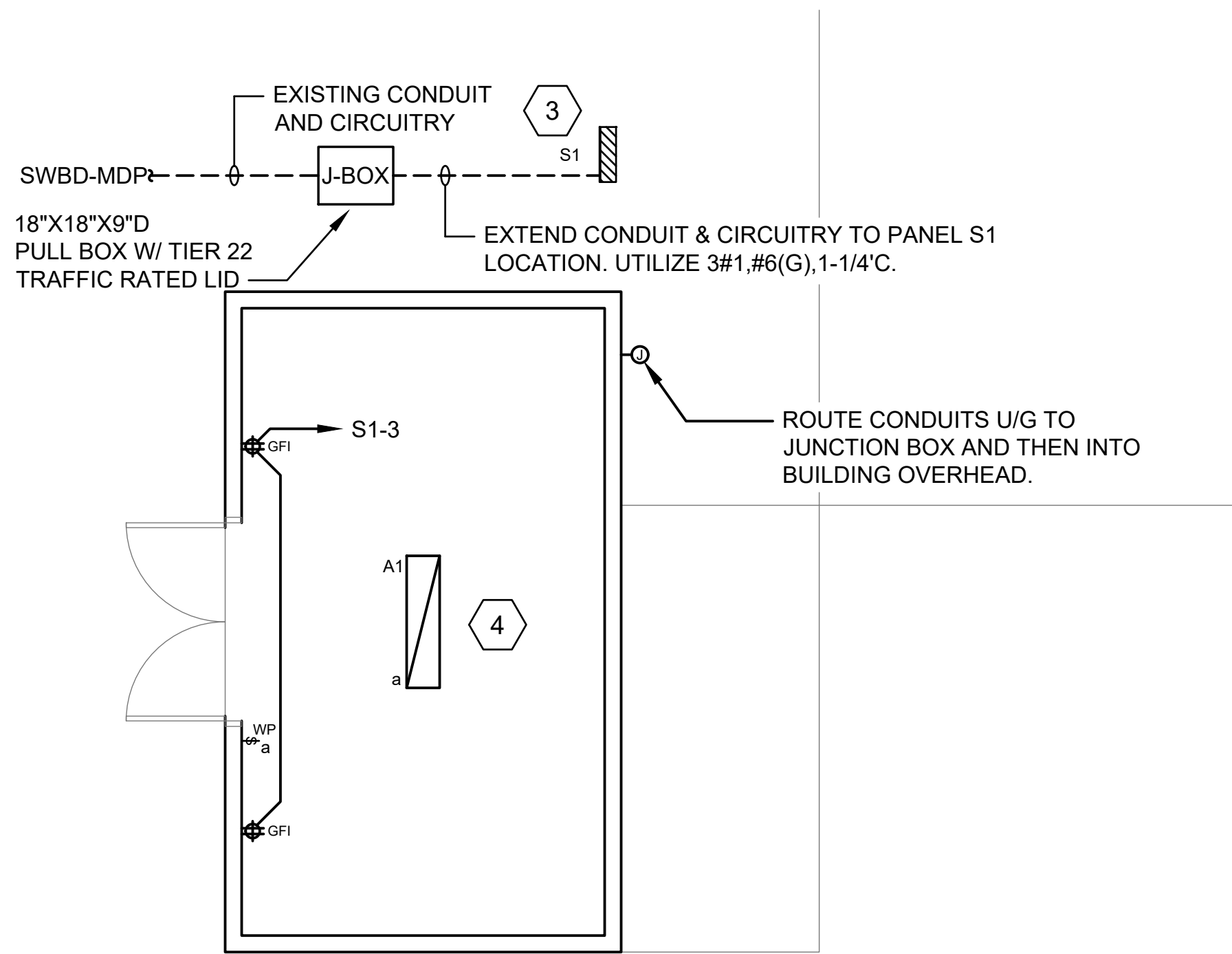
PANEL S1		LOCATION: EXTERIOR VOLT: 208/120V, 1Ø, 3W BUS: 100A		LUG LOCATION: MAIN BUS: MOUNTING:		BOTTOM FEED 100A MAIN BREAKER SURFACE				NEMA 3R ENCLOSURE PANELBOARD A/C RATING (A): 10,000	
CIRCUIT NO.	BREAKER		DESCRIPTION	PHASE LOAD (KVA)				DESCRIPTION	BREAKER		CIRCUIT NO.
	AMPS	POLES		L1	L2				AMPS	POLES	
1	20	1	LIGHTING	0.1	0.0			SPARE	20	2	2
3	20	1	GENERAL USE RECS.			0.4	0.0		-	-	4
5	20	1	SPARE	0.0	0.0			SPARE	20	1	6
7	20	1	SPARE			0.0	0.0	SPARE	20	1	8
9	20	1	SPARE	0.0	0.0			SPARE	20	1	10
11	20	1	SPARE			0.0	0.0	SPARE	20	1	12
13	20	1	SPARE	0.0	0.0			SPARE	20	1	14
15	20	1	SPARE			0.0	0.0	SPARE	20	1	16
17	20	1	SPARE	0.0	0.0			SPARE	20	1	18
19	20	1	SPARE			0.0	0.0	SPARE	20	1	20
21	20	1	SPARE	0.0	0.0			SPARE	20	1	22
23	20	1	SPARE			0.0	0.0	SPARE	20	1	24
TOTAL				0.1		0.4		* GFCI BREAKER			



2
EP103

BID OPTION #5 - LEESBURG FARM ENLARGED PESTICIDE STORAGE ELECTRICAL PLAN

SCALE: 1/4" = 1'-0"



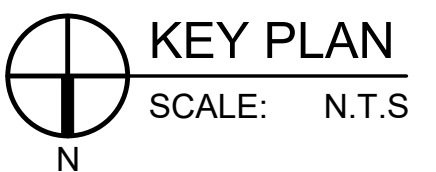
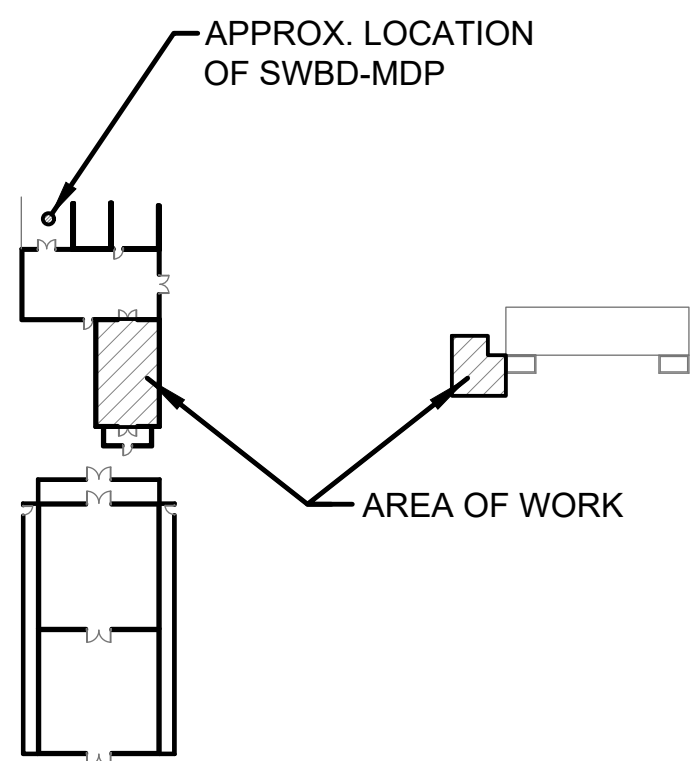
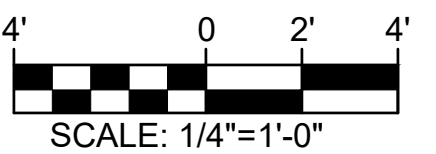
GENERAL NOTES

- ALL CONDUIT MOUNTED OVERHEAD SHALL BE UV RESISTANT, SCHEDULE 80 PVC CONDUIT.
- ALL BRANCH CIRCUITS SHALL BE XHHW-2 CONDUCTORS.
- ALL SHALL BE MOUNTED TO UNISTRUT IN THE HOOP STYLE GREENHOUSE.
- ALL JUNCTION BOXES ASSOCIATED WITH THE GREENHOUSES SHALL BE WATER-TIGHT POLYCARBONATE BOXES WITH WEATHERPROOF GASKETS AND FITTINGS.
- ALL MOUNTED RECEPTACLES SHALL BE GFI TYPE OR FED WITH A GFI BREAKER.

KEYED NOTES

- RECONNECT EXISTING CIRCUITRY TO NEW PANEL. SEE PANEL SCHEDULE FOR EXACT INFORMATION. MOUNT PANEL IN MANNER AS NOT TO DAMAGE GREENHOUSE SIDE PANELING. COORDINATE GREENHOUSE CONTROLLER LOCATION PRIOR TO ROUGH-IN PROVIDE NEMA 4X BOX FOR CONTROLLER AND CONTACTORS.
- WATER-TIGHT JUNCTION BOX, SOW CORD, AND WATER-TIGHT L5-20R RECEPTACLE. SEE DETAIL 3/E-002. (TYPICAL OF ALL OVERHEAD MOUNTED RECEPTACLES)
- APPROX. LOCATION OF NEW PANEL S1. COORDINATE EXACT LOCATION PRIOR TO INSTALLATION. PANEL SHALL BE MOUNTED ON GALVANIZED METAL STRUT FRAME. EXTEND EXISTING CONDUIT AND CIRCUITRY TO NEW PANEL LOCATION. UTILIZE UL LISTED POLARIS LUGS AS REQUIRED.
- LIGHTING FIXTURES WITHIN THIS ROOM SHALL BE SERVED FROM "S1-1".
- LIGHTING FIXTURES WITHIN THIS ROOM SHALL BE SERVED FROM "G1-1".

GRAPHIC SCALE

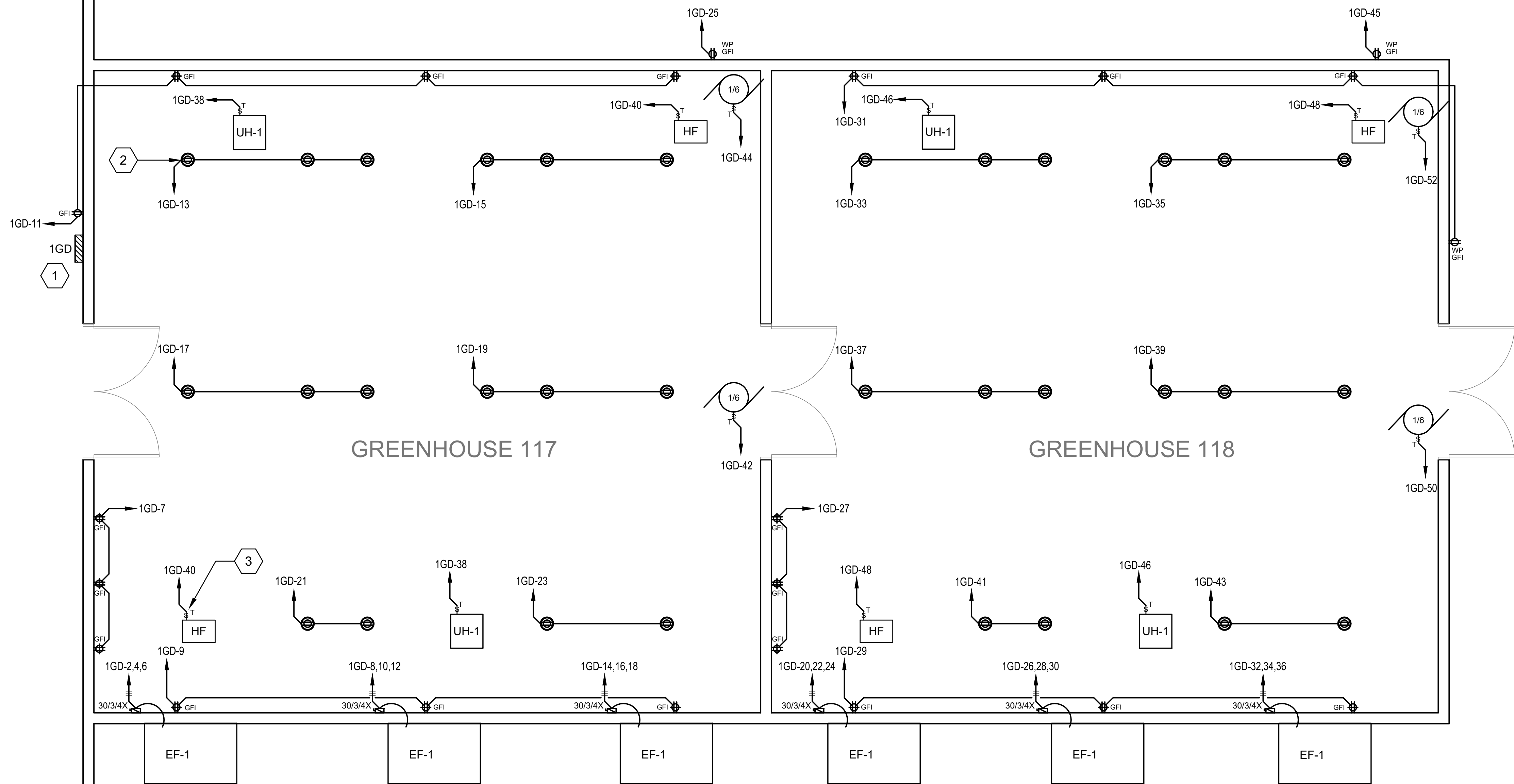


KEY PLAN
SCALE: N.T.S.

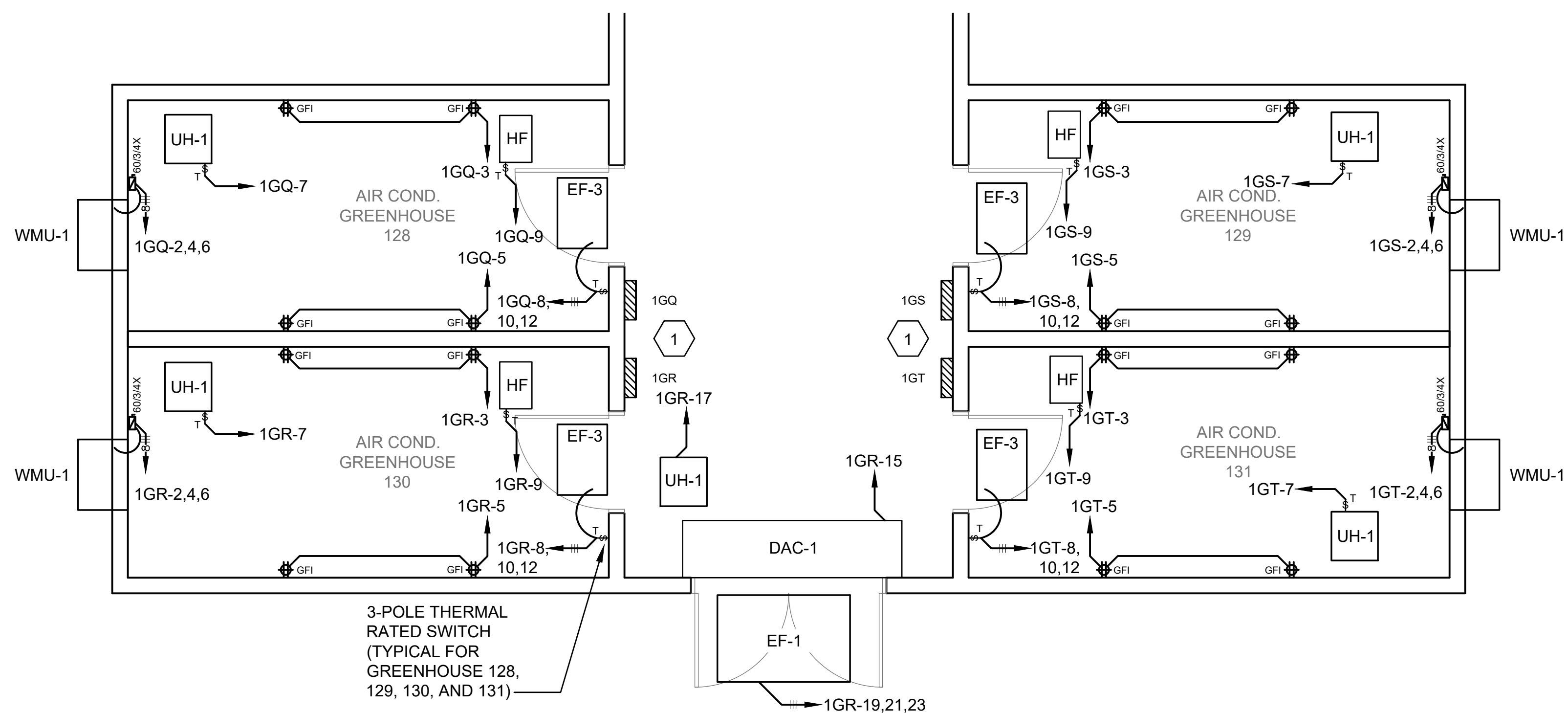
		NO.	DATE	BY	DESCRIPTION
A-E FIRM		USDA			
PROJECT MANAGER	BP	EPM	XXX		
DESIGNER	EH	PPM	XXX		
CHECKED BY	BTD	SAFETY & HEALTH	XXX		
JOB #	4825	REAL PROPERTY	XXX		
PROJECT NO.		AGRICULTURAL RESEARCH SERVICE		SHEET	
DATE		04/27/2026		FT. PIERCE, FL	
				EP103	
				105 OF 108	



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4/27/2026 9:55am



1 ENLARGED TYPICAL GREENHOUSE POWER PLAN
SCALE: 1/4" = 1'-0"



2 ENLARGED AIR CONDITIONED GREENHOUSE POWER PLAN
SCALE: 1/4" = 1'-0"

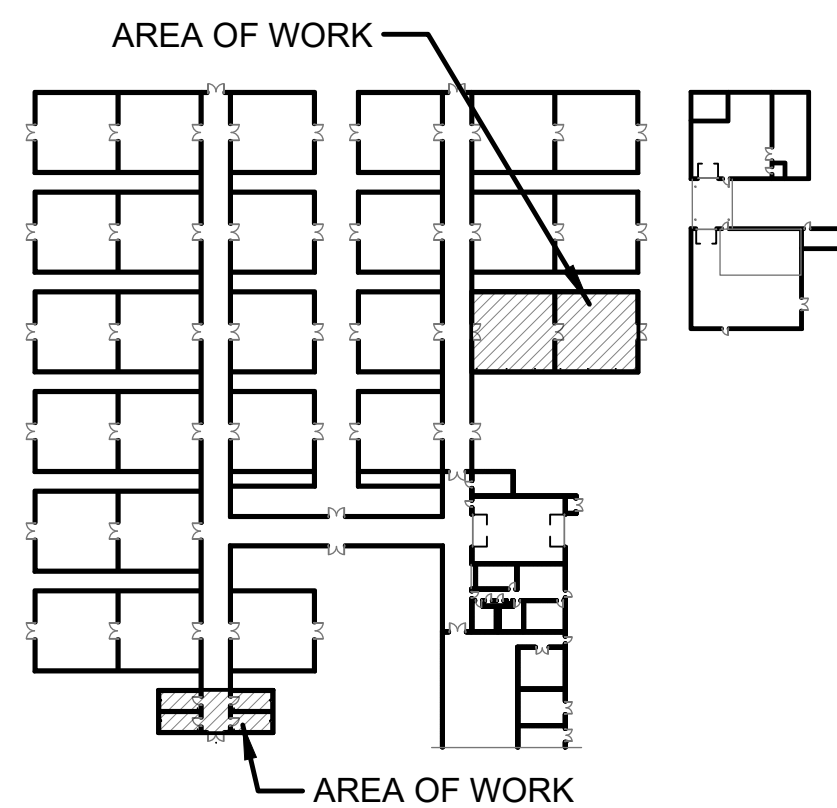
GENERAL NOTES

- ALL CONDUIT ASSOCIATED WITH THE GREENHOUSES SHALL BE UV RESISTANT, SCHEDULE 80 PVC.
- ALL CIRCUITRY SHALL BE XHHW-2.
- ALL JUNCTION BOXES ASSOCIATED WITH THE GREENHOUSES SHALL BE WATER-TIGHT POLYCARBONATE BOXES WITH WEATHERPROOF GASKETS AND FITTINGS.
- ALL RECEPTACLES SHALL HAVE A WEATHERPROOF COVER AND BE GFI TYPE OR FED WITH A GFI BREAKER.
- BREAKERS SERVING EQUIPMENT SHALL BE CAPABLE OF BEING LOCKED IN THE OPEN POSITION.

KEYED NOTES

- SEE GREENHOUSE PANEL DETAILS ON SHEET E-002 FOR TYPICAL PANEL AND MICROGROW CONTROLLER LAYOUTS.
- PROVISIONS FOR OWNER PROVIDED GROW LIGHTS. THE CONTRACTOR SHALL PROVIDE WATER-TIGHT JUNCTION BOX, SOW CORD, AND WATER-TIGHT L5-20R RECEPTACLE. SEE DETAIL 3/E-002. (TYPICAL OF ALL OVERHEAD MOUNTED RECEPTACLES)
- NEMA 3R MOTOR-RATED SWITCH AS LOCAL DISCONNECT. MOUNT ADJACENT TO UNIT. (TYPICAL)

GRAPHIC SCALE



KEY PLAN
SCALE: N.T.S.

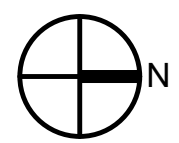
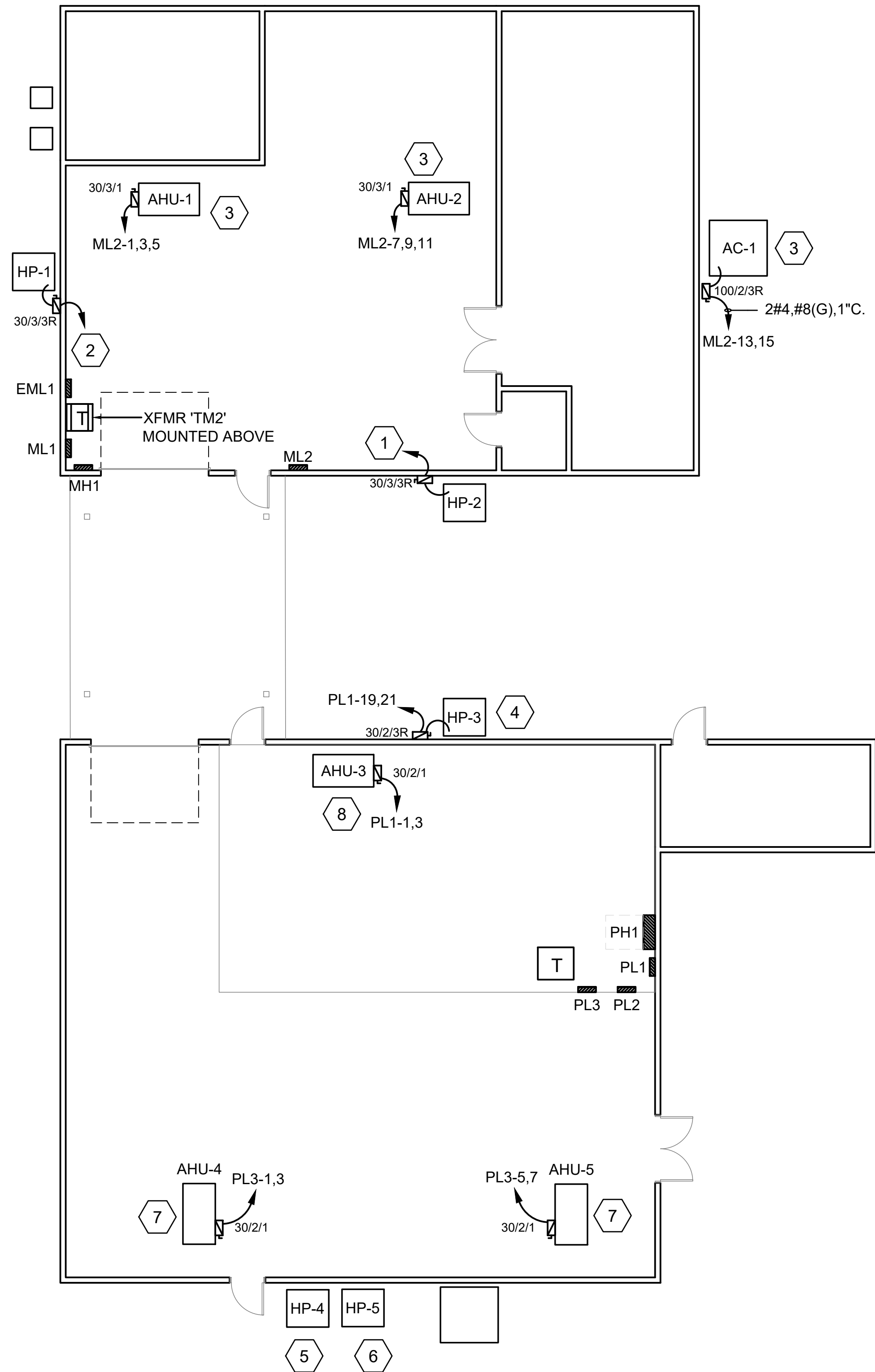
Johnson & Adams, Inc.
ELECTRICAL ENGINEERS • PLANNING • SURVEYING • MANAGEMENT
P.O. BOX 513, GREENWOOD, MISSISSIPPI 39242-0513

PROJECT MANAGER	EPM
DESIGNER	PPM
CHECKED BY	SAFETY & HEALTH
JOB #	REAL PROPERTY

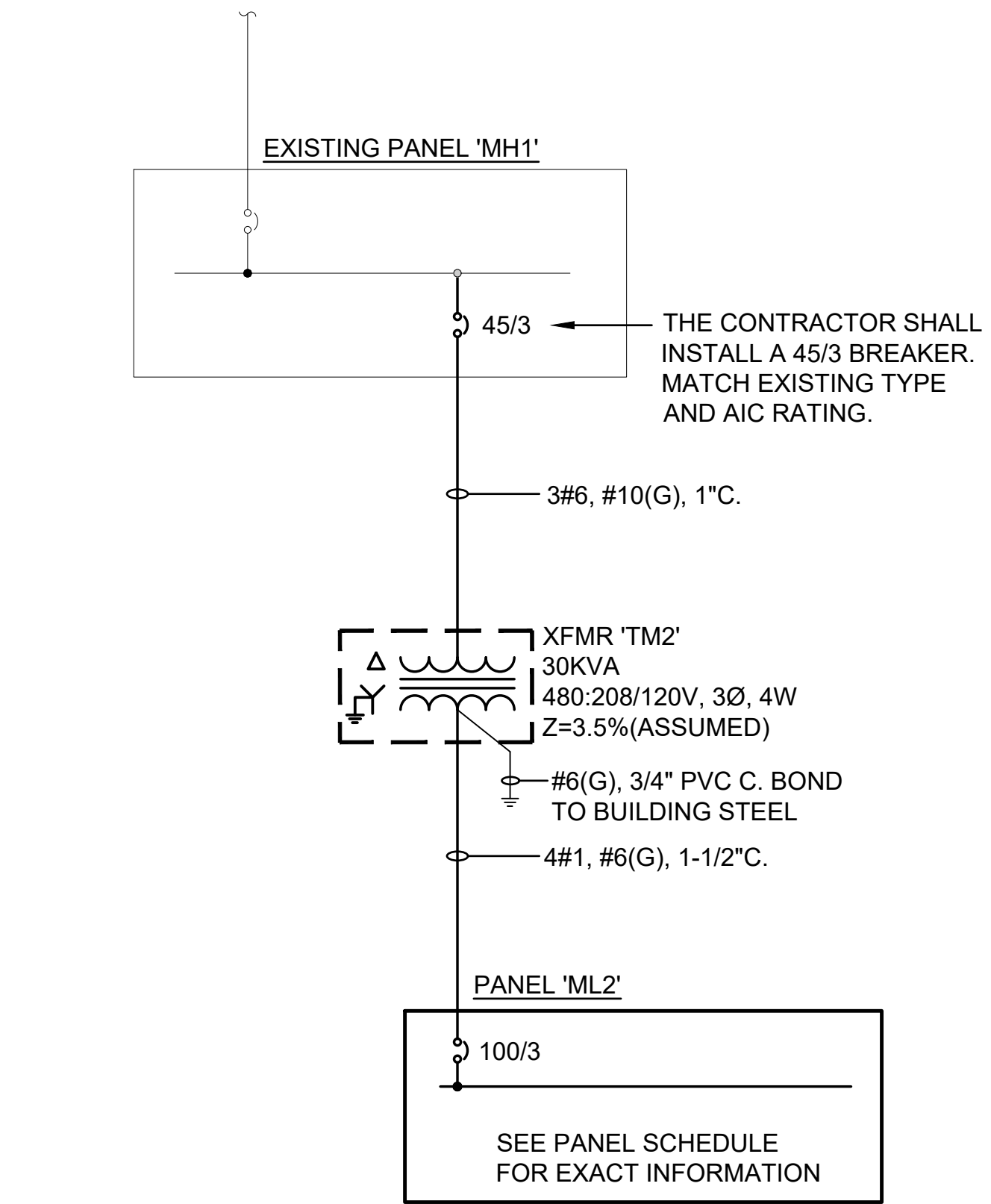
NO.	DATE	BY	DESCRIPTION
REVISIONS			
USDA Agricultural Research Service			
USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL			
ENLARGED GREENHOUSE POWER PLAN			
PROJECT NO.	AGRICULTURAL RESEARCH SERVICE	SHEET	
DATE	FT. PIERCE, FL	EP401	
4825	04/27/2026	106 OF 108	



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Apr 27, 2026 9:56am



1 ENLARGED MAINTENANCE GARAGE POWER PLAN
EP402 SCALE: 1/8" = 1'-0"



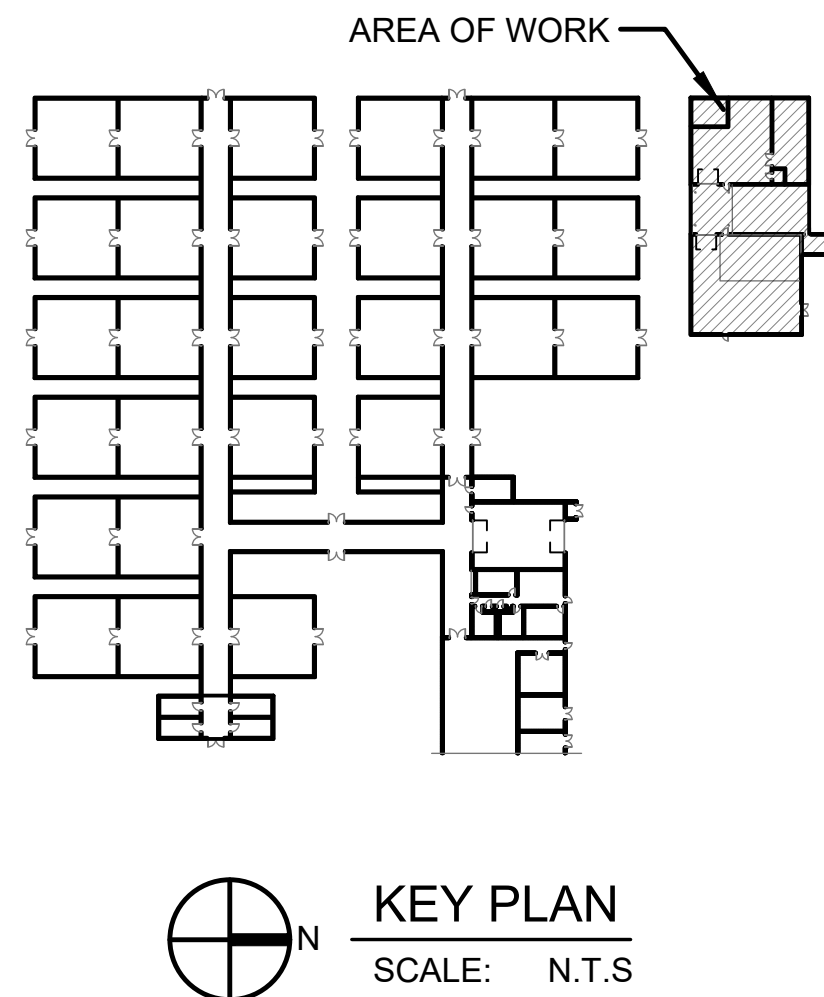
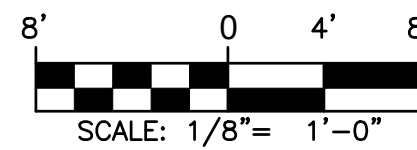
2 MAINTENANCE GARAGE SINGLE LINE DIAGRAM
EP402 SCALE: NONE

PANEL ML2			LOCATION: VOLT: 208Y/120V, 3Ø, 4W BUS: 100A		LUG LOCATION: MAIN BUS: MOUNTING:		TOP FEED 100A MAIN BREAKER SURFACE		PANELBOARD SCCR RATING (A): 10,000					
CIRCUIT NO.	BREAKER		DESCRIPTION	PHASE LOAD (KVA)						DESCRIPTION	BREAKER		CIRCUIT NO.	
	AMPS	POLES		A		B		C			AMPS	POLES		
1	30	3	AHU-1	0.7	0.0					SPARE	30	3	2	
3	-	-	-			0.7	0.0			-	-	-	4	
5	-	-	-			-		0.7	0.0	-	-	-	6	
7	30	3	AHU-2	0.7	0.0					SPARE	35	3	8	
9	-	-	-			0.7	0.0			-	-	-	10	
11	-	-	-					0.7	0.0	-	-	-	12	
13	70	3	AC-1	7.1	0.0					SPARE	15	3	14	
15	-	-	-			7.1	0.0			-	-	-	16	
17	-	-	SPARE					0.0	0.0	-	-	-	18	
19	20	1	SPARE	0.0	0.0					SPARE	20	1	20	
21	20	1	SPARE			0.0	0.0			SPARE	20	1	22	
23	20	1	SPARE					0.0	0.0	SPARE	20	1	24	
25	20	1	SPARE	0.0	0.0					SPARE	20	1	26	
27	20	1	SPARE			0.0	0.0			SPARE	20	1	28	
29	20	1	SPARE					0.0	0.0	SPARE	20	1	30	
31	20	1	SPARE	0.0	0.0					SPARE	20	1	32	
33	20	1	SPARE			0.0	0.0			SPARE	20	1	34	
35	20	1	SPARE					0.0	0.0	SPARE	20	1	36	
37	20	1	SPARE	0.0	0.0					SPARE	20	1	38	
39	20	1	SPARE			0.0	0.0			SPARE	20	1	40	
41	20	1	SPARE					0.0	0.0	SPARE	20	1	42	
TOTAL				8.5		8.5		1.4	* GFCI BREAKER					

KEYED NOTES

1. REUSE EXISTING CIRCUITRY MADE AVAILABLE DURING DEMOLITION. PROVIDE NEW DISCONNECT AND EXTEND CONDUIT AND CIRCUITRY TO UNIT.
2. PROVIDE A 35/3 BREAKER IN EXISTING PANEL ML1 (CKT 31,33,35). UTILIZE EXISTING CONDUIT AND 3#10, #10(G).
3. PROVIDE NEW CONDUIT AND CIRCUITRY FROM PANEL ML2.
4. REPLACE EXISTING 25/2 BREAKER IN PANEL PL1 (CKT 19, 21) WITH A 20/2 BREAKER. REUSE EXISTING CONDUIT AND CIRCUITRY.
5. REPLACE EXISTING 35/3 BREAKER IN PANEL PL1 (CKT 25, 27, 29) WITH A 20/2 BREAKER AND (1) SPACE. UTILIZE EXISTING CONDUIT AND 2#12, #12(G).
6. REPLACE EXISTING 35/3 BREAKER IN PANEL PL1 (CKT 31, 33, 35) WITH A 30/3. UTILIZE EXISTING CONDUIT AND CIRCUITRY.
7. PROVIDE A NEW 15/2 BREAKER IN PANEL PL3. UTILIZE 2#12, #12(G), 3/4"C.
8. REPLACE EXISTING 30/2 BREAKER IN PANEL PL1 (CKT 1,3) WITH A 15/2 BREAKER. REUSE EXISTING CONDUIT AND CIRCUITRY.

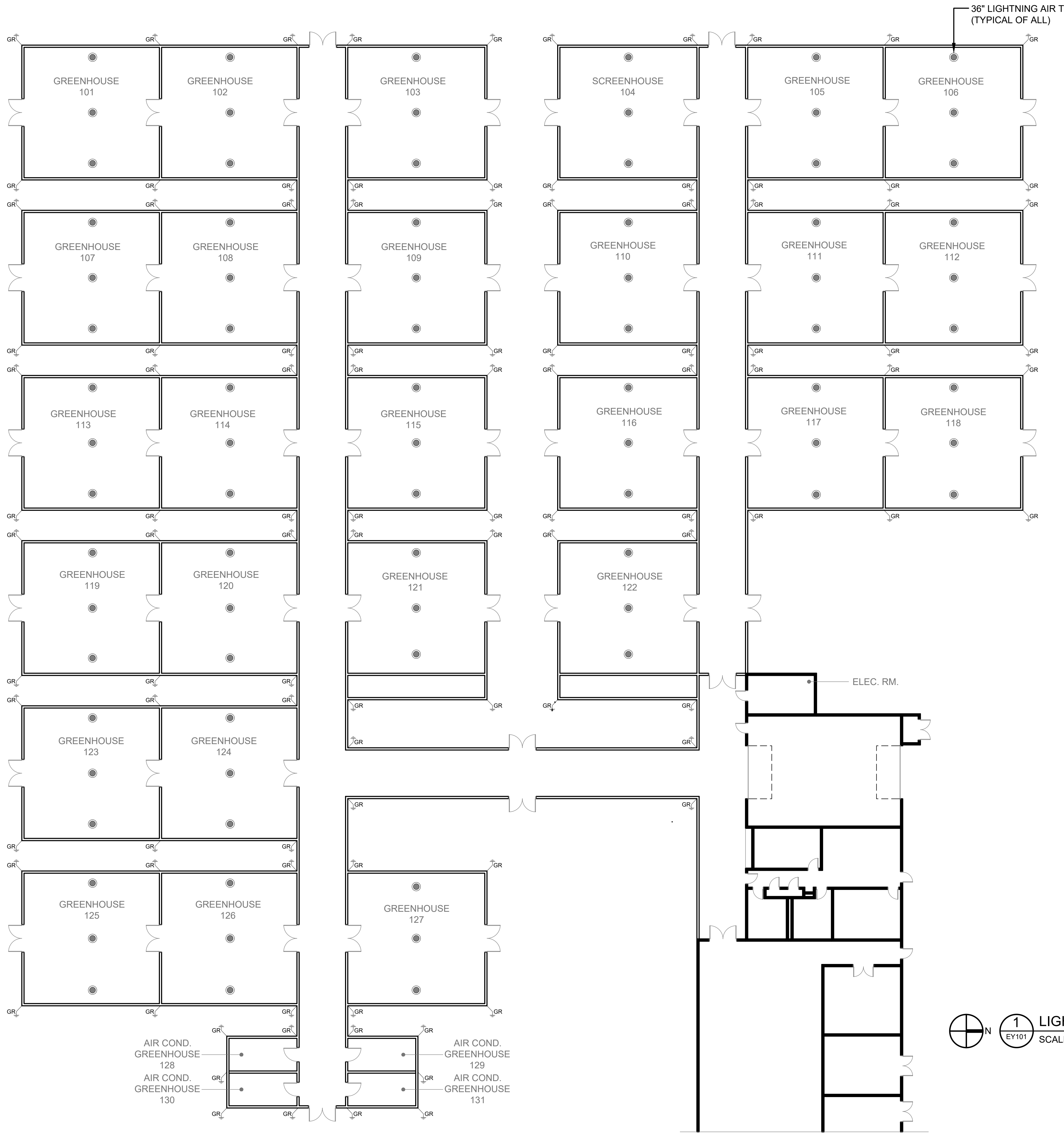
GRAPHIC SCALE



The Johnson & McAdams Firm PROJECTS • DESIGN • CONSTRUCTION • MAINTENANCE P.O. BOX 513, GREENWOOD, MISSISSIPPI 39242-0513		NO.	DATE	BY	DESCRIPTION
REVISIONS					
USDA	A-E FIRM		USDA		
	PROJECT MANAGER	EPM	XXX		
	DESIGNER	EH	PPM		
	CHECKED BY	BTD	SAFETY & HEALTH XXX		
USDA	JOB #	4825	REAL PROPERTY XXX		PROJECT NO. XXX
	DATE		04/27/2026		AGRICULTURAL RESEARCH SERVICE
FT. PIERCE, FL		SHEET		EP402	
107 OF 108					

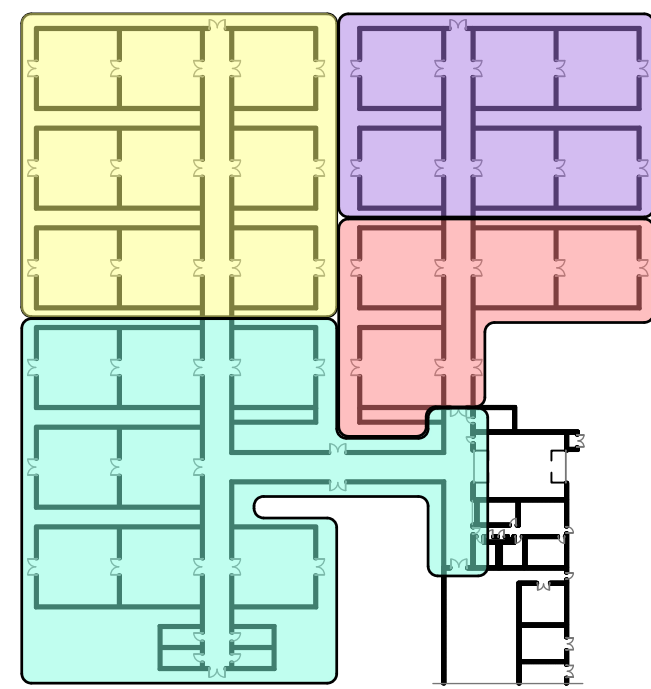


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Apr 27, 2026 9:56am



BID OPTION LEGEND

- BASE BID BID OPTION #2
BID OPTION #1 BID OPTION #3

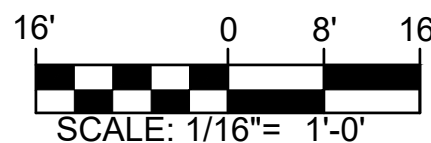


3 KEY PLAN
EY101 SCALE: N.T.S

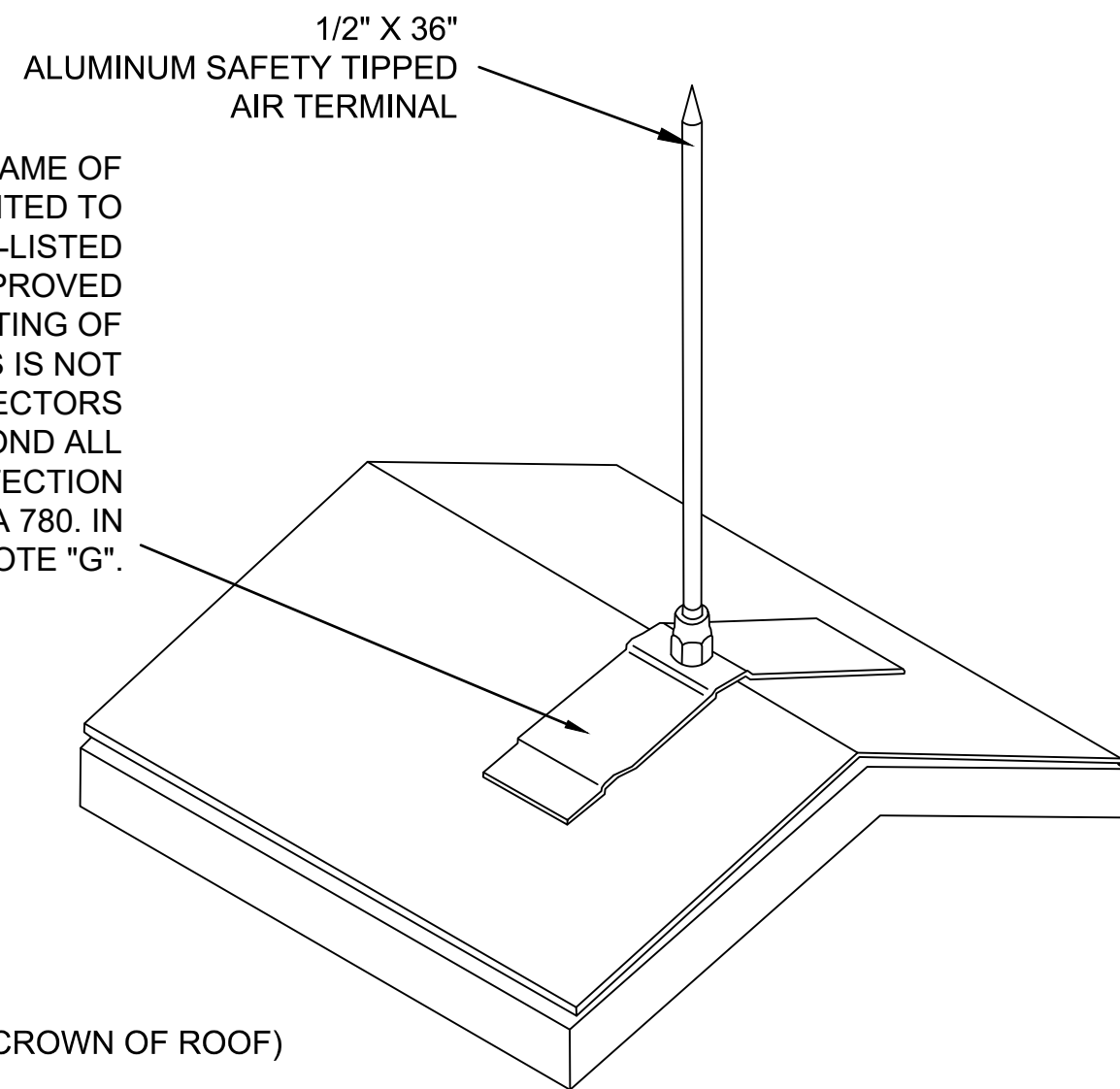
GENERAL NOTES

- LIGHTNING PROTECTION SYSTEM SHALL COMPLY WITH NFPA 780 AND SPECIFICATION 26 41 00.
- LIGHTNING AIR TERMINALS SHALL BE BONDED DIRECTLY TO ALUMINUM FRAME, ON TOP OF ROOF RIDGE.
- BOND ALUMINUM FRAME OF EACH GREENHOUSE TO A 3/4" x 10' COPPER CLAD GROUND ROD WITH A #1/0 AL DOWN CONDUCTOR PER NFPA 780.
- BOND ALUMINUM DOWN CONDUCTORS, ALUMINUM GREENHOUSE FRAME, AND EXISTING REBAR IN CONCRETE SLAB TO GROUND RODS PER NFPA 780.
- BOND LIGHTNING PROTECTION SYSTEM TO THE EXISTING GROUNDING ELECTRODE SYSTEM PER NFPA 780.
- EQUIPMENT AND ASSOCIATED WORK IN AREAS WILL BE DEFINED BY BID OPTION NUMBER SHOWN IN LEGEND AND KEY PLAN.
- COORDINATE WITH THE GREENHOUSE MANUFACTURER ON THE LOCATION AND MOUNTING METHOD OF AERIALS TO AVOID CONFLICT WITH THE OPERABLE VENTS.

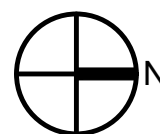
GRAPHIC SCALE



BASE IS DIRECTLY SECURED TO ALUMINUM FRAME OF ROOF RIDGE. AIR TERMINALS MUST BE MOUNTED TO GREENHOUSE ALUMINUM FRAMING USING UL-LISTED NON-PENETRATING BEAM CLAMPS OR APPROVED ADHESIVE BASES. DRILLING OR PENETRATING OF STRUCTURAL MEMBERS OR ROOFING PANELS IS NOT PERMITTED. PROVIDE BI-METALLIC CONNECTORS WHERE DISSIMILAR METALS OCCUR. BOND ALL METALLIC FRAMING TO THE LIGHTNING PROTECTION SYSTEM IN ACCORDANCE WITH NFPA 780. IN ADDITION, REFER TO GENERAL NOTE "G".



2 AIR TERMINAL MOUNTING DETAIL
EY101 SCALE: NONE



1 LIGHTNING PROTECTION PLAN
EY101 SCALE: 1/16" = 1'-0"

		NO.	DATE	BY	DESCRIPTION
		REVISIONS			
A-E FIRM		USDA			Agricultural Research Service
PROJECT MANAGER		EPM		USDA AGRICULTURAL RESEARCH SERVICE US HORTICULTURAL RESEARCH LABORATORY REPAIR TORNADO DAMAGE FT. PIERCE, FL	
BP		XXX			
DESIGNER		PPM			
EH		XXX			
CHECKED BY		SAFETY & HEALTH		LIGHTNING PROTECTION PLAN	
BTD		XXX			
JOB #		REAL PROPERTY			
4825		XXX			
		PROJECT NO.	AGRICULTURAL RESEARCH SERVICE		
		DATE	SHEET		
		04/27/2026	EY101		
			108 OF 108		
			FT. PIERCE, FL		